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European Genomic Data Infrastructure

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Framework for data governance

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1. Executive Summary

This document establishes the data governance framework for the Genome EDIC Secondary Use Framework, defining the rights and responsibilities of all actors throughout the data lifecycle for secondary use of genomic and health data. The framework identifies and delineates roles for key actors including the Genome EDIC governance structure (Assembly of Members, Central Coordination, and 1+MG Data Access Committee), data providers, 1+MG Data Holders, national DACs, IT infrastructure providers, National Coordination Points (1+MG NCP), and users with their hosting organisations. All recommendations have been consolidated into a comprehensive table designed to provide clarity on each actor's responsibilities within the framework. The governance model is built on two fundamental principles: data sovereignty (data remain in countries) and national responsibility (governments ensure necessary structures are in place). The framework balances national implementation flexibility with harmonisation requirements, particularly ensuring a homogeneous user experience across Member States. Clear distinction is made between the Genome EDIC's governing level (Assembly of Members setting principles and minimum requirements) and operational level (Central Coordination acting as one-stop-shop and ensuring quality adherence). Member States bear responsibility for establishing the required competencies in genomic and health data management, secure IT infrastructure operations, and appropriate tools for data subject interaction. The Genome EDIC Central Coordination will streamline operations by handling tasks more efficiently at the central level, avoiding duplication of efforts across countries. The framework is designed for integration with the European Health Data Space, with detailed recommendations and an overview table provided to support targeted implementation by stakeholders.

2. Contribution towards project outcomes

With this deliverable, the project has reached or the deliverable has contributed to the following project outcomes:

[Select 'Yes' (at least one) if the deliverable contributed to the key result, otherwise select 'No'. For more details of project outcomes, see here]

	Contributed
Outcome 1 Secure federated infrastructure and data governance needed to enable sustainable and secure cross border linkage of genomic data sets in compliance with the relevant and agreed legal, ethical, quality and interoperability requirements and standards based on the progress achieved by the 1+MG initiative.	Yes

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Outcome 2 Platform performing distributed analysis of genetic/genomic data and any linked clinical/phenotypic information; it should be based on the principle of federated access to data sources, include a federated/multi party authorisation and authentication system, and enable application of appropriate secure multi-party and/or high-end computing, AI and simulation techniques and resources.	No
Outcome 3 Clear description of the roles and responsibilities related to personal data and privacy protection, for humans and computers, applicable during project lifetime and after its finalisation.	Yes
Outcome 4 Business model including an uptake strategy explaining the motivation, patient incentives and conditions for all stakeholders at the different levels (national, European, global) to support the GDI towards its sustainability, including data controllers, patients, citizens, data users, service providers (e.g., IT and biotech companies), healthcare systems and public authorities at large.	No
Outcome 5 Sustained coordination mechanism for the GDI and for the GoE multi-country project launched in the context of the 1+MG initiative.	No
Outcome 6 Communication strategy – to be designed and implemented at the European and national levels.	No
Outcome 7 Capacity building measures necessary to ensure the establishment, sustainable operation, and successful uptake of the infrastructure.	No
Outcome 8	No

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Financial support to the relevant stakeholders to enable extension, upgrade, creation and/or physical connection of further data sources beyond the project consortium or to implement the communication strategy and for capacity-building.	

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3. Method

The data governance document builds on the work performed in the B1MG and B1MGplus project. Use case workshops were performed and workshops held with ethical and legal experts to capture the practical, legal and ethical requirements towards the data use.

The results were compiled for the GDI Pillar I governmental representatives (Pillar I GOV group), providing justifications for the chosen recommendations and discussing the recommendations in the context of the European Health Data Space. The recommendations were subsequently discussed and adjusted in the Pillar I GOV group.

The data governance was addressed in an incremental way, starting with the data inclusion governance and the subsequent data governance for research, comprising the data access and the data use governance. Subsequently, other areas of secondary use were addressed, comparing it with the data governance for research, starting from the healthcare reuse use case as the most different to the research use cases, then the data governance considerations for policy development for the health system and quality management in healthcare were added, equally comparing them with the other, already established data governance documents.

The resulting documents were shared with Pillar II and presented and analysed in various subsequent workshops including technical and ELSI experts, which led to a revision to adjust to the needs of the project and/or to clarify the recommendations. A further update was needed following the change of the legal basis for the data disclosure to consent and a related vote that the legal responsibility for the access decision should remain on the national level.

The outcome of this process was then integrated into a single document, highlighting in the text (including through colour) the applicability of the recommendations.

4. Work accomplished

This document summarises the recommendations for the data governance of the Genome EDIC Secondary Use Framework. The data governance describes the rights and responsibilities of the different actors along the data life cycle for secondary use.

Input from the various sources were compiled, intensive discussions took place on the content and workshops on the content held or participated in. This led to various revisions of the document. Finally, the content that was distributed over different documents was integrated into a single document and an overview table of all the recommendations compiled. Flowcharts were developed to help the reader to understand the workflows.

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5. Results

The resulting document can be found in the Annex below.

6. Next steps

The data governance is a “living document” that will be further discussed and revised in the run of the project, based on input by Pillar I, Pillar II and Pillar III as well as the 1+MG Working Groups in general.

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7. Annex: Data Governance master document

I. Reading Guide

This document summarises the recommendations for the data governance of the Genome EDIC Secondary Use Framework. The data governance describes the rights and responsibilities of the different actors along the data life cycle for secondary use.

The main actors are

- the Genome EDIC, which is in this document subdivided into the Assembly of Members as the governing board, the Central Coordination (Genome EDIC CC) and the 1+MG DAC, the data access committee established by the Genome EDIC,
- the data providers who bring in data to the Genome EDIC's secondary use framework
- the 1+MG Data Holders who are responsible for the access decision,
- the national DAC who supports the 1+MG Data Holders in their access decision,
- IT infrastructure providers on the national level,
- the National Coordination Point (1+MG NCP) who coordinates activities on the national level and is the main contact partner for the Genome EDIC as well as for the stakeholders on the national level
- the users and their hosting user organisation.

Additional roles are defined depending on functionalities, but these are overlapping in parts.

All recommendations are summarised in a table that should allow the different actors to understand their role in this Framework. Overall, the data governance follows the principles that the data remain in the countries and that in each country, the government ensures that the necessary structures are in place. Flexibility is given as much as possible for the national implementation while harmonisation and in particular homogeneity towards the user remains the guiding principle.

This means competence in the management of genomic and health data and the operation of a secure IT infrastructure must be provided in the country. Interaction with the data subjects will also fall into the national prerequisite and suitable tools must be in place that allow information of data subjects and, where applicable, tools for obtaining consent or permitting an objection. It is the responsibility of the Member States to make sure that the required IT infrastructure, personnel and tools are available for the Genome EDIC and that the inclusion of data into the Genome EDIC Secondary Use Framework is incentivised.

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For the Genome EDIC, a difference must be made between the governing level, which is formed by the Assembly of Members and the operational level of the central coordination. The Assembly of Members will take all the decisions on the principles for the operations; it will set the minimum requirements for the harmonisation in the Genome EDIC Secondary Use Framework and the rules according to which data can be included and made available for secondary use.

The central coordination of the Genome EDIC has the role to coordinate the activities and ensure quality and adherence to the 1+MG Secondary Use Framework. It will act as a one-stop-shop towards the user. In addition, it will be responsible for the tasks that can be pursued more efficiently on the central level, e.g., to avoid multiplying the work where this otherwise would have to be done in parallel in all or several Genome EDIC Member Countries.

The recommendations are always considered in the context of an integration into or service provision for the European Health Data Space. These considerations are listed in the detailed part of this document where all the recommendations are commented. An overview table is compiled that should allow a more targeted reading that aims to support implementation.

COLOUR CODING OF DISCUSSION DOCUMENT

Font colour: indication of the type of data reuse covered

All types of use: black
Several types of use covered: grey
Specific for Research: blue
Specific for Healthcare reuse: red
Specific forQM in Healthcare: brown
Specific for Policy making: green

II. INTRODUCTION

II.1. Context

The 1+MG Declaration foresees a harmonised data governance for cross-border access to harmonised genomic and health data. This document comprises the recommendations for this data governance. Three independent phases around the data life cycle for secondary use were identified:

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1. Establish formal and legal availability in accordance with 1+MG minimum requirements
2. Access application and granting access to users (data disclosure)
3. Data use by user in 1+MG SPEs

Each phase has its own controller(s) for the processing, where each controller needs to establish its own legal basis. The scenarios that are considered for the disclosure considers two different cases, one where the legal responsibility for the decision-making is on the national level (1+MG compliant data) and one where the legal responsibility for the decision-making is with the Genome EDIC legal entity. The different options for the responsibility for the disclosure are reflected in alternative scenarios.

The rights and obligations of each actor in the Genome EDIC Secondary Use Framework are described and summarised in overview tables that cover the specifics of the different phases of data inclusion, data access request management and data use as well as the aspects that apply equally to all of these phases. The entries in the overview tables each contain a link to a section about that recommendation with the detailed justification as well as the relation to the European Health Data Space (EHDS). The data governance recommendations here are principles; a translation into concrete implementation rules, technical requirements, risk and quality management measures etc. has to be worked out by the responsible working groups and committees. As an example, in this document you may see that "the introduction of minimum data models and metadata models is required". The concrete data and metadata models are not part of this document but have to be determined through WG3.

The recommendations of the data governance are based on initial workshops that identify relevant ELSI aspects along the processing chain, use case workshops and in-depth workshops on individual subjects. The conclusions were discussed in the 1+MG ELSI Working Group before translating them into the rights and obligations as listed below. The recommendations were then shared with the group as well as with GDI Pillar II. Pillar I has voted on an earlier version of the recommendation, but changes have now been introduced to address feedback and because of changes to the implementation of the access decision: consent as a legal basis and decentralised responsibility.

II.2. Principles guiding the data governance

The data governance was based on principles that are derived from the 1+MG Declaration as well as principles of scalability towards large amounts of data and a large number of users. These principles are:

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- Compliant:
  - **GDPR-compliant**: Building on data protection and privacy by design and default, the infrastructure should be intrinsically compliant with GDPR in accordance with recognised ethical standards, and facilitate compliance with subject rights.
  - **Secure**: IT infrastructure and processes must be vetted for appropriate levels of security so that the infrastructure in different member countries can be put together and form a coherent federation.
  - **EHDS-proof**: The infrastructure should be possible to integrate seamlessly into the EHDS, ensuring compatibility while adding functionality that cannot be achieved with the EHDS alone.
- Flexible and scalable:
  - **Continuous improvement**: Member countries together continue to define the scope, conditions, minimum requirements and workflows of the Framework to work on increasing realisation of the 1+MG ambitions.
  - **Implementation freedom**: Member countries have flexibility in how they incorporate the infrastructure in the national infrastructure, to account for national and regional differences, as long as their implementation satisfies requirements and implements the common functionalities.
  - **Scalable**: Implementations and procedures should allow the operations to scale to large data volumes and high numbers of users.
- Homogeneous data that remains federated:
  - **A Virtual Cohort**: The infrastructure is set up in such a way that its data behaves as much as possible as a single unified cohort.
  - **Federated**: Personal data are held in a federated IT infrastructure from which users can only download non-personal results from analysis and/or aggregation.

II.3. Scope of secondary use

The 1+MG Initiative aims to enhance personalised health and European competitiveness by fostering innovation in prevention and healthcare through a collaborative approach of pooling data across borders - not physically but through an approach towards harmonisation of data, IT infrastructure and data governance. Achieving this requires a harmonised, joint European effort to provide researchers and innovators, healthcare providers, and policymakers with access to sufficient, high-quality genomic and health data. The Genome EDIC will operate and continue to innovate a secure data infrastructure for cross-border access to genomic and related health data for secondary use. Current governance foresees use of the data in scientific research, healthcare including quality

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management and policy making in the health system in accordance with the 1+MG Declaration. All these purposes advance personalised medicine.

The following activities are covered by the four types of data reuse:

- Research

Research projects supported by 1+MG must aim to further a better understanding of diseases and/or aim at new product development with the aim to ultimately promote prevention, diagnosis and treatment of disease. The infrastructure also supports the recruitment of participants to clinical studies by facilitating the contact between R&D-performing organisations and (genetically) suitable data subjects.

Research in this text should be understood as scientific research in accordance with the GDPR Recital 159 which should be interpreted in a broad manner including technological development and demonstration, fundamental research, applied research and privately funded research. We further assume in accordance with the European Data Protection Board that scientific research projects must implement relevant sector-related methodological and ethical standards. The European Science Foundation (ESF) has published guidance on good scientific practice, which emphasises that scientific research should always aim at creating knowledge based on valid and concrete scientific questions and allow reproducibility. An important element of scientific research is the publication in a peer-reviewed journal.

- Policy development

Policy development in the health system aims to provide the government or relevant public authorities with information that helps to improve the health system, which includes access to healthcare, delivery of healthcare, cost and reimbursement approaches.

Where access to data of 1+MG is requested for policy development, the aim is to provide input into the decision making of relevant policy making bodies in the health system for the forming of policies. Applicant / user can in this case be either a policy maker directly or any entity that has the mission and sufficient legal empowerment to provide input for policy making in the health system.

The difference to scientific research is that the driving factor for policy making is the strategic and/or operational decision making towards improving generally citizen health, the health care system's delivery and performance rather than generating generalisable knowledge. Studies with a policy development goal are not always published and if they are published, they are rarely appearing in peer-reviewed journals. The results are meant to be used directly by stakeholders in the healthcare system. If a peer-reviewed publication in a scientific journal is pursued, this is a sign that not only stakeholders in the health system are

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targeted but also the scientific community. In this case it is suggested that access should take place under the governance for research.

The processing for policy development is expected to be limited to statistics as different outcomes are to be compared to come to conclusions suitable for a recommendation. Where more complex processing is pursued, such as training of a model, this should take place under the governance for scientific research.

- Quality management in healthcare

In quality management in healthcare, the outcome of a type of diagnosis or treatment that takes place in a defined healthcare organisation or for prevention recommendations given is scrutinised for potential improvement. Data from other stakeholders in healthcare as made available through 1+MG can be used in quality management as a benchmark to investigate and recommend the scope for improvement.

An access request for quality management may be initiated by a user organisation, which is a healthcare provider, seeking to benchmark its own practices against others or identify evidence on best practices, all with the aim of improving its own healthcare provision. Access may also be sought by a service provider on behalf of a healthcare provider.

The aim of quality management projects is to generate knowledge that is in the first place relevant locally, based on indicators on internal and external performance. The purpose is to make decisions on the treatment of categories of patients rather than individual patients.

Finding evidence means that technically, generating statistics will likely be enough. However, the clinical / phenotypic data must be of the highest quality, as they will be used to inform treatment recommendations.

- Healthcare reuse

Healthcare reuse consists of comparing information on patients like an actual current patient, to find evidence or at least inspiration on how to provide healthcare. Applications considered can be preventive, diagnostic or treatment.

Finding evidence means that technically, generating statistics is a suitable methodology for this purpose. However, statistics alone may not be sufficient as, e.g., in rare diseases the number of patients is low, and the specifics of a patient may not be captured. For those cases, record level access may be needed.

Sometimes, information is needed beyond record level access, such as the further disease trajectory of a patient which is not captured in the data available in the infrastructure. It was expressed as a specific user need that the possibility to reach out to the treating healthcare professional is important.

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These three types of access to patient information (i.e., having relevant statistics, having record-level information and having access to treating healthcare professionals directly) as communicated by the Use Case Working Groups are reflected in this recommendation on data access for healthcare reuse.

II.4. Elements not covered in the data governance

While this document describes a comprehensive framework of the governance of the data life cycle regarding data from individuals, covering all steps from inclusion to deletion, it is important to realise also what is not subject of the described data governance.

- Data not related to individuals

This data governance focuses on giving access to data about individuals. What is not covered in the recommendation is access to any data not related to individuals, such as reference genomes or pre-computed statistics (e.g. allele frequencies). These will be addressed in a separate data governance, which is expected to to be developed in the Genome of Europe project.

- Secondary findings

Also not covered is the screening of 1+MG data for "secondary findings", which is a deliberate screening to identify people who could benefit from certain treatment. The disclosure of data for this use case was not recommended by the 1+MG ELSI Working Group and explicitly rejected by some members of the 1+MG Special Group. As secondary findings are not explicitly arranged in unified European law, a screening for secondary findings can only happen under national healthcare legislation and will therefore be limited to national datasets. Such an application can be pursued entirely on the national level and would not need 1+MG to facilitate cross-border access.

- Third countries

It was agreed in both the 1+MG Special Group as well as the GDI Pillar I GOV Working Group that 1+MG would be open for third countries beyond the EEA. This includes that, subject to suitable safeguards and a thorough risk assessment, data could be made available to users from third countries if the envisaged use will be of general interest. Also, members from third countries are considered if they meet the qualification of a participation in an EDIC and adhere to EU ethical standards and human rights. However, neither users from third countries nor Genome EDIC members from third countries are considered in this data governance document.

- Data from countries not member to the Genome EDIC

GDI project receives funding from the European Union's Digital Europe Programme under grant agreement number 101081813.

Furthermore, this data governance only considers 1+MG compliant data or 1+MG cohort data from Genome EDIC Member Countries. Subject to the decision of the Genome EDIC Member Countries decision, also models for the inclusion of data from non-Member Countries could be considered.

III. DEFINITIONS AND QUALIFICATION OF STAKEHOLDERS

Data subject – a natural person whose data are made available in the Genome EDIC. By principle, we assume that these persons are identifiable, even though that may not be true in all cases or a person may already be deceased.

1+MG Data Provider – an organisation that, individually or jointly with other 1+MG Data Providers, can form at least one category of 1+MG minimum datasets from data it processes as controller for its own purposes, and makes these data available within the Genome EDIC Secondary Use Framework in accordance with the 1+MG Data Governance. The actual datasets may be broader and contain more data types than the respective 1+MG minimum dataset.

A 1+MG Data Provider may be, but is not necessarily, the same entity as the 1+MG Data Holder. A 1+MG Data Provider may also be a health data holder in the EHDS. 1+MG Data Providers must be based in a country that is a full member of the Genome EDIC.

They must be able to demonstrate a legal basis to make the data available in 1+MG. Such a legal basis can be established through consent under Art. 6.1 and 9.2 GDPR or through a legislative act on the country/region or Union level that explicitly mandates them for such a data sharing.

1+MG Data Holder – an organisation in a Genome EDIC Member Country that is responsible for the access decision to disclose 1+MG compliant data to Users.

1+MG Data Host – an entity that physically holds 1+MG compliant data or 1+MG cohort datasets in 1+MG compliant local IT infrastructure on behalf of the 1+MG Data Holder or the Genome EDIC who hold the data legally.

1+MG National Coordination Point (1+MG NCP) – the national node within 1+MG Member Countries that is responsible for, among other things, coordinating national stakeholders and serving as the point of contact for the country, both for the Genome EDIC and for the national stakeholders, including the public. 1+MG NCPs must have been mandated by the Genome EDIC Member Country. They must have sufficient knowledge on the legal situation in the country to advise 1+MG Data Providers. They may also be a provider of the local 1+MG IT infrastructure.

GDI project receives funding from the European Union's Digital Europe Programme under grant agreement number 101081813.

Genome EDIC – the central legal entity incorporating the 1+MG infrastructure. It is established through a Commission Implementing Decision as a European Digital Infrastructure Consortium implementing the multi-country project based on the 1+MG Declaration.

Genome EDIC Assembly of Members – the decision-making body of the Genome EDIC.

Genome EDIC Member Country - a country that has become a full member of the Genome EDIC. Genome EDIC Member Countries must have established a 1+MG NCP and the relevant 1+MG IT infrastructure locally or have access to relevant 1+MG IT infrastructure.

Genome EDIC Central Coordination (Genome EDIC CC) – support staff in the Genome EDIC; the Genome EDIC CC is responsible for operationalising the overarching elements of the Genome EDIC. It serves as a one-stop-shop for Users and a support for EU-level questions on the infrastructure also towards 1+MG Data Providers. The Genome EDIC CC is listed separately to the Genome EDIC Assembly of Members to separate more clearly responsibilities within the Genome EDIC.

1+MG DAC – Data Access Committee, which may include several domain-specific sub-committees, provided by the Genome EDIC CC that receives and reviews access requests and moderates consensus discussions on access requests where necessary.

Local DAC – group of natural or legal persons that reviews the access requests and the opinion by the 1+MG DAC relevant for the datasets for which it is in charge to come to a recommendation on the access request for the data it oversees.

1+MG IT infrastructure provider – an entity that makes an IT environment on the central, national and local level, as applicable, that may host data and metadata of 1+MG and enables local or federated workflows through suitable tools. The 1+MG IT infrastructure must be accredited for service provision in the Genome EDIC. [The details are to be defined jointly with 1+MG WG5/Pillar II.]

Genome EDIC national entities – entities established in a Genome EDIC Member Country that is contributing to the operations under the Genome EDIC Secondary Use Framework

User – the individual that is seeking and achieving access to data in 1+MG.

User organisation – the legal entity under which the User is operating.

Genome EDIC Secondary Use Framework (EDIC-SUF) – rules and requirements for data inclusion, data access management and decision and data use specifying the responsibilities of the various actors over the data life cycle as defined in the data governance and adopted by the Genome EDIC Assembly of Members

GDI project receives funding from the European Union's Digital Europe Programme under grant agreement number 101081813.

1+MG compliant datasets – datasets that can be disclosed to Users by 1+MG Data Holders as controllers in accordance with the 1+MG data governance.

1+MG cohort datasets – datasets that can be disclosed to Users by the Genome EDIC as controller in accordance with the 1+MG data governance.

Externally governed datasets – datasets that should meet certain inclusion criteria on quality and ELSI, based on which they are made findable in the and will be made available to Users in the 1+MG IT infrastructure but not according to the 1+MG data governance.

1+MG compliant local IT infrastructure – IT infrastructure used to process (including store) data for the operations under the Genome EDIC Secondary Use Framework that is compliant with the requirements for functionalities, security and performance set out by the Genome EDIC General Assembly.

Sleeping Period – times where data are removed temporarily from an active secure processing environment where the user does not need to have access. This can save costs and increase data protection.

GDI project receives funding from the European Union's Digital Europe Programme under grant agreement number 101081813.

I. OVERVIEW/SUMMARY

I.1. Aspects relevant for all processes

Data protection, risk and quality management						
	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
	1. must ensure the national implementation has the necessary resources, expertise and infrastructure to fulfil the tasks under the Genome EDIC Secondary Use Framework on the national level. 2. is responsible that any potential conflict of interest is mitigated.	1. must adopt a risk and quality management policy framework that guides the service provision and underlying IT infrastructure used, including tools. This framework should guide the data protection and cyber security implementation in accordance with the GDPR and other relevant laws as well as the risks to business continuity, performance of the services provided, loss of trust, and any other relevant factor that can impact the Genome EDIC's operations. 2. must adopt relevant security and ELSI policies for the protection of data subjects and their data.	1. must provide decision points to the Assembly of Members to adopt policies guided by advice from relevant committees and advisory boards. 2. must implement the risk and quality management policy framework, ELSI, and security policies adopted by the Assembly of Members for its work and procedures including, where applicable, external auditing or certification. 3. must provide the minimum requirements for an auditing and certification framework that ensures the implementation of the risk and quality	1. must implement secure platforms in accordance with the risk and quality management policies 2. must provide documentation for local operations, including local SOP's and risk management practices, and provide information for compliance controls by the Genome-EDIC. 3. must, at minimum, be audited on risk management and data protection for their operations in accordance with the framework defined by the Genome EDIC CC by suitable external entities and/or strive for certification of risk management system in		

		3. must ensure the establishment of an Information Security Management System that identifies, assesses, and mitigates threats that could harm the organisation's objectives, financial stability, or reputation. 4. must ensure the Genome EDIC CC has necessary resources, expertise and infrastructure to fulfil the tasks assigned to the Genome EDIC CC.	management framework, ELSI, and information and cyber security policies adopted by the Assembly of Members. These minimum requirements must be implemented by the Genome EDIC national entities such as 1+MG NCPs, 1+MG IT infrastructure providers, 1+MG Data Holders, and other relevant service providers. 4. acts as a single source of truth for all relevant information across the federated system, including providing a self-hosted documentation system. 5. Must provide a framework of contracts covering rights and responsibilities of all actors but also additional information and training materials. 6. must introduce controls for adherence monitoring beyond audits and certifications where necessary. 7. should drive an implementation of tools	accordance with ISO27001, NIST or similar. 4. must obtain the certification mentioned above if it provides an SPE that is used to temporarily pool data from other Member Countries in cases where the intended data use is not possible/feasible in a federated manner. 5. must support the performance of a DPIA by the relevant controllers. 6. must implement suitable technical and organisational measures (TOMs) that reflect the high risk of the operations. Such TOMs include secure storage and processing environments, secure and robust authentication and authorisation protocols (ensuring reliable authentication of natural persons by identity as well as by their status - e.g., as employee or affiliate of a User Organisation), access logging, encryption, reduction of identifiability of tabular		

			and services that builds on a data protection by design and default approach. 8. must consider the requirements for EHDS participation by the Genome EDIC and its national entities when pursuing the above listed tasks.	data in accordance with the ELSI, and security and risk management policies.		
Data Subjects' Rights						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
1. must ensure that data subjects are informed before the data inclusion about their principal rights and any exceptions that may apply if their data are included in the Genome EDIC Secondary Use Framework (e.g., rights to transparency, access,	1. must enable, to the extent possible and required, the information and exercise of data subjects' rights through the provision or instigation of the respective organisational framework. This includes the possibility to re-establish the link between identity and key-coded identifiers. Ideally, a data subject information portal should allow data subjects to authenticate themselves and communicate their requests in an easy but	1. must establish standard policies to ensure respect for individual rights of information rectification, withdrawal, objection, and/or erasure, as well as any standard limitations, e.g., where data cannot be removed once access is granted, or once analyses are completed or archived.	1. must support the establishment and, where applicable, the operation of IT tools that enable the compliance with data subjects' rights under the GDPR. 2. should support (with tools) the possibility of establishing a national-level information portal where data subjects can be informed about the use of their data and set the parameters for how they want to be informed. This portal should also facilitate an easy	1. must enable the implementation of the changes resulting from the exercising of data subjects' rights, via IT tools and workflows. 2. must prepare for dataset versioning for cases where data subjects have exercised their rights to rectification, withdrawal of consent to data use under the Genome EDIC Secondary Use Framework, withdrawal of consent or objection to (certain) processing under the Framework, or	1. must, where applicable, translate information for data subjects into a language they understand, at minimum where this is one of the official languages of the country. While this can be supported by automated translation, quality checks will be necessary to ensure that no wrong or misleading information is provided. 1+MG NCPs may collaborate on this task.	1. must communicate to 1+MG, through the 1+MG User Portal and as early as possible, the extent to which or the point at which their data use no longer allows the exercising of data subjects' rights, where the exercising of rights changes the dataset available for use. Examples for affected rights are the rectification or erasure from the research dataset.

rectification, portability, withdrawal of consent and objection). 2. must ensure that data subjects are informed on their potential limitations of rights in the downstream use of data. 3.	nevertheless reliable way. 2. should provide a nationally-established information portal where data subjects can be informed about the use of their data and set the parameters for their information. This portal should also facilitate an easy exercise of rights and the possibility to contact a person to ask questions. The implementation is a national competency, where each country is free to choose the measures most suited nationally or locally. It is recommended to build on pre-existing solutions or, where these do not yet exist, to strive for common solutions with the EHDS implementation on the national level, where applicable. 3. should aim for synergies with the national implementation of the EHDS.		exercise of rights, including the right to object and provide a simple means of contacting a person for asking questions. Where consent is the legal basis for the disclosure, the portal should also support the consenting process to decide upon the use of their data. 3. must provide suitable APIs that allow the communication of relevant information between the central information database and national/local information portals. 4. must provide information clauses that allow to inform data subjects in advance of a consent to the use of their data on a possible limitation to possibility to object or any other limitation of their rights that may happen during the specific project for which data are intended to be disclosed. This includes information on the limitation of the right to erasure when data are stored for reproducibility	erasure. This ensures that it is possible to keep track of datasets used where Users continue using the dataset disclosed to them and are unable to use the updated datasets.	2. should serve as a contact point for data subjects in cases where they cannot or do not want to exercise their rights and communicate through an information portal (e.g. also where they are interested in general information on the initiative rather than wanting to exercise their data subjects' rights) and in cases where this task is not performed by another Genome EDIC national entity (e.g. the 1+MG Data Provider or the 1+MG Data Holder).	2. must accept that the datasets may change during data use, due to the exercising of data subjects' rights, where no requirement for a stable, i.e. fixed, dataset can be justified.

			obligations of a research project. 5. must provide a communication channel to inform Users about any changes affecting their dataset due to the exercising of data subjects' rights during use or any other relevant communication regarding the data subjects' rights.			

I.2. Data Inclusion Governance

→ OVERVIEW DATA INCLUSION GOVERNANCE

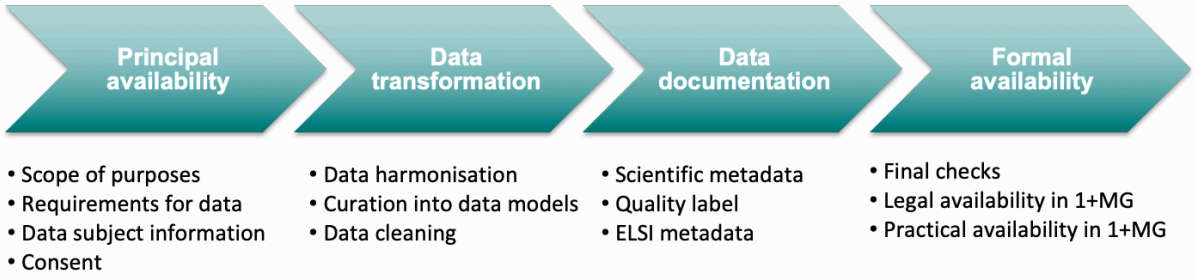


Figure 1. Steps to make data available in 1+mg/data inclusion

Establish Principal Availability						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
1. **Sign the agreement** provided by Genome EDIC CC and, where applicable, 1+MG NCP in accordance with Genome EDIC Secondary Use Framework. 2. Ensure data can **comply with the inclusion criteria** of data for the Genome EDIC. 3. **Generate information and consent sheets** based on information and guidance provided by Genome EDIC and the 1+MG NCP with **1+MG**	1. Decide on a **national strategy for the responsibility of downstream data disclosure** and nominate, where applicable, 1+MG Data Holder(s) in the country.	1. Define the **scope of purposes** for which data will be shared with Users in 1+MG and the **corresponding data governance**. 2. Adopt **technical and organisational measures (TOMs)** to comply with data protection by design and default, applicable to all actors in, contributing to, or involved in, 1+MG infrastructure. 3. Adopt the **data models and ontologies for datasets** as well as **scientific metadata**	1. Provide an **agreement that specifies the rights and obligations** of the 1+MG Data Providers, the 1+MG Data Holders (in case of 1+MG compliant data where the 1+MG Data Holder is not also the 1+MG Data Provider), and the Genome EDIC for making data available for secondary use through the Genome EDIC Secondary Use Framework. [*Moved up from below*] 2. Coordinate the definition of TOMs, data models and ontologies for datasets and		1. Provide **information and consent/opt-in/opt-out clauses as given by Genome EDIC in relevant languages** for the respective country. 2. Provide, where applicable **additional clauses** that are **specific** for the **situation in the country.** 3. Provide **relevant information to 1+MG Data Providers on legal and/or ethical framework** to be considered in the country such as	

Data Provider and 1+MG Data Holder specific information. 4. Define jointly with the 1+MG NCP, 1+MG Data Holder and the Genome EDIC an approach to obtain consent, which mitigates any potential imbalance of power as it can be found when consenting in the healthcare context. 5. Ensure that there is a **positive ethics assessment** by a relevant ethics body in the country of the 1+MG Data Provider on the inclusion of data for future availability through 1+MG. The 1+MG Data Provider has to obtain the ethics assessment from a competent ethics body unless such assessment is obtained on an overarching level. As an example, the government of a country could obtain an ethics assessment by a competent national ethics board		**models and ontologies**, at least for the minimum datasets that will be accepted in 1+MG. This should include **quality labels and quality metadata** to describe the genomes to be included in the 1+MG infrastructure. Minimum specifications for datasets of high impact based on Art. 80 EHDS must be considered. 4. Adopt minimum **ELSI metadata** models relevant for the data characterisation to allow a harmonised and machine readable description of ELSI-relevant information about the included data, Users, and types of data use. These metadata must allow integration into the EHDS data catalogue but can go beyond EHDS defined dataset descriptions.	metadata through the relevant Genome EDIC committees. 3. Coordinate and/or provide input to EHDS-related projects and to the EHDS Board on suitable data with regard to the minimum amount (depending on the field), how data should be structured (data models), how the meaning of could be structured for semantic interoperability (ontologies) and how data should described (metadata). Such an input could be given from the point of view of genomics research and genomic health, to be considered for the EHDS dataset descriptions and quality labels. 4. Provide the **phrasing of consent options** relevant for data inclusion in the Genome EDIC Secondary Use Framework. These options are to be offered in an opt-in/opt-out procedure or in a consent as the legal basis under the GDPR.		possible legal basis for data inclusion into 1+MG and potential **conditions that need to be considered**, in particular information on where/whether an **ethics assessment** is **performed** on the **national/regional level** or to be **obtained by the 1+MG Data Provider but also legislation, where applicable, on what happens to data from deceased people**.	

for the legislation that is put in place. 6. Perform a **data protection impact assessment (DPIA)** for the inclusion of data in the EDIC and do a prior consultation of the Data Protection Authorities (DPAs) on the DPIA before including any data subjects. 7. Provide information on how future communication with the data subject is established (information portal/consenting app). **8. Provide information** and **obtain** from data subjects one of the **following types of consents**, whatever is applicable: GDPR consent as a legal basis or an opt-in/opt-out depending on the legal basis on which the data can be made available within the Genome EDIC. The information must cover all relevant			5. Provide **clauses in English** that can be included in Genome EDIC Secondary Use Framework information and consent sheets for data inclusion, in accordance with the transparency and consent guidance. 6. Provide information and consent clauses for the initial consent to collect data for a later disclosure in the context of data use by third party Users, as well as for any other relevant consents, such as those related to the return of incidental findings and other information to the data subjects. The principal availability can only be established where the subsequent 1+MG Data Holder/Genome EDIC have a legal basis to actually hold the data.			

information related to the consent. 9. Provide information on and obtain consent for the 1+MG Data Holder and/or the Genome EDIC, as applicable, to hold the data for re-use and subsequent disclosure for a) supporting healthcare for other patients; b) being findable and informed on a potential participation in clinical trials or observational studies; c) allowing subject-level data discovery for research projects, quality management in healthcare and policy development. 10. **Record the consent preferences and/or, where applicable, objections** of data subjects as ELSI metadata in accordance with the definition by the Genome EDIC. This						

information must be made available alongside the data to be queried for the future use.						
Data Transformation						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
1. **Transform** data into respective **data models,** i.e. the agreed structure of the data, **and standards** as adopted by the Genome EDIC Assembly of Members and, where applicable, **clean** the data.			1. Provide **information on minimum datasets** and **data models** accepted for inclusion into the Genome EDIC Secondary Use Framework. 2. Provide **suitable written instructions**, if deemed helpful also as video, training courses and, where applicable, tools to support the data processing into these data models, for cleaning of data. The tools and information are documented in English.	1. **Support for the transformation** may be provided through the staff in the 1+MG NCP or the 1+MG IT Infrastructure Provider where it acts as Data Host. Also, an altogether different entity on the national level could be mandated. It is for the **Member Country to decide** if and who would provide such support in accordance with relevant EU and national law.	1. **Support for the transformation** may be provided through the staff in the 1+MG NCP or the 1+MG IT Infrastructure Provider where it acts as Data Host. Also, an altogether different entity on the national level could be mandated. It is for the **Member Country to decide** if and who would provide such support in accordance with relevant EU and national law.	

Data Documentation

1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
1. Produce the **relevant scientific metadata and quality label** as adopted by the Genome EDIC Assembly of Members. This may require collecting information from other parties where the 1+MG Data Provider is not the initial controller. 2. Provide **scientific metadata including a quality label** and **metadata related to ethical, legal and social aspects** (ELSI metadata) in the format adopted by the Genome EDIC Assembly of Members, including any special access and use conditions, information on incidental findings to be returned etc. 3. Provide a mock (synthetic) version of the dataset that reflects the variables and formats of the dataset to be included. [*Moved here from Data Access Governance*]	1. **May support the creation or adaptation of metadata** by the 1+MG Data Provider through funds or direct support by tasking the staff in the 1+MG NCP or the 1+MG Data Host or an altogether different entity on the national level with active support in the data documentation. It is for the Member Country to decide if and who would provide such support in accordance with relevant national and EU law. 2. Should create incentives or direct support through tools for 1+MG Data Providers to generate metadata as much as possible automatically at the time of data collection.		1. Provide **information on metadata, including quality labels for genomes and/or information in accordance with the EHDS quality label** and the corresponding **metadata models** and **standards** as required for inclusion into the Genome EDIC Secondary Use Framework. 2. Provide suitable **written instructions**, if deemed helpful also as video, **training** courses and, where applicable, **tools** to collect or transform existing metadata as well as for quality control. The tools and information are documented in English. 3. Provide a **central data catalogue** covering the relevant metadata provided by 1+MG Data Providers.		1. The **1+MG NCP provides guidance and relevant tools, such as checklists and software,** to 1+MG Data Providers on how to **ascertain and communicate additional data-specific access and use conditions**, e.g., providing checklists on national legal or ethics requirements.	

4. Provide **transparency on any special procedures** to be maintained that may **influence the data access review process**. 5. Where several 1+MG Data Providers **contribute to the same dataset** to create the required minimum dataset or a bigger joined dataset available under the same 1+MG Data Holder or the Genome EDIC, they must **coordinate with each other** to generate a full and consistent metadata set for this dataset.						

Establish Formal Availability						
1+MG Data Provider	Genome EDIC Member Countries	1+MG Data Holder	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
1. Accept that data will be made available according to the Genome EDIC Secondary Use Framework, which defines how access will	1. Strive to **modify unnecessarily restrictive and/or complicated conditions** stemming from national or regional practice or even modify legislation	1. Conclude an **agreement with the Genome EDIC** on the participation as a 1+MG Data Holder in the Genome EDIC	1. **Publish the metadata** in the Genome EDIC data catalogue subsequent to the compliance confirmation by the 1+MG NCP.	1. **Host datasets** provided by 1+MG Data Provider. (NB: the 1+MG Data Provider may also act as 1+MG Data Host.)	1. 1+MG NCP, or entities on national or regional level on behalf of the 1+MG NCP, must **verify compliance** of the data with **1+MG and national/regional**	

be granted to data and use be pursued within the Genome EDIC Secondary Use Framework (e.g., relating to access timelines, conditions of procedural fairness (justified refusals/appeals), IP conditions, appropriate fees (or lack thereof), and publication conditions acknowledging the 1+MG Data Provider). 2. Make metadata available to 1+MG NCP for checking and ultimately and, where applicable, data to 1+MG Data Host. Where the1+MG Data Provider extends the dataset made available through the Genome EDIC Secondary Use Framework with further related data, e.g. longitudinal extensions, this has to be made part of the agreement signed.	applicable to making data available for secondary use in a pan-European data infrastructure. 2. Aim to establish a **legal basis with suitable safeguards** to enable 1+MG Data Providers to make data available through the Genome EDIC Secondary Use Framework, covering the principal availability as well as ideally, where the data disclosure to the User is controlled locally, legislation that implements the 1+MG data governance for data disclosure and data use. 3. Provide suitable **IT infrastructure/tools** on the national level that allows a **reliable linking** between the **identity of the data subject** and **key-coded identifiers** for data held in under the Genome EDIC Secondary Use Framework. Genomic data and clinical/phenotypic/demographic data should be stored separately	Secondary Use Framework. 2. **Receive and subsequently hold data legally** for secondary use in accordance with the **Genome EDIC Secondary Use Framework**, which defines according to which criteria and use conditions 1+MG Data Holders grant access to data within the Genome EDIC Secondary Use Framework (e.g., relating to access timelines, conditions of procedural fairness (justified refusals/appeals), IP conditions, appropriate fees (or lack thereof), and publication conditions). 3. Provide information to be published in the data catalogue with whom the data access application will be shared as part of the data access approval procedure.		2. The infrastructure of the 1+MG Data Host must fulfil the requirements for data protection by design and default including the quality and risk management approach of the Genome EDIC Secondary Use Framework. 3. Where the 1+MG Data Host is not a 1+MG Data Holder for the data held, it must sign a data processing agreement.	**requirements**. This includes: Assessment of **legal and ethical compliance of establishing availability**. Assessment of adherence of data to the **defined data models and quality of the data**. Assessment of **completeness** and **correctness** of provided **metadata**. 2. communicate compliance to the Genome EDIC CC. This includes documentation of checks performed and the transmission of the metadata for the Genome EDIC's data catalogue (e.g. through a FAIR data point). 3. For 1+MG Cohort Data: provide information to be published in the data catalogue with whom the data access application will be shared for the data	

	and ideally be linked to a different keycoded identifier.				access approval procedure.	

I.3. Data Access Governance

→ OVERVIEW DATA ACCESS GOVERNANCE

To be created based on a more stable document - this will provide a tabular overview of responsibilities per stakeholder

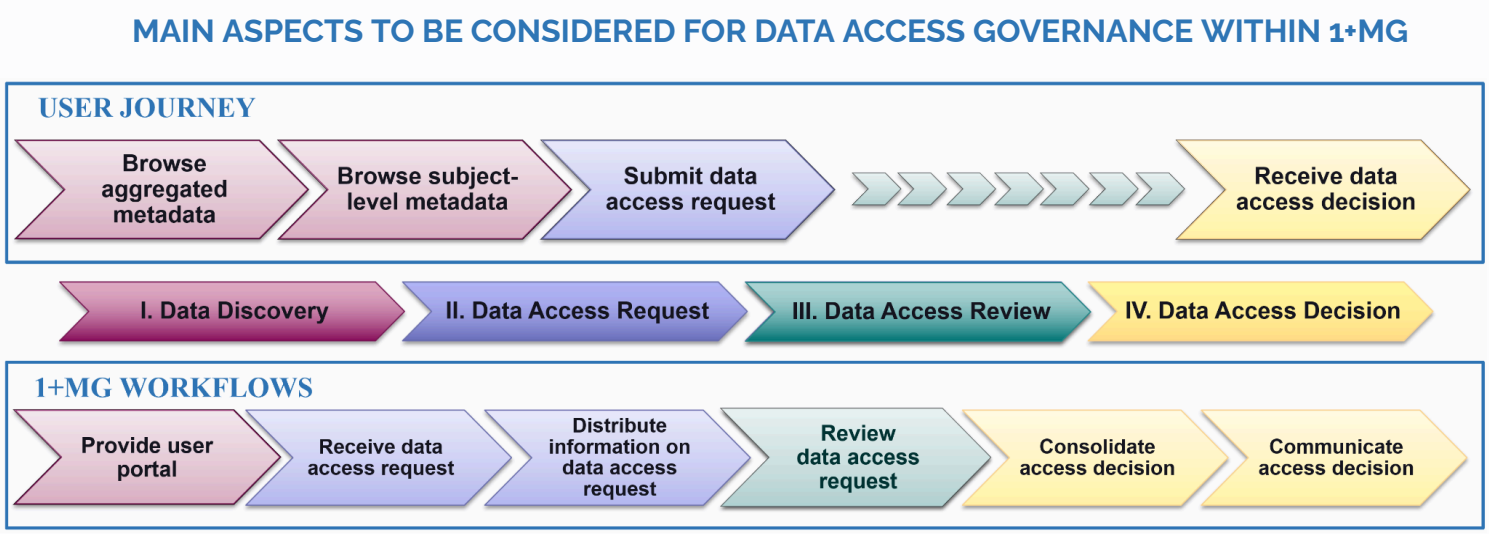


Figure 2. Required steps of the user journey and in the 1+MG workflows covering the entire data access procedure from data discovery to data access decision in the areas of scientific research, policy development in the health system and quality management in healthcare.

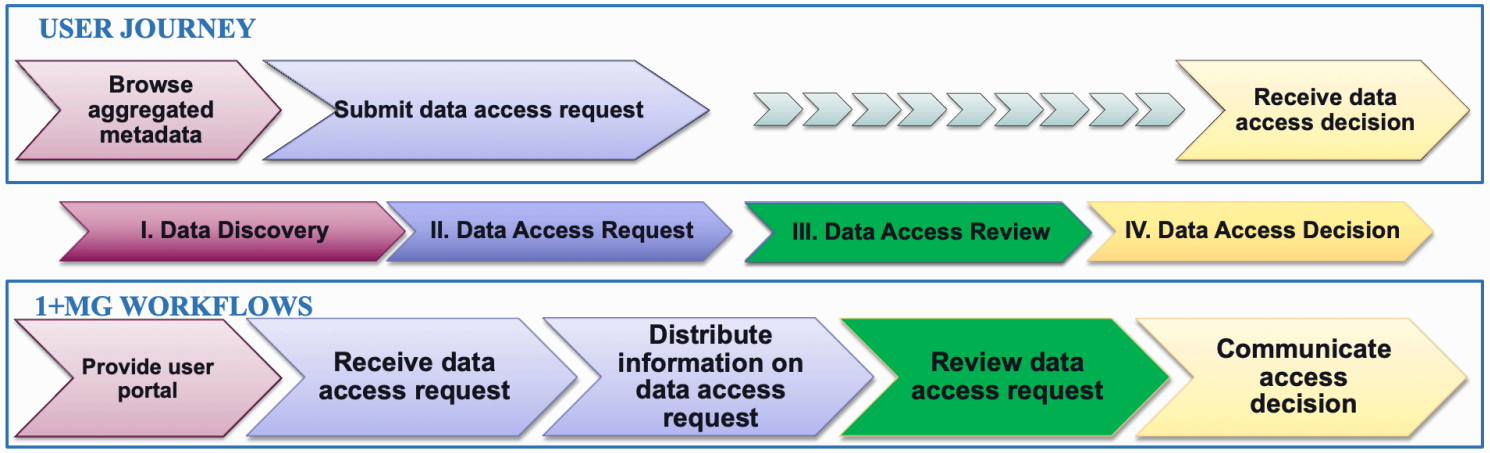


Figure 3. Required steps of the user journey and in the 1+MG workflows covering the entire data access procedure from data discovery to data access decision in the areas of healthcare reuse.

Prior Registration of the User Organisation 1+MG Data Provider						
1+MG Data Provider	Genome EDIC Member Countries	1+MG Data Holder	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	User organisation
	1. **Enable the vetting** of healthcare organisations and/or health professionals with a **valid license**, **acting under medical secrecy**.	1. **Delegate responsibility of User Organisation administration and vetting to Genome EDIC, including the signature of relevant agreement as part of the registration.**	1. Establish a **User Organisation registry** and support the **registration process** through a form in the 1+MG User Portal and by online sharing of **relevant information** with the appropriate stakeholders through			1. **Register** in the Genome EDIC User Registry **before** any **subject-level data discovery** (where applicable) and/or the **submission of any access request** with relevant information on the organisation, such as

			pre-existing forms and instructions. 2. Provide the **terms of use** and the **framework data use agreement for healthcare reuse** as well as other relevant information in the 1+MG User Portal. 3. **Vet User Organisation** with regard to the **validity** of its information and for the **scope of access** it has registered for. 4. Sign the **relevant agreements** with the User Organisation.			its legal representatives and other relevant contacts, administrative information including financial information, its legal setup, the scope of activities that can legally be pursued and the corresponding legal basis the processing will be based on. 2. Agree to the **terms of use**, which include the **joint controller arrangements** with the controller(s) for the data disclosure for subject-level data discovery. *[Research, policy development, QM]* 2. Agree to the **terms of use** and sign a **framework data use agreement for healthcare reuse** covering the principal terms for the later access by healthcare professionals and staff under their supervision. [*Healthcare*] 3. Provide **information on Users**

						as employees or otherwise persons acting under the supervision of the User Organisation **permitted to pursue subject level data discovery** and/or **access requests** (where applicable). *[Research, policy development, QM]* 3. Provide **information on healthcare professionals or other staff** acting under **medical secrecy** that are authorised to access data under this framework agreement; information must be verifiable real-time. 4. Keep all information **up to date**.

Data Discovery

1+MG Data Provider	Genome EDIC Member Countries	1+MG Data Holder	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
1. Shall **respond within a reasonable timeframe to questions by the 1+MG NCP** regarding the data collections they have made available through the Genome EDIC Secondary Use Framework if any clarification is required to progress with the data access procedures. However, where data were initially **collected by another entity** than the 1+MG Data Provider, this initial collector **can only be motivated** by the 1+MG Data Provider (or, where relevant the 1+MG NCP) to provide any missing information **but not obliged** to provide the necessary input.		1. Perform a **data protection impact assessment (DPIA)** for the principal procedures of subject-level data discovery with **prior consultation of the data protection authorities**. The DPIA should also consider that results returned to the User could aggregate information obtained across several 1+MG Data Hosts. Where the Genome EDIC is found to be a joint controller with the 1+MG Data Holder for 1+MG compliant data, the Genome EDIC may prepare and pursue the DPIA for the 1+MG Data Holder.	1. Provide a **1+MG User Portal** with **information** on the Genome EDIC Secondary Use Framework in general as well as on the data collections based on own knowledge or after retrieving such information from the national level through the respective 1+MG NCP. For information on the **secure processing environment (SPE)**, a **direct discussion** of the User should with the **1+MG SPE Provider** should be possible where information **needs go beyond the capabilities of Genome EDIC CC**. 2. Host and make available through the 1+MG User Portal the **central Genome EDIC data catalogue** characterising different data collections with rich aggregated metadata; each collection should include a **mock**	1. If acting as a 1+MG Host, provide the **IT infrastructure and/or services** necessary for the User to pursue a **subject-level data search** on the relevant scope of data available for the purpose of the User. 2. If acting as a 1+MG Host, deploy a **search engine that is linked to the 1+MG User Portal** and that has **built-in safeguards** in line with the DPIA for sufficient cybersecurity management and to mitigate the risk of a malicious or accidental compromising of the privacy of data subjects. 3. If acting as a 1+MG Host, **log the subject-level searches** and use suitable measures to **detect potential suspicious behaviour** of the User.	1. Act as a **gateway between the Genome EDIC CC and 1+MG Data Providers** where additional information by the 1+MG Data Providers is needed.	1. When pursuing subject-level data discovery, provide **in advance basic information** about the **nature of their access request** (research, policy development or QM), a **brief summary of the purpose** together with a justification for the need of subject-level data discovery and the **relevant inclusion criteria (which features should data subjects have)**, which determine the search to be performed. 2. **Agree to Terms of Use** for the subject-level data discovery. This includes in particular that the search should **not be used for any other aim than to find out about the availability** of

			(synthetic) version of the dataset. 3. Offer an **option for subject-level data discovery** to be **launched from the Genome EDIC Data Catalogue**. Accompany the discovery tool with **information on the terms of use** relevant for the User. *[Research, Policy Development, QM in Healthcare]* 4. Offer the possibility to find in the central Genome EDIC data catalogue **externally governed datasets** that are **clearly labelled as external** to the Genome EDIC Secondary Use Framework. 1. Perform a **data protection impact assessment (DPIA)** for the principal procedures of subject-level data discovery with **prior consultation of the data protection authorities**. The DPIA should also consider that results returned to the User could			sufficient data for an access request.

			aggregate information obtained across several 1+MG Data Hosts. Where the Genome EDIC is found to be a joint controller with the 1+MG Data Holder for 1+MG compliant data, the Genome EDIC may prepare and pursue the DPIA for the 1+MG Data Holder.			

Data Access Request						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	User Organisation	1+MG NCP	Users
			1. Provide **comprehensive online information** and **tutorials** on the data access request procedures. 2. Provide a **help desk** for dealing with the questions of Users; perspectively, consider **offering a chat bot** as a first line of communication. 3. Provide through a **user-friendly central**	Research, Policy Development, QM in Healthcare] **1. Confirm the access request of a User** and/or **provide information on whitelisted Users always authorised to submit requests** where such data shall always be kept up to date.	**1. Channel the information on the data access request to the relevant stakeholders, e.g., 1+MG Data Holders for 1+MG Compliant Data, where applicable, 1+MG Data Providers and/or a Local DAC as defined on the national level.**	[Research, Policy Development, QM in Healthcare] 1. build for their access request on the existing information in the User Organisation Registration database. 2. must provide information in English about the **context of the access request**, the

			1+MG User Portal the possibility to submit a data access request. The 1+MG User Portal shall capture structured and free-text information on the access request in a single access request form for the entire data available in the Genome EDIC. Access request forms have to be specific for the different applications of research, policy making, QM and healthcare reuse. The 1+MG User Portal should have features that allow guiding the applying User in the choice of the right form. The access request form must integrate into the central data access request form of the EHDS but may go beyond if additional assessments are made or if the EHDS provides for a less structured information capture. 4. Provide for an efficient possibility in the 1+MG User Portal that an access request on the same project can be submitted by different User Organisations			**User Organisation**, the natural and legal **persons** that will access the data under this request; information on their **project as well as the context** (e.g. research project from competitive funding or a policy development project following a concrete governmental request), including a **scientific and a lay summary** and a **controlled-vocabulary** characterisation; a **data analysis plan** and, where applicable, **statements on AI methodologies**; the **inclusion criteria** for data subject profiles and a justification for the **requested data types**; where applicable, information about **data to be imported**, **software needed** including **bespoke/proprietary software** to be imported, not readily available on

			acting as Joint Controllers, where a User Organisation in the application process "invites" additional User Organisations to its application. 5. Offer a staged access request procedure for healthcare reuse where Users can progressively ask for more intrusive information. 6. **Detect**, through suitable settings in the 1+MG User Portal, **obvious mistakes** (missing/wrong values) and **prevent submission** in such cases. 7. Where the Genome EDIC CC has the mandate, **reject an access request** based on **obvious mistakes** (e.g. wrong or outdated documents) detected and inform the 1+MG NCP and/or 1+MG Data Holders; **without such a mandate**: at least, **inform applicant** about the likely rejection. 8. Request **confirmation of the validity of an**			the platform; **improved datasets** that may be generated; estimated **duration of access**, any **expected sleeping periods** where no access to the data will be needed for a longer time, and requirements on the **stability of the dataset** and/or the need for **longitudinal updates** during the project; justification where for the same question, access will be pursued over a longer timeframe and potentially updated/additional data (this may in particular be applicable to healthcare QM); an **ethics assessment** by a relevant ethics body about the planned work in case of a research or a policy development project; and, where applicable, information about the **funding**

			access request to the User Organisation. 9. Ideally: Use **suitable technology to flag inconsistencies** that need **to be examined** in the data access review by the data access committee(s). 10. **Channel the information** on a **valid data access request** to the **1+MG DAC** as well as the affected 1+MG NCPs. 11. Provide the possibility for interested **Users** to **contact directly Data Providers** for high value externally governed datasets to request access **outside the Genome EDIC Secondary Use Framework**. 12. The Genome EDIC CC should have **in place suitable IT infrastructure** that allows it to support the above tasks.			**organisation** and any **peer-reviews**. 3. may invite collaborators on the same data access request where these are to act as joint controllers on the project and will want to access the data for the same project. **1. Confirm the access request of a User** and/or **provide information on whitelisted Users always authorised to submit requests** where such data shall always be kept up to date. *[Healthcare reuse]* 1. Provide information about the **context of the access request**, the **User Organisation**, the **persons** that will access the data under this request; information on **context of their access request**

						based on a **controlled-vocabulary characterisation** (general information **the medical issue** prompting access); information on **data types** and **categories of data subjects** to be investigated with a **brief justification for the requested data types**; information about **software** needed; estimated **duration of access**. 2. Follow, where applicable a 3-step approach: 1. Step: apply to gain information based on **statistical data**. 2. Step: in case statistical information is not sufficient, submit a **follow up request** under the same application to access individual level data to be able to review individual records of patients. Extend the

						original data access request with a justification to **explain why statistics based on these records are not sufficient** for them to achieve the purpose. 3. Step: in case the review of individual records does not provide sufficient information, submit a **second follow up request** under the same application to be able to get **information on individual patients** beyond the data available through the Genome EDIC Secondary Use Framework itself. Extend the original data access request with a **justification to explain why the additional information on the patient is needed** to achieve the purpose.

Data Access Review						
1+MG Data Provider	Genome EDIC Member Countries	1+MG Data Holder	Genome EDIC CC	User Organisation	1+MG NCP	Users
[1+MG Cohort Data only] [Research, Policy Development, QM in Healthcare] 1. **Review the recommendation on the access request decision including, where applicable, the opinion of an additional ethics board for vulnerabilities.** 2. **Communicate, if there is no agreement with the opinion of the 1+MG DAC, a veto together with a justification why the 1+MG DAC opinion is not suitable within 10 working days.** 3. **Optional: communicate an agreement with the 1+MG DAC opinion within the time frame of 10 working days.**		[1+MG Compliant Data] [Research, Policy Development, QM in Healthcare] 1. **Perform a review of the data access request based on an internal or external local DAC that has no conflict of interest, considering, among others, the opinion of the 1+MG DAC and, where applicable, an additional ethics board for vulnerabilities.** 2. **Communicate the outcome of the data access request review within 15 working days where no ethical challenges requiring extra scrutiny were identified and within 25 working days if additional consulting on ethical challenges is necessary.**	1. **Provide the personnel for a 1+MG DAC to review the data access request. The review shall cover the appropriateness of the chosen application track (i.e., is it a research project, a policy development project, a QM project or potentially healthcare reuse), the consistency of information provided, the suitability of the chosen data for the intended project and the detection of any vulnerability of subjects that should be considered in case of research or policy development projects, which is not already flagged through the ethics assessment provided.** 2. **Ensure that the 1+MG DAC finishes the initial assessment in**	[Healthcare reuse] 1. Confirm the ongoing validity of the credentials of the requestor affiliated to them within 1 working day or through reliable authorisation tools.	[Research, Policy Development, QM in Healthcare] **1. Channel the information on the data access request and subsequently the information on the 1+MG DAC opinion to the relevant stakeholders, e.g., 1+MG Data Holders for 1+MG compliant data, 1+MG Data Providers and, where applicable, a Local DAC as defined on the national level.** **2. Communicate the feedback received from the stakeholders on the national level to the 1+MG DAC within not more than 1 working day.** **3. Send reminders to the relevant**	

[Consequences of a veto – see below under access decision]			principle within 5 working days and transmit the outcome of the review, which includes an opinion for the decision on the access request to the 1+MG NCPs who can channel it to 1+MG Data Holders and/or relevant bodies on the national level (see below). *[NB: the feasibility of a timeframe of 5 working days needs to be reviewed after some experience with the reviewing process has been collected. In addition, a check of the technical feasibility may lead to a suspension of the review.]* 3. Ensure that all entities on the national level and all external experts involved in the review through the 1+MG DAC have signed a confidentiality agreement. 4. Give reminders to 1+MG Data Holders and/or the 1+MG NCP before the deadline for feedback is reached.		stakeholders before the deadline is reached. [Healthcare reuse] **1. For 1+MG Compliant Data: Channel the information on the data access request to the 1+MG Data Holders immediately.** **2. For 1+MG Cohort Data: channel the information on the data access request following the 1+MG DAC decision to the relevant stakeholders or databases within not more than 1 working day.** **3. For 1+MG Compliant Data: channel the information on the outcome of the checks performed by the 1+MG DAC immediately.**	

			In exceptional and justified cases an extension of the time frame may be given. [additional for Research, Policy Development] 1. In cases where the 1+MG DAC found the planned project may affect vulnerable subjects or groups, and /or may lead to discrimination, call on an additional review committee for vulnerabilities with relevant experts including a representation of affected vulnerable groups, involving where applicable, the 1+MG Data Providers. Another reason for calling on an ethics review committee is where other special ethical issues are identified, the provided ethics assessment does not give sufficient information, or a User may not have been able to obtain an			

			external ethics review at all. Obtaining an assessment by the additional review committee should not take longer than 10 working days. [Healthcare reuse] **1. Establish whether the User's healthcare professional status has remained unchanged since the signature of the framework agreement either through the User Organisation or following the verification procedures established by the 1+MG Member Country.** **2. Provide the personnel for a 1+MG DAC to review the data access request. This review shall cover the appropriateness of choice to submit the request under healthcare reuse, consistency of information provided and the suitability of the requested data for**			

			the described healthcare application. This applies to the entire step-wise approach, where applicable: the initial request as well as to any request for more information through record-level access and subsequently to reach out to the treating healthcare professional that may subsequently be raised by the User. 4. Even for this "fast track" procedure a "4-eyes" principle must be implemented to prevent corruptibility of the process. This means that in case of 1+MG Cohort Data, it has to be ensured that at minimum two people have to check the appropriateness of the access request and in case of 1+MG Compliant Data arrangements have to be made with the 1+MG Data Holder on who should co-review the data access request in the short			

			timeframe on top a single person in the 1+MG DAC. ***[NB: the feasibility of a timeframe of 1-3 working days needs to be reviewed after some experience with the review process has been collected.]***			

Data Access Decision						
1+MG Data Holder	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	1+MG DAC	1+MG NCP	Users
[1+MG Compliant Data] 1. **Perform a principal DPIA and regular updates for the data disclosure procedure and related workflows and IT infrastructure with a prior consultation of the data protection authorities before the first access is granted. Perform a specific DPIA in cases that exhibit a high risk related to the data disclosure due to the nature of the**	[Research, Policy Development, QM in Healthcare] 1. **Enable the possibility of data subjects to consent to or reject the use of their data in the planned project where the legal basis for the disclosure is consent. There may be a limited time window only within which their feedback can be considered.** **Have the possibility to object against the disclosure where the**	1. **Shall implement a right to re-dress procedures for User Organisations.**	1. **should offer modular tools to 1+MG Member Countries that support an information portal, a consenting app and a related identification management service. The tools should allow for customisation by the data subjects according to their preferences in notification and include the possibility for giving automated responses for consent requests, subject to suitable safeguards.**	[Research, Policy Development, QM in Healthcare] 1. **Integrate the feedback obtained (approvals by 1+MG Data Holders on 1+MG compliant data; national veto on 1+MG Cohort Data, where applicable; consents by data subjects/objections in case of disclosure based on legislation).** 2. **Aim to find consensus on the data access request in case of**	[Research (except for recruitment into clinical studies), Policy Development, QM in Healthcare] 1. **Inform the data subjects in case of a positive access decision about the planned project providing the lay summary and other key information as required under the GDPR within not more than 1 working day through a reliable**	

processing and/or the amount and types of data involved. *[Research, Policy Development, QM in Healthcare]* 2. **Adopt and document the access decision based on the opinions given by the local and the 1+MG DAC.** 3. **Distribute the information through the 1+MG NCP to the relevant stakeholders of the Genome EDIC Secondary Use Framework.**	**disclosure is based on legislation. Such objections should be communicated within 12 working days.** *[Research (except for recruitment into clinical studies), Policy Development, QM in Healthcare]* 2. **Establish a mediation body to help the 1+MG DAC and a local DAC to find consensus in case of deviating opinions on a data access request.** 3. **Review annually the quality of the reviews (accuracy, timeframe) and take action where necessary to mitigate bottlenecks and/or lack of expertise.** 4. **Provide 1+MG IT infrastructure on the national level that allows a reliable linking between the key-coded identifiers used in the context of the Genome EDIC Secondary Use Framework and the identity of the data subject for the subject of information provision to the data subject and**		2. **must perform a DPIA with regular updates on the information portal components and a prior consultation of the data protection authorities on the information portal and consenting app. Where applicable, support the DPIAs and prior consultation of 1+MG Data Holders.** 3. **must support the 1+MG Data Holder with the DPIA and the prior consultation of DPA in case of 1+MG Compliant Data or perform DPIA with prior consultation of DPA in case of 1+MG Cohort Data.** *[Only for 1+MG Cohort data use in Research, Policy Development, QM in Healthcare]* 4. **must adopt and document the access decision on the disclosure of relevant 1+MG Cohort data based on the opinions given by the local and the 1+MG DAC and distribute the information through**	**conflicting opinions received from the national levels or insufficient justification of exercised vetoes. If still no consensus can be reached, even with a mediation body, the respective data on which no consensus on the disclosure could be achieved will not be released.** 3. **Inform, where applicable, other 1+MG Data Holders/local DACs on the reason for the change of the original opinion and allow them to reconsider their own opinion.**	**information workflows and based on suitable 1+MG IT infrastructure including a identity management tool/service swapping between key-coded identifiers and identity as well as an information portal where data subject can find the specific information on the use of their data (for more details see below for the Genome EDIC CC).** 2. **Collect and channel consent information or information on objection to the relevant parties on the national level as well as the 1+MG DAC.**	

	the possibility to obtain consent or information on the objection of the data subject against the disclosure in case of disclosure based on legislation. 5. Strive for a legal basis to allow the use of the entity that handles the transfer between key-coded identifiers of the Genome EDIC Secondary Use Framework and identifiers in the EHDS to also serve for the purposes in the context of the Genome EDIC Secondary Use Framework.		the 1+MG NCP to the relevant stakeholders of the Genome EDIC Secondary Use Framework. 5. **must notify the User and the User Organisation on the data access decision and, in case of entirely or partially rejected access, the reasons for the rejection with a possibility to resubmit, where appropriate, or, in case the of approved access, the conditions (e.g. prices, technical setup) to the extent they had not been established before submission.** 6. **if the request is accepted, must issue a data use agreement covering all data sources and [where applicable, after approval by 1+MG Data Holders on 1+MG compliant data], encourage the User Organisation to promptly sign the data use agreement.** 7. **must have suitable IT infrastructure in place that can**			

			automatically store the information about all access requests, request assessments and decision-making steps. In case of 1+MG Compliant Data, local copies should be available for all 1+MG Data Holders through suitable APIs. 8. must publish regularly aggregate statistics on access and maintain a transparent register of access requests and approvals. This register should include only general information about the request (category of purpose; categories of User Organisations; countries of origin of User Organisation; countries from which data are requested; in case of rejection, category of reason for rejection; time to decision making).			

I.4. Data Use Governance

→ OVERVIEW DATA USE GOVERNANCE

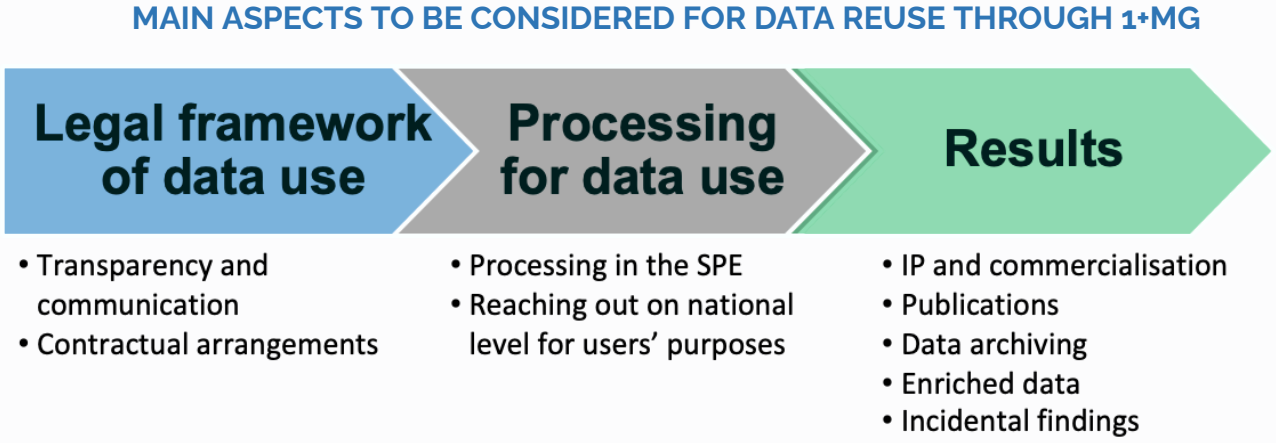


Figure 4. Required steps for data reuse through 1+MG

Transparency and Communication						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
			1. must prepare the suitable information capture through the 1+MG User Portal, ideally directly in the application form to capture the information elements		1. must provide translated information to data subjects in the language of choice in addition to the English versions	1. must make all information elements that are required by Art. 14 GDPR available through the 1+MG User Portal at the

			required under the GDPR for the information of data subjects. 2. should have End-to-end Information Management Tools, i.e. automated workflows minimising manual intervention. This should include a common European portal with IT tools and APIs that captures the relevant information from the User during the access procedures in the 1+MG User Portal and channels the information to both central and national information portals, including the data subject information portal, as well as, where applicable, 1+MG Data Holder's information management systems. Allows auditable administration and linking of all relevant information related to ongoing and completed projects: approval process, contracts, logging of access etc. Detailed requirements for the information management will be provided through a		provided by the Genome EDIC CC, at minimum, the choice should cover the official languages of the respective country. Other relevant languages of the country may be offered as a choice as well. 2. should publish information on projects and results of relevant projects on information portals in the country in the relevant language of the country. 3. must consider privacy and quality aspects when using natural language processing tools to support any translation in the context of 1+MG. 4. should strive to improve the performance of natural language processing tools, e.g. through glossaries, where such tools are used for the translations	time of submission of the data access request to ease the retrieval of information to be made available to the data subject. 2. must provide a plain language summary of the results obtained that can be published. In case of healthcare reuse, this should include, at minimum, whether the access was useful for clinical decision making. 3. must report any achievements, based on the results, to the Genome EDIC, such as policies adopted in the health system, based entirely or partially on the access to 1+MG data. The Genome EDIC should be allowed to reference these achievements if published.

			separate DPbDD characterisation of the IT infrastructure. 3. must allow the User Organisation to fulfil its transparency requirements towards the data subject through the communication tools of 1+MG; 4. must comply with the transparency requirements applicable to controllers for the data disclosure of 1+MG Cohort Data and applicable to processors for the processing in the data use context, where it acts as processor for the User Organisation. 5. must support the transparency requirements of 1+MG Data Holders for 1+MG Compliant Data. 6. should publish a transparent register of all authorised projects. 7. should support feedback collection by Users on output achieved through suitable communication channels and forms that permit a targeted		performed in the context of 1+MG.	4. must report back to 1+MG EDIC the benefits achieved for quality management based on results from data access in 1+MG. Where relevant benchmarking data on diagnosis or treatment is obtained where this evidence could be of interest for other healthcare providers as well, this information should be fed back to the Genome EDIC; the Genome EDIC reserves the right to make this data available as potentially valuable for other countries. 5. must adhere to the rules of the publication policy as also further specified below. 6. must collaborate with Genome EDIC CC on the communication of results.

			collection of relevant information. 8. should communicate results of data use to the data subjects participating whose data were used for the respective project/data use. 9. should communicate results of data use to the general public, including providing special features of a more detailed reporting of results with the support of communication experts, based on a selection of projects recommended by an advisory board including representatives of patients and the general public.			
Data use agreement						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
			1. must establish **standard Data Use Agreement templates** that includes also flexible elements for the specific information about the			1. have the right to pursue the proposed use as specified in the approved data application request

			requestor, project, and data requested, including special conditions that can be derived of the specific information, the controller for the disclosure, the fees applicable to the data use as well as general conditions applicable to data use through the Genome EDIC Secondary Use Framework that are derived from the data governance, 2. must sign a data processing agreement with the User specifying the relevant subprocessors and the TOMs established based on the Genome EDIC Secondary Use Framework, 3. must conclude **(sub)processing agreements** with all entities providing the federated 1+MG IT Infrastructure for data use, 4. must make the relevant **SPE audit reports** available to Users/User Organisation,			with 1+MG data as long as it complies with the data use agreement. 2. must agree to the conditions of the standard Data Use Agreement, in particular to the condition that the processing takes place in the 1+MG secure processing environment (SPE) or, where applicable, the SPE provided by the European Commission under the EHDS Regulation, to use 1+MG data for the approved purpose only, to respect confidentiality and not make any attempt to trace or match the identification of data subjects, to not pursue any record linkage without explicit permission, not to export any personal data, to report any errors found in the data and to report any

						data breach in accordance with the contractual provisions to the Genome EDIC CC; 3. may extend a data access request to ask for additional information beyond the already approved request, subject to the Genome EDIC Secondary Use Framework approval mechanisms.
Processing in the SPE						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
	[Research only] 1. may choose to adhere to the approach of federated data use only; in this case the data always remain with the national 1+MG SPE provider(s); alternatively countries allow data streaming and/or enable temporary data pooling for data use		1. **where applicable, should orchestrate the provision of SPEs** and **ensure that the conditions** offered in the Data Use Agreement are fulfilled. 2. must provide in the 1+MG User Portal for the possibility to request	1. when acting as 1+MG Data Host, must make data available swiftly on behalf of the 1+MG Data Holder or the Genome EDIC as required based on the Data Use Agreement or any update of the Data Use Agreement, 2. when acting as 1+MG Data Host, must ensure		1. must agree to the Genome EDIC's subprocessors to take measures to minimise the risk to the rights and freedoms of data subjects, such as to remove duplicates originating from data linkage, and/or hardening infrastructure and

	where the purpose of the processing cannot be achieved otherwise or would be too complicated to implement. 2. The choice must be clearly described and communicated as ELSI metadata to the Genome EDIC CC and the national 1+MG NCP to include into the relevant dataset descriptions and to inform the 1+MG Data Providers as this information may be relevant for the data subjects' decision to have their data included. The national policy may differ on the level of categories of sources of data (i.e., healthcare data may follow different rules to data from research).		sleeping periods or stable datasets. 3. has the power to penalise User Organisations for infringements and also **revoke legally the access** given based on the Data Use Agreement if **the conditions of the Data Use Agreement were violated**, in particular purpose limitation or confidentiality. The Genome EDIC CC will also have the power to prohibit future access by the User and/or User Organisation for a certain period depending on the nature of the infringement. The details on potential penalties will be worked out as part of the EDIC contractual and procedural framework.	only those selected data, which are covered in the Data Use Agreement, are made available to authorised Users. 3. when acting as 1+MG Data Host, must perform the steps needed for data minimisation or engage a processor who implements data minimisation 4. must provide secure processing environments (SPEs) for the data use meeting the needs of Users as agreed in the data use agreement; the SPE should be compatible with the requirements for authorised participants in the EHDS (see below for more details) and the audit/certification framework set out under the Genome EDIC Secondary Use Framework, 5. when providing an SPE, must enable the import of externally governed data from the User directly, from data providers of externally		software, i.e. limiting the potential for information breaches, where applicable. 2. should make arrangements for times when access provision is actually needed as Users may benefit from a "sleeping" period, i.e. periods where access to data is ceased during a certain timeframe when it is not necessary to access the data but access can be resumed on request later on, e.g. for validations; 3. must agree to the access scheme that requires to confirm continued need for access each year; 4. may bring in their own data and/or own software (including commercial and open-source software) subject to sufficient

				governed high value datasets or from HDABs, 6. must take steps to limit the risk to the privacy of data subjects (e.g. removing duplicates) resulting from data linkage from different sources or engage a processor who takes care of these steps, 7. when providing an SPE, should allow efficient analysis of data from multiple 1+MG Data Providers/Member Countries, ideally in a federated manner across national secure processing environments. 8. when providing an SPE, offer a suitable IT platform in compliance with the risk and quality management framework of the Genome EDIC Secondary Use Framework; the IT platform is part of a federated system on top of which tools are deployed and into which data can be imported, where applicable to allow Users the pursuit of their purpose in an		documentation and a security review by the 1+MG IT Infrastructure to prevent data breaches and potential contamination. This includes data obtained through HDABs based on the EHDS Regulation and externally governed high value datasets under other authorisation regimes. 5. may be able to study pseudonymised individual records of data subjects or reach out to the treating healthcare professional where this is necessary for their purpose and where this has been approved by the 1+MG DAC (healthcare only).

				efficient manner across data from different sources, where applicable; this includes tools for federated data analysis, which are of guaranteed quality, security and suitable licensing and should also plan for high-performance computing to attract Users. 9. when providing an SPE, must technically revoke access in case of a suspected breach of the purpose limitation or confidentiality by the User, inform immediately Genome EDIC CC (as well as other affected 1+MG SPE providers where applicable – this should ideally be enabled automatically through the 1+MG communication infrastructure) in such a case and align with Genome EDIC CC on the communication with the User and User Organisation.		

Special Processing in [healthcare reuse only]						
1+MG Data Provider	Genome EDIC Member Countries	1+MG Data Holder	Genome EDIC CC	Treating Healthcare Professionals	1+MG NCP	Users
		1. must enable that data subjects can consent to their treating healthcare professional to disclose additional data to an approved User for purposes of prevention, diagnosis or treatment; this consent should be provided prior to any specific request or access by the User and must be incorporated into the 1+MG metadata; 2. must establish mechanisms in the country that data subjects may revoke the consent given to the treating healthcare professional at any time with effect on any future disclosure.	1. must provide in the implementation of the Genome EDIC Secondary Use Framework for the possibility of Users pursuing healthcare purposes to reach out to treating healthcare professionals and establish through its central and local IT infrastructure to do so in a GDPR compliant way; 2. must ensure through the Genome EDIC Secondary Use Framework that there will be no direct contact data or identities of treating healthcare professionals or data subjects given to Users. While the overarching possibility for healthcare reuse is defined on the level of the Assembly of Members, it is the Genome EDIC CC that must provide for the practical implementation. An element of that implementation is that for healthcare reuse	1. may consent to being contacted for providing additional patient information if their patient has consented to this information; 2. may disclose additional data to Users based on the consent given by their patients, i.e. disclosing information only within the scope covered by the consent; 3. shall not disclose directly identifying information on their patient and strive to minimise indirectly identifying information to what is necessary; 4. should send an information of rejection through the 1+MG communication system if they intend not to reply to the User's request.	1. must have an information chain to reach out to treating healthcare professionals; in this case the information chain is to inform them about the additional information need of a User and allow the treating healthcare professionals to get into contact with the User for a secure information exchange or for the treating healthcare professionals to reject the information request; 2. must be able to inform the treating healthcare professional about the identity of the respective data subject in accordance with the identity management used by the treating	1. should have the possibility to request additional information from treating healthcare professionals via a request lodged in the 1+ MG communication infrastructure for update, clarification and/or other additional information if consented by the treating healthcare professionals and the data subjects; 2. must not have direct contact data or identities of the data subjects.

			purposes of the User. Users should never get in contact with data subjects themselves but only with their treating healthcare professionals if such contacting has been consented by the treating healthcare professionals and the data subject. However, also here, no unnecessary disclosure of personal data should happen. 1. must enable that data subjects can consent to their treating healthcare professional to disclose additional data to an approved User for purposes of prevention, diagnosis or treatment; this consent should be provided prior to any specific request or access by the User and must be incorporated into the 1+MG metadata; *2. must establish mechanisms in the country that data subjects may revoke the consent given to the treating healthcare professional at*		healthcare professional.This may require the possibility to link between the key-coded identifier in accordance with the Genome EDIC Secondary Use Framework and the identifier used on the side of the treating healthcare professional.	

			any time with effect on any future disclosure.			

Reaching Back To Data Subjects						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
1. [see also data access governance]: must have an information chain to reach out to data subjects with a reliable linking of project-specific identifiers and real identity link to a contacting possibility (ideally through a data subject information portal that can also be used for information about individual projects and the possibility for data subjects to exercise rights); in this case the information chain is to inform them about a possible participation in observational	1. may make legal provisions to oblige 1+MG Data Providers or, where these have not collected the data from data subjects directly, original data collectors to provide feedback on certain questions related to the data (i.e. additional metadata);		1. Must provide for an efficient and reliable information handling on consents obtained from data subjects and treating healthcare professionals on the option for a User to reach out to the treating healthcare professional. 2. Must build into the 1+MG User Portal a possibility for Users to reach out and communicate with treating healthcare professionals, linking to communication channels established on the national level. 3. Must not disclose the identity of the treating		1. [see also data access governance]: must have an information chain to reach out to data subjects with a reliable linking of project-specific identifiers and real identity link to a contacting possibility (ideally through a data subject information portal that can also be used for information about individual projects and the possibility for data subjects to exercise rights); in this case the information chain is to inform them about a possible	1. [Research only] should have the right to reach out to data subjects via the 1+ MG communication infrastructure to recruit them into research projects in accordance with the preferences expressed by the data subjects. 2. should have the right to reach out to 1+MG Data Providers for additional data on data subjects beyond the data available under the Genome EDIC Secondary Use Framework [if

studies and clinical trials; 2. should also enable going back to 1+MG Data Providers, or, where applicable, original data collectors, with the possibility to gain additional information outside 1+MG Compliant Data or 1+MG Cohort Data (updated data; new data; clarifications). This may require the possibility to link between a project specific identifier and an identifier on the 1+MG Data Provider's side. Any provision of additional information/data that the subject was not aware may be shared on them will also require a renewed information of data subjects and likely consent to provide a legal basis for this data sharing. 3. must establish with relevant actors on the national level a communication chain to reach the treating healthcare professional			healthcare professional to the User. 4. Should consider to structure exchanges with treating healthcare professionals and Users as much as possible to minimise the risk of accidental data disclosures beyond the information covered by the consent.		participation in observational studies and clinical trials; 2. should also enable going back to 1+MG Data Providers, or, where applicable, original data collectors, with the possibility to gain additional information outside 1+MG Compliant Data or 1+MG Cohort Data (updated data; new data; clarifications). This may require the possibility to link between a project specific identifier and an identifier on the 1+MG Data Provider's side. Any provision of additional information/data that the subject was not aware may be shared on them will also require a renewed information of data subjects and likely consent to provide a legal basis for this data sharing.	agreed by 1+MG Data Providers]. 3. [Healthcare reuse only] should have the right to study individual records of data subjects (that have been de-identified as much as possible) where this has been approved by the 1+MG DAC. 4. [Healthcare reuse only] should have the right to reach out to the treating healthcare professional for additional information on the data subject [if agreed by treating healthcare professional and consented by the data subject; the additional information must be limited to the information covered in the consent by the data subject and approved by the 1+MG DAC] 5. must not have any direct contact

and enable a communication exchange between the treating healthcare professional and the User without disclosing the identity of the treating healthcare professional to the User. [Healthcare reuse only]					3. must establish with relevant actors on the national level a communication chain to reach the treating healthcare professional and enable a communication exchange between the treating healthcare professional and the User without disclosing the identity of the treating healthcare professional to the User. [Healthcare reuse only]	data with the data subjects. 6. Must mandate the Genome EDIC and/or its national entities to perform a risk analysis and limit any additional information provision on data subjects that increases too much the risk to the data subjects.

IP and Commercialisation						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	User Organisations	1+MG NCP	Users
1. [research only] Must agree not to make any IP claims on results derived from the data. 2. Must agree not to pursue IP protections that would prevent or block access to or use			[research only] 1. Should record information about the results commercial exploitation of results pursued though IP	[research only] 1. May pursue IP claims on downstream discoveries or inventions;		

of any data or results drawn directly from data.			protection and publish this information;	2. Must not pursue IP claims that limit the use of 1+MG data itself; 3. Must inform the Genome EDIC CC if any IP claims were made in the context of the data use.		

Publications						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
		1. must adopt a publication policy and make it public; 2. must adopt a legal framework allowing peer reviewers' access as required by the respective journal where the work is published;	1. should follow up with Users and User Organisations on their obligation to publish and inform on publications; 2. should ensure compliance of the publication with 1+MG rules e.g. open access, acknowledgement, respect for embargos, public reporting of results; 3. should perform regular publication searches and report on published findings to identify		1. Provide translated communication into official languages of the country at least for those cases where data of the respective country were used.	1. must publish their findings where these result from a research project pursued with 1+MG data or where the results are compiled into a reports (e.g., case report for healthcare reuse, best practice findings in QM or a report to inform policy making) unless there are good reasons that a publication does not make sense or the report would

			publications that may not have been notified to it; 4. should record information about the publications in relevant databases and publish these information on the portals; 5. should provide professional communication expertise on a representative selection of publications for the general public in English, based on the recommendation of a communication advice panel involving citizens.			be protected (to be specified in a publication policy); 2. are permitted to extend their data use agreement for the access of peer-reviewers from outside the User Organisation where necessary to pursue publications; 3. must respect the 1+MG publication policy for publishing, which comprises among others the elements listed below; 4. must provide a lay summary of the publication in English as well as the language of the publication if not published in English. 5. must respect any publication embargos, and not publish during the embargo period without written confirmation of a justified

						derogation from the embargo (as part of the use conditions in the contract or as amendment to the contract); 6. must acknowledge all data sources as well as and the Genome EDIC, where applicable, the 1+MG Data Holder; 7. must report the results of completed projects (e.g., publications) to Genome EDIC CC to add to the access register, the public information portal and, where applicable, data subject information portals; 8. must publish manuscripts or reports on the results with 1+MG data as open access.

						9. must not use the publication of results from a QM project as advertising for their own organisation and/or its services; [Healthcare QM only] 10. must support the Genome EDIC in the communication of results that are specially featured for the communication with the general public.

Archiving						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
	1. Genome EDIC Member Countries must accept that archiving of data used for research projects will be required for a certain period of		1. Must develop strategies together with the Genome EDIC national entities for how an efficient archiving of data and software for	1. Must have technical data and information management systems in place to create project-specific data archives or have		1. Have the right to the archiving of data used to generate the results used for publication or IP

	time after the completion of the analysis (e.g., 10 years) even if they withdraw the data from (future) availability through the Genome EDIC Secondary Use Framework, e.g. when withdrawing from the Genome EDIC;		reproducibility can be offered to Users to make up for the prohibition of a download of personal data These strategies must strive to avoid having to store multiple copies of datasets derived from the same core data resources. . 2. Must provide the contractual implementation of data processing for reproducibility on behalf of the User Organisation.	otherwise suitable data management that allow reproducibility. Preference should be given to referencing data used rather than storing the data in an archive. This will require explicit dataset versioning. 2. Must ensure that, where data were held for project archiving only, these are deleted at the end of the last archiving period. 3. Must make provisions that analyses can be re-run with the same software also up to 10 years after the pursuit of a project. This includes archiving of analyses environments and backward-compatibility of the system.		protection where archiving is necessary for reproducibility reasons. 2. Must provide information on any required archiving periods. 3. Must archive any own data used in the project.

Enriched and Improved Data						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users

			1. Must include specific details in the data use agreement templates addressing the return of specific enriched or improved data including a reference to the legal basis for such return, where applicable.			1. must specify in access requests any enriched or improved data (e.g., annotated, harmonised, corrected) that will be generated as part of the data use that could have value for future Users; 2. must have the possibility to correct mistakes in the data even where no improvement was planned for. 3. if lawfulness can be established, must legally transfer enriched or improved data of potential value for future Users to the Genome EDIC or 1+MG Data Holders for further sharing free of charge at the end of the project; 4. must provide suitable metadata as required by the Genome EDIC

						Secondary Use Framework that describes the changes on the data if enriched or improved data are made available in the Genome EDIC.

Data Subject Related Data [healthcare only]						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
			1. Must include specific details in the data use agreement templates addressing the reporting of observations and the obligation to ask their patient for data inclusion into the Genome EDIC Secondary Use Framework. 2. Must have a dedicated communication channel in place on the 1+MG User Portal to allow the reporting of the observations from healthcare secondary use.			1. must report back to the Genome EDIC their observations related to additional variants based on their own patient for a potential further use through the Genome EDIC (only if the connection between the User and the observation can be severed, ensuring that the observation no longer qualifies as personal data

						related to the patient; otherwise consent has to be asked). 2. must offer to their patients the option of becoming part of the 1+MG initiative and making their data available if the data can meet the necessary inclusion criteria.
Incidental Findings						
1+MG Data Provider	Genome EDIC Member Countries	Genome EDIC General Assembly	Genome EDIC CC	IT Infrastructure Provider	1+MG NCP	Users
1. should establish policies and processes to responsibly handle the incidental findings, subject to national policies and to the individual's consent/express wishes where this is not regulated by law. (see 1+MG Incidental Findings Policy). Where the 1+MG Data Provider is not in direct contact with the data subject, e.g. when data were obtained from another			1. Must establish, based on relevant guidances, a list of findings to be reported by Users and regularly update it. 2. Must provide for a possibility for Users to report incidental findings and engage with the 1+MG NCPs on the best possible solution to establish a communication chain. 3. Should implement the possibility of a "no return"		1. Must establish jointly with the Genome EDIC CC and relevant entities on the national level a communication chain to inform 1+MG Data providers on any requirements for incidental findings in the context of 1+MG on the national level.	1. Must report certain clinically actionable incidental findings as defined within 1+MG to Genome EDIC if a flag for the return of incidental findings is set.

entity, the entity that collected the data directly from the data subject may have to take over some of the above tasks. 2. obtain consent by data subjects on the communication of incidental findings.			flag that interrupts or even prevents the reporting where reporting is not wanted.			

V. GENERAL ASPECTS

V.1. Measures for data protection, risk and quality management

V.1.1. Data protection, risk and quality management – Responsibility: Genome EDIC Assembly of Members

1. must adopt a risk and quality management policy framework that guides the service provision and underlying IT infrastructure used, including tools. This framework should guide the data protection and cyber security implementation in accordance with the GDPR and other relevant laws as well as the risks to business continuity, performance of the services provided, loss of trust, and any other relevant factor that can impact the Genome EDIC's operations.

2. must adopt relevant security and ELSI policies for the protection of data subjects and their data.

3. must ensure the implementation of an Information Security Management System that identifies, assesses, and mitigates threats that could harm the organisation's objectives, financial stability, or reputation.

4. must ensure the Genome EDIC CC has necessary resources, expertise and infrastructure to fulfil the tasks assigned to the Genome EDIC CC.

Justification: Approaches to operational risk and quality management are determined by the organisation's highest decision-making body. Risk management in this context includes data protection, information and cyber security, but also extending to include all relevant risks to their operations. In addition to risk management approaches, policies that focus specifically on the implementation of data protection and cyber security risks will also be required.

Part of the risk management is also to provide the prerequisite that the Genome EDIC CC is in a position to fulfil its tasks.

EHDS Relevance: The EHDS Regulation does not address a risk and quality management framework specifically. This may be addressed on the level of Member States for the operations of HDABs. Regarding data protection and security, many of the related risks will be dealt with through the implementing acts under Art. 75 EHDS. Policies adopted by the Genome EDIC Assembly of Members should consider compatibility of its own policies with the requirements set out in these implementing acts.

V.1.2. Data protection, risk and quality management – Responsibility: Genome EDIC CC

1. must provide decision points to the Assembly of Members to adopt policies guided by advice from relevant committees and advisory boards.
2. must implement the risk and quality management policy framework, ELSI, and security policies adopted by the Assembly of Members for its work and procedures including, where applicable, external auditing or certification.
3. must provide the minimum requirements for an auditing and certification framework that ensures the implementation of the risk and quality management framework, ELSI, and information and cyber security policies adopted by the Assembly of Members. These minimum requirements must be implemented by the Genome EDIC national entities such as 1+MG NCPs, 1+MG IT infrastructure providers, 1+MG Data Holders, and other relevant service providers.
4. acts as a single source of truth for all relevant information across the federated system, including providing a self-hosted documentation system.
5. Must provide a framework of contracts covering rights and responsibilities of all actors but also additional information and training materials.
6. must introduce controls for adherence monitoring beyond audits and certifications where necessary.
7. should drive an implementation of tools and services that builds on a data protection by design and default approach.
8. must consider the requirements for EHDS participation by the Genome EDIC and its national entities when pursuing the above listed tasks.

Justification: It is the task of the central coordination to ensure the implementation of all the policies, either through their own measures or by equipping the Genome EDIC national entities with a framework that is to be implemented and the relevant controls. Considering data protection requirements from the design stage will lead to a more efficient implementation. Data protection by design and default is a key component of good data protection practices.

EHDS Relevance: Requirements for the EHDS will largely come from Implementing Acts and must be considered when drawing up the more precise requirements for the Genome EDIC services.

V.1.3. Data protection, risk and quality management – Responsibility: 1+MG IT Infrastructure Provider

1. must implement secure platforms in accordance with the risk and quality management policies
2. must provide documentation for local operations, including local SOP's and risk management practices, and provide information for compliance controls by the Genome-EDIC.
3. must, at minimum, be audited on risk management and data protection for their operations in accordance with the framework defined by the Genome EDIC CC by suitable external entities and/or strive for certification of risk management system in accordance with ISO27001, NIST or similar.
4. must obtain the certification mentioned above if it provides an SPE that is used to temporarily pool data from other Member Countries in cases where the intended data use is not possible/feasible in a federated manner.
5. must support the performance of a DPIA by the relevant controllers.
6. must implement suitable technical and organisational measures (TOMs) that reflect the high risk of the operations. Such TOMs include secure storage and processing environments, secure and robust authentication and authorisation protocols (ensuring reliable authentication of natural persons by identity as well as by their status - e.g., as employee or affiliate of a User Organisation), access logging, encryption, reduction of identifiability of tabular data in accordance with the ELSI, and security and risk management policies.

Justification: Data processing in secure platforms/SPEs forms an important element to create and maintain trust. External audits and certifications will support the legitimacy and trust level needed externally and ensure the integrity and professionalisation of risk management practices internally.

EHDS-Relevance: The EHDS Regulation provides for the Art. 9.2 GDPR permission for health data users of the EHDS to process health and genetic data, which means the Regulation must provide suitable safeguards. This will include requirements for the SPE as part of the necessary safeguards. The Genome EDIC SPE(s) will need to be compliant with SPE rules described for the EHDS if the SPE(s) are supposed to be usable in the EHDS. This is particularly the case where the Genome EDIC becomes an authorised participant of the EHDS or provides SPE services for actors in the EHDS.

The requirements for 1+MG may be more stringent though than strictly required by the EHDS Regulation, particularly where they prescribe and facilitate federation instead of pooling of

data in a Commission provided SPE as an approach to cross-border data processing. Also external audit or certification are envisaged where EHDS Regulation requires only an auditing of the SPE, etc. Compatibility of technical requirements will have to be ensured (see above).

V.1.4. Data protection, risk and quality management – Responsibility: 1+MG Member Countries

1. must ensure the national implementation has the necessary resources, expertise and infrastructure to fulfil the tasks under the Genome EDIC Secondary Use Framework on the national level.
2. is responsible that any potential conflict of interest is mitigated.

Justification: Same as above, these elements are important to contain risks. Also on the national level, in particular the NCP who has an important compliance role, the entities need to be able to fulfil their tasks. In particular where an NCP has a responsibility to confirm compliance, it cannot be that it assesses its own work. Also 1+MG Data Holders may have conflicts of interest if it is at the same time a 1+MG IT infrastructure provider or a 1+MG Data Provider engaged in research with these data.

EHDS Relevance: Where Genome EDIC national entities also take a role in the national EHDS implementation, the ability to fulfil their tasks is equally important. Art. 55.2 EHDS also requires that HDABs are sufficiently staffed and equipped. The conflict-of-interest management is addressed in Art. 55.3 EHDS, recommending segregation between different tasks of HDABs such as the assessment of health data access requests and the provision of SPEs. National entities under the Genome EDIC Secondary Use Framework can contribute to a segregation by providing independent components for the HDAB tasks in the Member States.

V.2. DATA SUBJECTS' RIGHTS

V.2.1. Data Subjects' Rights – Responsibility: Genome EDIC Assembly of Members

1. must establish standard policies to ensure respect for individual rights of information rectification, withdrawal, objection, and/or erasure, as well as any standard limitations, e.g., where data cannot be removed once access is granted, or once analyses are completed or archived.

Justification: The rights of data subjects must be respected, which means rules for implementation have to be foreseen. However, it may not always be possible to grant these rights (as also provided for in the GDPR). To properly manage such situations and to make them transparent, a policy should be established that guides how the exercising of rights must be handled for an efficient and reliable implementation and under which circumstances and conditions must be in place where limitations of the rights may apply. For example, the Genome EDIC CC must provide the framework that allows it to guarantee a stable datasets. Data subjects' rights can be limited if this is necessary for the processing (or in some cases, if the exercising of rights leads to a disproportionate effort, such as in direct information obligations when data were not collected from the data subject)..

EHDS Relevance: Data subjects' rights of the GDPR apply also under the EHDS Regulation. Practical considerations on the data subjects' rights will be addressed in the JA TEHDAS2, where we will aim for synergies. However, ultimately the national policies will determine the implementation. Each Genome EDIC Member Country and Genome EDIC national entities should aim to align for any synergies (see also below).

V.2.2. Data Subjects' Rights – Responsibility: Genome EDIC CC

1. must support the establishment and, where applicable, the operation of IT tools that enable the compliance with data subjects' rights under the GDPR.
2. should support (with tools) the possibility of establishing a national-level information portal where data subjects can be informed about the use of their data and set the parameters for how they want to be informed. This portal should also facilitate an easy exercise of rights, including the right to object and provide a simple means of contacting a person for asking questions. Where consent is the legal basis for the disclosure, the portal should also support the consenting process to decide upon the use of their data.
3. must provide suitable APIs that allow the communication of relevant information between the central information database and national/local information portals.
4. must provide information clauses that allow to inform data subjects in advance of a consent to the use of their data on a possible limitation to possibility to object or any other limitation of their rights that may happen during the specific project for which data are intended to be disclosed. This includes information on the limitation of the right to erasure when data are stored for reproducibility obligations of a research project.
5. must provide a communication channel to inform Users about any changes affecting their dataset due to the exercising of data subjects' rights during use or any other relevant communication with regard to the data subjects' rights.

Justification: The procedures around data subjects' rights should be supported by tools that enable an efficient and reliable execution of data subjects' information and/or requests, when required.

Where data subjects cannot be served through a suitable portal or other measures on the national level, the Genome EDIC should offer its own information portal as an alternative solution. However, this should be seen as an exception.

Any limitation of rights must be communicated to the data subjects to allow them to take a fully informed decision on whether they are comfortable with their data being made available for a specific project and give consent or, in case of the disclosure being based on legislation, not to exercise an objection. For this reason, the Genome EDIC CC receives the information from the User on the requirements for dataset stability as part of the application process.

As the main interface to the User, it is the responsibility of the Genome EDIC CC to communicate any changes made related to the dataset, even if the information on these changes is generated locally.

EHDS Relevance: Specific for the role of the Genome EDIC and its responsibilities, which is not governed by the EHDS Regulation. However, the national implementation of the EHDS could benefit from the considerations made and tools provided through the Genome EDIC. This would particularly be the case where a Genome EDIC national entity acts as an HDAB or on behalf of an HDAB.

V.2.3. Data Subjects' Rights – Responsibility: 1+MG Data Provider

1. must ensure that data subjects are informed before the data inclusion about their principal rights and any exceptions that may apply if their data are included in the Genome EDIC Secondary Use Framework (e.g., rights to transparency, access, rectification, portability, withdrawal of consent and objection).

2. must ensure that data subjects are informed on their potential limitations of rights in the downstream use of data.

Justification: The 1+MG Data Providers are those closest in contact with the data subject and are, therefore, responsible for supporting the communication with the data subjects, at least when introducing them to the initiative. – See also: Data Inclusion Policy

EHDS Relevance: Specific for the role of the 1+MG Data Providers and their responsibilities, which are not governed by the EHDS Regulation but only under the GDPR.

V.2.4. Data Subjects' Rights – Responsibility: Genome EDIC Member Country

1. must enable, to the extent possible and required, the information and exercise of data subjects' rights through the provision or instigation of the respective organisational framework. This includes the possibility to re-establish the link between identity and key-coded identifiers. Ideally, a data subject information portal should allow data subjects to authenticate themselves and communicate their requests in an easy but nevertheless reliable way.

2. should provide a nationally established information portal where data subjects can be informed about the use of their data and set the parameters for their information. This portal should also facilitate an easy exercise of rights and the possibility to contact a person to ask questions. The implementation is a national competency, where each country is free to choose the measures most suited nationally or locally. It is recommended to build on pre-existing solutions or, where these do not yet exist, to strive for common solutions with the EHDS implementation on the national level, where applicable.

3. should aim for synergies with the national implementation of the EHDS.

Justification
Data protection by design and default requires that the processing is designed in a way that enables data subjects to easily exercise their rights. The introduction of a data subject information portal seems to be the simplest solution, particularly as it could also allow data subjects to be informed in the way they wish. – See also: Data Access Policy

EHDS Relevance
The requirements of the GDPR also apply to processing activities under the EHDS, and countries need to develop strategies for how these requirements can be realised within their national EHDS implementation. In addition, going back between key-coded identifiers and identities is needed in the context of significant findings. It would therefore make sense that countries strive for identity management solutions that can be used for both EHDS and Genome EDIC related key-coding in a compatible manner.

V.2.5. Data Subjects' Rights – Responsibility: 1+MG NCP

1. must, where applicable, translate information for data subjects into a language they understand, at minimum where this is one of the official languages of the country. While this can be supported by automated translation, quality checks will be necessary to ensure that no wrong or misleading information is provided. 1+MG NCPs may collaborate on this task.

2. should serve as a contact point for data subjects in cases where they cannot or do not want to exercise their rights and communicate through an information portal (e.g. also where

they are interested in general information on the initiative rather than wanting to exercise their data subjects' rights) and in cases where this task is not performed by another Genome EDIC national entity (e.g. the 1+MG Data Provider or the 1+MG Data Holder).

Justification: Controllers are obligated to provide data subjects with information in a language that they understand. As the language of the application is English, this, in many cases, will require translation and should ideally be done by native speakers. In the interest of efficiency, there should be one entity responsible for translation and, where possible, efforts should be minimised via cross-border collaboration.

As the 1+MG NCP for the participation and the operations of the country in the Secondary Use Framework, it is the country's main contact point.

EHDS Relevance: The requirements of the GDPR also apply to processing under the EHDS, and countries need to come up with strategies on how this can be realised in their national EHDS implementation. In addition, reverting from key-coded identifiers to identities is needed in cases of significant findings. It would, therefore, make sense that countries strive for solutions that work in the context of both the EHDS and the Genome EDIC Secondary Use Framework.

V.2.6. Data Subjects' Rights – Responsibility: 1+MG IT Infrastructure Provider

1. must enable the implementation of the changes resulting from the exercising of data subjects' rights, via IT tools and workflows.
2. must prepare for dataset versioning for cases where data subjects have exercised their rights to rectification, withdrawal of consent to data use under the Genome EDIC Secondary Use Framework, withdrawal of consent or objection to (certain) processing under the Framework, or erasure. This ensures that it is possible to keep track of datasets used where Users continue using the dataset disclosed to them and are unable to use the updated datasets.

Justification
The technical implementation of the policies outlined above must guarantee the proper execution of data subjects' rights and address the impact of any limitations of these rights. There must be sufficient control on why and where datasets are changed and what the consequences are for the processing.

EHDS Relevance
Also, in accordance with the EHDS, it must be possible to adapt datasets under use where the stability of the dataset is not required by the health data user. However, the versioning of datasets by health data holders or HDABs is not required by the EHDS where datasets are

assembled from data held by health data holders for each project and no archiving is provided for.

V.2.7. Data Subjects' Rights – Responsibility: User

1. must communicate to 1+MG, through the 1+MG User Portal and as early as possible, the extent to which or the point at which their data use no longer allows the exercising of data subjects' rights, where the exercising of rights changes the dataset available for use. Examples for affected rights are the rectification or erasure from the research dataset.
2. must accept that the datasets may change during data use, due to the exercising of data subjects' rights, where no requirement for a stable, i.e. fixed, dataset can be justified.

Justification
1+MG respects the data subjects' rights, however, certain data use cases may require a stable dataset. In instances where such a stable dataset is necessary, the interest of use will typically outweigh the interest of the data subjects, in line with Art. 21 GDPR, in particular, where the processing occurs for research. Such a situation would likely arise at the latest, for example, once results are published and the raw data needs to be kept for reproducibility reasons. Communication of when such a limitation applies is required to implement the corresponding data management rules. This ensures that the reaction to the request can be timely and efficient, without having to individually confirm the situation with each User in case of a request.

EHDS Relevance
These considerations could also be relevant for the national EHDS implementation to make the exercising of data subjects' rights more efficient.

→ IN THE FOLLOWING: DETAILED JUSTIFICATION DATA INCLUSION GOVERNANCE

VI. DATA INCLUSION GOVERNANCE

VI.1. ESTABLISH PRINCIPAL AVAILABILITY

VI.1.1. Establish Principal Availability – Responsibility Genome EDIC General Assembly

1. Define the **scope of purposes** for which data will be shared with Users in 1+MG and the **corresponding data governance**.
2. Adopt **technical and organisational measures (TOMs)** to comply with data protection by design and default, applicable to all actors in, contributing to, or involved in, 1+MG infrastructure.
3. Adopt the **data models and ontologies for datasets** as well as **scientific metadata models and ontologies**, at least for the minimum datasets that will be accepted in

1+MG. This should include **quality labels and quality metadata** to describe the genomes to be included in the 1+MG infrastructure. Minimum specifications for datasets of high impact based on Art. 80 EHDS must be considered.

4. Adopt minimum **ELSI metadata** models relevant for the data characterisation to allow a harmonised and machine readable description of ELSI-relevant information about the included data, Users, and types of data use. These metadata must allow integration into the EHDS data catalogue but can go beyond EHDS defined dataset descriptions.

VI.1.2. Establish Principal Availability – Responsibility Genome EDIC CC

1. Provide an **agreement that specifies the rights and obligations** of the 1+MG Data Providers, the 1+MG Data Holders (in case of 1+MG compliant data where the 1+MG Data Holder is not also the 1+MG Data Provider), and the Genome EDIC for making data available for secondary use through the Genome EDIC Secondary Use Framework. [*Moved up from below*]
2. Coordinate the definition of TOMs, data models and ontologies for datasets and metadata through the relevant Genome EDIC committees.
3. Coordinate and/or provide input to EHDS-related projects and to the EHDS Board on suitable data with regard to the minimum amount (depending on the field), how data should be structured (data models), how the meaning of could be structured for semantic interoperability (ontologies) and how data should described (metadata). Such an input could be given from the point of view of genomics research and genomic health, to be considered for the EHDS dataset descriptions and quality labels.
4. Provide the **phrasing of consent options** relevant for data inclusion in the Genome EDIC Secondary Use Framework. These options are to be offered in an opt-in/opt-out procedure or in a consent as the legal basis under the GDPR.
5. Provide **clauses in English** that can be included in Genome EDIC Secondary Use Framework information and consent sheets for data inclusion, in accordance with the transparency and consent guidance.
6. Provide information and consent clauses for the initial consent to collect data for a later disclosure in the context of data use by third party Users, as well as for any other relevant consents, such as those related to the return of incidental findings and other information to the data subjects. The principal availability can only be established where the subsequent 1+MG Data Holder/Genome EDIC have a legal basis to actually hold the data.

Justification
The goal of the 1+MG initiative is to create an interoperable federated cross-border cohort with a harmonised governance. Therefore, the decisions on data and metadata as well as the data governance need to be agreed upon on the level of the overall 1+MG infrastructure. Decisions taken that affect the cost and modality of data inclusion should be taken by the Genome EDIC Member Countries jointly, thus, through the Assembly of Members.

Support relevant for all members should be provided centrally, including on matters of transparency and consent option (based on EU law) and recognised ethical standards. The introduction of ELSI metadata in 1+MG helps to operationalise the data governance and automate or semi-automate workflows. The rights and responsibilities of the different stakeholders are agreed upon as part of the Genome EDIC Secondary Use Framework. Therefore, the Genome EDIC CC must also provide a corresponding agreement for the data inclusion.

The downstream holding of data and subsequent disclosure to Users must be consented for where 1+MG Data Holders and/or the Genome EDIC cannot rely on legislation as a legal basis for the data disclosure. [NB: legislation to provide for the disclosure of data would have to provide for the Genome EDIC Secondary Use Framework to permit participation; a legal basis derived from legislation with a different data governance would not allow the integration as 1+MG compliant data.]

EHDS-Relevance
There will be an overlap between the metadata and quality labels defined by the Genome EDIC Secondary Use Framework and the EHDS. Any definition of metadata by the Genome EDIC must, therefore, allow the integration with the EHDS, the possibility to reference the Genome EDIC data catalogue in the EHDS EU dataset catalogue. Where relevant, the specifications for datasets of high impact for secondary use, as defined in Art. 80 EHDS, must be considered to allow maximum interoperability with datasets outside the Genome EDIC Secondary Use Framework. At the same time, the Genome EDIC and its national entities/community can provide valuable input for the EHDS implementing acts that can support and improve the EHDS interoperability and suitability for genomics applications.

VI.1.3. Establish Principal Availability – Responsibility Genome EDIC Member Country

1. Decide on a **national strategy for the responsibility of downstream data disclosure** and nominate, where applicable, 1+MG Data Holder(s) in the country.

Justification: The data inclusion planning and information must cover who will be the entity responsible for the downstream data disclosure to Users as this becomes part of the

consent for data inclusion. Therefore, the decisions have to be made in advance to any data inclusion.

EHDS Relevance: Where the 1+MG compliant dataset held by the 1+MG Data Holder should be made available also under the mechanisms or the EHDS Regulation, the 1+MG Data Holder should be chosen in a way that it can also become a health data holder in the EHDS for these data. However, even if the Genome EDIC is made the controller for the downstream data disclosure, data could be held on the national level for other purposes as well that make them subject to the EHDS Regulation under Art. 51 EHDS. It is to be considered that to be a health data holder, an entity has to be an actor in the health system or to pursue research and that it is a health data holder only for those data that are processed by the entity as a controller for purposes in the health system, statistics, policy making and/or research. Holding data for secondary use through the Genome EDIC Secondary Use Framework by themselves does not qualify for the obligation under Art. 51 EHDS, irrespective of who is holding them.

VI.1.4. Establish Principal Availability – Responsibility 1+MG NCP

1. Provide **information and consent/opt-in/opt-out clauses as given by Genome EDIC in relevant languages** for the respective country.
2. Provide, where applicable **additional clauses** that are **specific** for the **situation in the country.**
3. Provide **relevant information to 1+MG Data Providers on legal and/or ethical framework** to be considered in the country such as possible legal basis for data inclusion into 1+MG and potential **conditions that need to be considered**, in particular information on where/whether an **ethics assessment** is **performed** on the **national/regional level** or to be **obtained by the 1+MG Data Provider but also legislation, where applicable, on what happens to data from deceased people**.

Justification: Each country may have different requirements from their legal or policy background that will influence the conditions for the inclusion of data into the Genome EDIC, including whether to collect the data with an opt-in or an opt-out. We want to leave as much discretion to the Member Countries as possible to allow for accommodating to the national or regional situation. This means, however, that there will be information specific for each country. In addition, where 1+MG compliant data are disclosed by the 1+MG Data Holder, certain national laws apply, i.e. those that are applicable even where the processing is based on consent. This is for example legislation on deceased people.

The 1+MG NCP must be the knowledge repository for any nationally relevant conditions as such specific information can only be provided nationally. There should be one clear contact point for all relevant questions by the 1+MG Data Providers, which is naturally the 1+MG NCP. However, the 1+MG NCP can of course subcontract certain tasks if needed.

EHDS Relevance: The 1+MG NCP may or may not be a health data access body in the EHDS. For many countries it will make sense to join functions. However, it is not yet clear how much flexibility will be given in the EHDS, so this information role may be specific for the Genome EDIC.

VI.1.5. Establish Principal Availability – Responsibility 1+MG Data Provider

1. **Sign the agreement** provided by Genome EDIC CC and, where applicable, 1+MG NCP in accordance with Genome EDIC Secondary Use Framework.
2. Ensure data can **comply with the inclusion criteria** of data for the Genome EDIC.
3. **Generate information and consent sheets** based on information and guidance provided by Genome EDIC and the 1+MG NCP with **1+MG Data Provider and 1+MG Data Holder specific information**.
4. Define jointly with the 1+MG NCP, 1+MG Data Holder and the Genome EDIC an approach to obtain consent, which mitigates any potential imbalance of power as it can be found when consenting in the healthcare context.
5. Ensure that there is a **positive ethics assessment** by a relevant ethics body in the country of the 1+MG Data Provider on the inclusion of data for future availability through 1+MG. The 1+MG Data Provider must obtain the ethics assessment from a competent ethics body unless such assessment is obtained on an overarching level. As an example, the government of a country could obtain an ethics assessment by a competent national ethics board for the legislation that is put in place.
6. Perform a **data protection impact assessment (DPIA)** for the inclusion of data in the EDIC and do a prior consultation of the Data Protection Authorities (DPAs) on the DPIA before including any data subjects.
7. Provide information on how future communication with the data subject is established (information portal/consenting app).
8. **Provide information** and **obtain** from data subjects one of the **following types of consents**, whatever is applicable: GDPR consent as a legal basis or an opt-in/opt-out depending on the legal basis on which the data can be made available within the Genome EDIC. The information must cover all relevant information related to the consent.
9. Provide information on and obtain consent for the 1+MG Data Holder and/or the Genome EDIC, as applicable, to hold the data for re-use and subsequent disclosure for
a) supporting healthcare for other patients.

b) being findable and informed on a potential participation in clinical trials or observational studies.
c) allowing subject-level data discovery for research projects, quality management in healthcare and policy development.
10. **Record the consent preferences and/or, where applicable, objections** of data subjects as ELSI metadata in accordance with the definition by the Genome EDIC. This information must be made available alongside the data to be queried for the future use.

Justification: Before the 1+MG Data Provider can take any action, it should have established its legal relations with the relevant parties (1+MG Data Holder, Genome EDIC, others as applicable) as there is otherwise no legal basis. E.g., the 1+MG Data Provider cannot consent for the inclusion in the Genome EDIC Secondary Use Framework if the possibility to include has not been established, including the support on the national level for the inclusion.

It should be for the 1+MG Data Providers to inform the data subjects, even if the inclusion is based on a legal provision, because they are the controller for establishing the availability. However, there may be an imbalance between the data subjects and the 1+MG Data Provider where for example consent would be obtained by the treating physician in the healthcare context. Therefore, a jointly agreed approach is needed that mitigates any such imbalance. Due to the sensitivity of the data and the broadness of secondary use, a DPIA will be necessary. Experience is that some DPAs expect to be consulted in case of very high risk processing and not leave the assessment of the risk mitigation to the controller so that a prior consultation of the DPAs should take place. Where reference is made to the 1+MG Data Holder as the subsequent controller for the disclosure, this can still be the same legal entity as the 1+MG Data Provider. As the legal basis for the data disclosure is now expected to be consent, the initial consent must be obtained at the level of the data inclusion.

For the consent recording, the same considerations apply as above: this is needed to allow efficient data management. It will also reduce the likelihood of mistakes.

EHDS Relevance: Only applicable to 1+MG as the availability is established outside the EHDS. Availability in the EHDS through EHDS mechanisms is defined in the EHDS Regulation, which is independent from the data that are brought in by authorised participants as defined in Art. 75 EHDS.

VI.2. DATA TRANSFORMATION

VI.2.1. Data Transformation – Responsibility Genome EDIC CC

1. Provide **information on minimum datasets** and **data models** accepted for inclusion into the Genome EDIC Secondary Use Framework.
2. Provide **suitable written instructions**, if deemed helpful also as video, training courses and, where applicable, tools to support the data processing into these data models, for cleaning of data. The tools and information are documented in English.

Justification: The provision of information and tools that are applicable across all Member Countries should always be provided centrally as much as possible because this is more efficient and cheaper.

EHDS Relevance: The EHDS Regulation does not provide for any curation of data for secondary use outside the necessity to respond to individual health data (access) requests. Nevertheless, the EHDS Regulation provides recommendations for minimum specifications for datasets of high impact for secondary use with which the curated datasets should be compliant (see also above).

VI.2.2. Data Transformation – Responsibility 1+MG Data Provider

1. **Transform** data into respective **data models,** i.e. the agreed structure of the data, **and standards** as adopted by the Genome EDIC Assembly of Members and, where applicable, **clean** the data.

Justification: Data curation and transformation is a requirement to fulfil the inclusion criteria for data into the Genome EDIC Secondary Use Framework. Therefore, the legal responsibility (controllership) for this processing is with the 1+MG Data Provider.

EHDS Relevance: the EHDS Regulation does aim to curate data into certain data models for secondary use. Therefore, these considerations have no equivalence in the EHDS Regulation. The Genome EDIC can provide added value by offering data that are interoperable across the EU, in accordance with the minimum specifications for datasets of high impact for secondary use, introduced in Art. 80 EHDS.

VI.2.3. Data Transformation - Responsibility 1+MG NCP or 1+MG IT Infrastructure Provider or other Genome EDIC national entity

1. **Support for the transformation** may be provided through the staff in the 1+MG NCP or the 1+MG IT Infrastructure Provider where it acts as Data Host. Also, an altogether different entity on the national level could be mandated. It is for the **Member Country to**

decide if and who would provide such support in accordance with relevant EU and national law.

Justification: The entities holding the data may not have the necessary expertise and resources in data transformation and may benefit from support. As the data may be in the national language and reflecting the local situation, such support cannot be provided centrally. It is up to each country to decide to which extent such support should be provided. The support is not obligatory if the necessary results can be achieved without.

EHDS Relevance: In the original proposal of the EHDS Regulation, any potential data curation takes place only to the extent necessary following an approved health data (access) request. The HDAB is responsible for such curation. It is down to the national strategy how this is implemented. Genome EDIC Member Countries could consider synergies by mandating the same entities to support HDABs and 1+MG Data Holders in such tasks.

VI.3. DATA DOCUMENTATION

VI.3.1. Data Documentation – Responsibility Genome EDIC CC

1. Provide **information on metadata, including quality labels for genomes and/or information in accordance with the EHDS quality label** and the corresponding **metadata models** and **standards** as required for inclusion into the Genome EDIC Secondary Use Framework.
2. Provide suitable **written instructions**, if deemed helpful also as video, **training** courses and, where applicable, **tools** to collect or transform existing metadata as well as for quality control. The tools and information are documented in English.
3. Provide a **central data catalogue** covering the relevant metadata provided by 1+MG Data Providers.

Justification: The provision of general information and tools through the central hub that could be used across all Member Countries is cost-effective and ensures a single source of truth centrally. However, countries are free to use their own tools as long as the result complies with the requirements for the inclusion into the Genome EDIC Secondary Use Framework.

The central data catalogue is necessary to provide a “one-stop-shop” for the User to find and request access to data.

EHDS Relevance: the metadata and quality label will be overlapping with the EHDS as the Genome EDIC data catalogue should integrate into the EU dataset catalogue of the EHDS.

Any tool and support developed in the context of 1+MG is likely suitable to support also the characterisation of datasets in the EHDS.

VI.3.2. Data Documentation – Responsibility 1+MG Data Provider

1. Produce the **relevant scientific metadata and quality label** as adopted by the Genome EDIC Assembly of Members. This may require collecting information from other parties where the 1+MG Data Provider is not the initial controller.
2. Provide **scientific metadata including a quality label** and **metadata related to ethical, legal and social aspects** (ELSI metadata) in the format adopted by the Genome EDIC Assembly of Members, including any special access and use conditions, information on incidental findings to be returned etc.
3. Provide a mock (synthetic) version of the dataset that reflects the variables and formats of the dataset to be included. [*Moved here from Data Access Governance*]
4. Provide **transparency on any special procedures** to be maintained that may **influence the data access review process**.
5. Where several 1+MG Data Providers **contribute to the same dataset** to create the required minimum dataset or a bigger joined dataset available under the same 1+MG Data Holder or the Genome EDIC, they must **coordinate with each other** to generate a full and consistent metadata set for this dataset.

Justification: Data documentation is part of the data inclusion process and therefore the 1+MG Data Provider's responsibility. It also requires a deep understanding of the data, which cannot be done without the entity who collected the data initially. Therefore, the responsibility for the data characterisation must be with the 1+MG Data Provider. As a dataset will be addressed later as a whole (genomic and phenotypic/other data) to allow a User to understand which data are available on the respective data subjects, the metadata must be consistent and describe the full assembled dataset. The mock dataset will enable the Users even more to find out in advance whether the data in 1+MG will be useful for them as they can test their data usage before going through the effort of requesting the real data.

As there is freedom with respect to the national and local workflows for the access review procedure, information about this is necessary to be documented. Also, any other knowledge relevant for the use and the ethic-legal compliance throughout the data life cycle is important to be documented.

EHDS Relevance: Under the EHDS Regulation, health data holders as the source of the data also have the responsibility to characterise the data. Experience potentially made and tools used through the engagement in the Genome EDIC can be useful for these organisations.

VI.3.3. Data Documentation – Responsibility Genome EDIC Member Country

1. **May support the creation or adaptation of metadata** by the 1+MG Data Provider through funds or direct support by tasking the staff in the 1+MG NCP or the 1+MG Data Host or an altogether different entity on the national level with active support in the data documentation. It is for the Member Country to decide if and who would provide such support in accordance with relevant national and EU law.
2. Should create incentives or direct support through tools for 1+MG Data Providers to generate metadata as much as possible automatically at the time of data collection.

Justification: The entities holding the data may not have the necessary expertise or capacities and maybe also limited motivation to produce the full metadata set. and may benefit from support. As the data may be in the national language and reflecting the local situation, such support cannot be provided centrally. However, the extent to which the creation of metadata is supported is a decision by the country. To make the data inclusion more scalable, the efforts for a metadata capture "by hand" should be as limited as possible. However, as long as a proper dataset description is achieved, it is up to each country to decide to which extent such support should be provided. The support is not obligatory.

EHDS Relevance: No such support is provided for in the EHDS Regulation, but it is also not excluded. Support built up in 1+MG can also support health data holders in the EHDS.

VI.3.4. Data Documentation – Responsibility 1+MG NCP

1. The **1+MG NCP provides guidance and relevant tools, such as checklists and software,** to 1+MG Data Providers on how to **ascertain and communicate additional data-specific access and use conditions**, e.g., providing checklists on national legal or ethics requirements.

Justification: The 1+MG NCP is the main contact point on the national level and should therefore hold and provide guidance (e.g., documents check-lists) and tools whereas the practical support can also be given by another entity.

EHDS Relevance: No such support is provided for in the EHDS Regulation, but it is also not excluded. Support built up in 1+MG can also support health data holders in the EHDS.

VI.4. ESTABLISH FORMAL AVAILABILITY

VI.4.1. Establish Formal Availability – Responsibility Genome EDIC Member Countries

1. Strive to **modify unnecessarily restrictive and/or complicated conditions** stemming from national or regional practice or even modify legislation applicable to making data available for secondary use in a pan-European data infrastructure.
2. Aim to establish a **legal basis with suitable safeguards** to enable 1+MG Data Providers to make data available through the Genome EDIC Secondary Use Framework, covering the principal availability as well as ideally, where the data disclosure to the User is controlled locally, legislation that implements the 1+MG data governance for data disclosure and data use.
3. Provide suitable **IT infrastructure/tools** on the national level that allows a **reliable linking** between the **identity of the data subject** and **key-coded identifiers** for data held in under the Genome EDIC Secondary Use Framework. Genomic data and clinical/phenotypic/demographic data should be stored separately and ideally be linked to a different keycode identifier.

Justification: Countries are asked to review and ideally adapt legal provisions or practices to support the inclusion of data into the Genome-EDIC framework. Personal genomic and health data can only be brought into the Genome EDIC if consent is obtained for this inclusion or if a legislation enables the inclusion of these data. Genome EDIC Member Countries are therefore asked to support the procedures for data inclusion and keep them as simple as possible.

Member Countries may also consider providing legislative provisions for 1+MG Data Holders to disclose data according to the Genome EDIC Secondary Use Framework based on national law rather than having to rely on consent as a legal basis.

Data held for secondary use under the Genome EDIC Secondary Use Framework must comply with data minimisation principles. This means, as much as possible, direct identifiers should be removed. Since subject level genomic data is a direct identifier, these data should be kept separately from other data. It also means that potentially existing key-coded identifiers used in a different context should be replaced by a new keycoded identifier.

EHDS Relevance: The EHDS Regulation does not cover transforming and holding data for secondary use independent of any User requests. Formal availability in the Genome EDIC Secondary Use Framework does not

VI.4.2. Establish Formal Availability – Responsibility Genome EDIC CC

1. **Publish the metadata** in the Genome EDIC data catalogue after the compliance confirmation by the 1+MG NCP.

Justification: The Genome EDIC serves as a one-stop-shop to the Users, which is why it holds a central data catalogue. However, only datasets that fulfil the requirements of the Genome EDIC Secondary Use Framework should be published as 1+MG compliant datasets. The Genome EDIC must ensure this through the 1+MG NCP.

EHDS Relevance: If the Genome EDIC wants to join the EHDS as an authorised participant, it must ensure that the data offered through its mechanisms and that will be published in the EU dataset catalogue of the EHDS fulfil the conditions set out by the Genome EDIC.

VI.4.3. Establish Formal Availability – Responsibility 1+MG Data Providers

1. **Accept** that data will be made available according to the **Genome EDIC Secondary Use Framework**, which defines how access will be granted to data and use be pursued within the Genome EDIC Secondary Use Framework (e.g., relating to access timelines, conditions of procedural fairness (justified refusals/appeals), IP conditions, appropriate fees (or lack thereof), and publication conditions acknowledging the 1+MG Data Provider).
2. **Make metadata** available to 1+MG NCP for checking and ultimately and, **where applicable, data** to 1+MG Data Host.

Where the1+MG Data Provider extends the dataset made available through the Genome EDIC Secondary Use Framework with further related data, e.g. longitudinal extensions, this has to be made part of the agreement signed.

Justification: An adherence to the Genome EDIC Secondary Use Framework is required to realise the objective of the 1+MG initiative for a harmonised data governance.

EHDS Relevance: Not applicable to EHDS. No interference with EHDS.

VI.4.4. Establish Formal Availability – Responsibility 1+MG NCP

1. 1+MG NCP, or entities on national or regional level on behalf of the 1+MG NCP, must **verify compliance** of the data with **1+MG and national/regional requirements**. This includes:
   - Assessment of **legal and ethical compliance of establishing availability**.
   - Assessment of adherence of data to the **defined data models and quality of the data**.
   - Assessment of **completeness** and **correctness** of provided **metadata**.

2. communicate compliance to the Genome EDIC CC. This includes documentation of checks performed and the transmission of the metadata for the Genome EDIC's data catalogue (e.g. through a FAIR data point).
3. For 1+MG Cohort Data: provide information to be published in the data catalogue with whom the data access application will be shared for the data access approval procedure.

Justification: Even though only the 1+MG Data Providers are responsible for the data inclusion in 1+MG, the Genome EDIC Member Countries must ensure that the data are suitable and that proper risk management systems and procedures are in place to protect the rights and freedoms of the data subjects. The quality of data and metadata must be ensured as part of the commitment of the Genome EDIC Member Country to provide access to high value data within the Genome EDIC. Therefore, procedures should be established on the national level how the compliance and quality of data and metadata can be ensured and confirmed. The 1+MG NCP as the verified contact point for the Genome EDIC should be the one giving the "all clear" but would not necessarily be the one who performs (all) the necessary checks.

It is also important for the Genome EDIC itself that data are of the envisaged quality. However, this check must be done on the country level as the Genome EDIC does not have access to the data. Also, the use of national language (e.g. in healthcare data) will not allow the Genome EDIC central services to support this process. Last not least, country specific conditions related to the ELSI setup are only known locally. The Genome EDIC CC nevertheless provides tools to support the process for both curation and quality vetting that the 1+MG NCP will build on for its tasks.

However, countries are free to decide which entity will provide this check.

It is also down to the country to decide who participates in the review process. This information may be important where, e.g., 1+MG Data Providers who are research organisations participate in the review and may receive information from competitors. Users will have to know before submission who will see their application. The 1+MG NCP as the entity coordinating and overseeing the procedures should communicate this information. However, where 1+MG Data Holders have the freedom to decide as controllers on the reviewing process, the information needs to come from them (see below).

EHDS Relevance: In the EHDS, a quality check of the metadata and quality label is performed by the health data access body. This suggests that sharing functions between Genome EDIC national entities and HDAB tasks can create synergies.

VI.4.5. Establish Formal Availability – Responsibility 1+MG Data Holders [1+MG compliant datasets]

1. Conclude an **agreement with the Genome EDIC** on the participation as a 1+MG Data Holder in the Genome EDIC Secondary Use Framework.
2. **Receive and subsequently hold data legally** for secondary use in accordance with the **Genome EDIC Secondary Use Framework**, which defines according to which criteria and use conditions 1+MG Data Holders grant access to data within the Genome EDIC Secondary Use Framework (e.g., relating to access timelines, conditions of procedural fairness (justified refusals/appeals), IP conditions, appropriate fees (or lack thereof), and publication conditions).
3. Provide information to be published in the data catalogue with whom the data access application will be shared as part of the data access approval procedure.

Justification: 1+MG Data Holders must agree to the harmonised rules if they want to receive and make available genomic and health data under the Genome EDIC Secondary Use Framework.

Where 1+MG Data Holders have the freedom to decide as controllers on the reviewing process, the information needs to come from them (see below).

EHDS Relevance: An adherence to the Genome EDIC Secondary Use Framework is necessary to make these data available within the EHDS under the authorised participant option.

VI.4.6. Establish Formal Availability – Responsibility 1+MG IT Infrastructure Provider

1. **Host datasets** provided by 1+MG Data Provider. (NB: the 1+MG Data Provider may also act as 1+MG Data Host.)
2. The infrastructure of the 1+MG Data Host must fulfil the requirements for data protection by design and default including the quality and risk management approach of the Genome EDIC Secondary Use Framework.
3. Where the 1+MG Data Host is not a 1+MG Data Holder for the data held, it must sign a data processing agreement.

Justification: The data may have to be hosted in an IT environment not available at the 1+MG Data Holder or 1+MG Data provider as it needs to be connected to the communication infrastructure for data discovery. Where the entity holding the data is not also responsible for the access decision (i.e. is not a 1+MG Data Holder), it will have to act as a processor with

the corresponding agreement. It should be up to the Genome EDIC Member Countries to decide how they organise the national setup. The choice should, however, meet the data protection by design and default requirements, which include the quality and risk management framework as well as an approach that unnecessary copies are to be avoided.

EHDS Relevance: The current setup of the EHDS Regulation implies that data remain at the data holder and are only submitted to the health data access body on request, requiring the data holder to extract the data every time data is requested. The role of health intermediation entities should lift the burden where these entities take over the tasks of health data holders. Where a Genome EDIC national entity holds the same data as a health data holder, it can support the data availability based on the EHDS Regulation.

→ IN THE FOLLOWING: DETAILED JUSTIFICATION – DATA ACCESS GOVERNANCE

VII. DATA ACCESS GOVERNANCE

VII.1. PRIOR REGISTRATION OF THE USER ORGANISATION

NB: the prior registration was previously not part of the considerations on research access requests but already introduced for all the other applications. For that reason, it is nevertheless coloured in grey to indicate it as a change to the research data access governance.

VII.1.1. Prior Registration of the User Organisation – Responsibility: 1+MG Data Holders [1+MG compliant datasets]

1. Delegate responsibility of User Organisation administration and vetting to Genome EDIC, including the signature of relevant agreement as part of the registration.

Justification: The delegation follows the principle that the User (Organisation) should not be concerned with the underlying complexity of the 1+MG implementation and federation. It also creates economies of scale if not every 1+MG Data Holder must establish its own registry. It should equally be avoided that a User Organisation would have to sign again and again agreements with each new 1+MG Data Holder joining. Apart from the burden of such a repeated signature, the complexity of its administration may also bear the risk that physical access is given by a 1+MG Data Host despite a missing signature.

EHDS Relevance: The EHDS Regulation does not cover any formal registration of health data users before requesting access. However, some kind of database will have to be established at least at the national level, likely under the responsibility of HDABs. A vetting will likely take place as part of the verification of the legal basis that is to be demonstrated by health data users. There will be a partial overlap between User Organisations vetted by the Genome

EDIC and health data users applying for access with an HDAB. Arrangements could be made, subject to agreement by the User Organisation/health data holder, to share information between the Genome EDIC and the HDAB.

VII.1.2. Prior Registration of the User Organisation – Responsibility: Genome EDIC CC

1. Establish a **User Organisation registry** and support the **registration process** through a form in the 1+MG User Portal and by online sharing of **relevant information** with the appropriate stakeholders through pre-existing forms and instructions.

2. Provide the **terms of use** and the **framework data use agreement for healthcare reuse** as well as other relevant information in the 1+MG User Portal.

2. **Vet User Organisation** with regard to the **validity** of its information and for the **scope of access** it has registered for.

3. Sign the **relevant agreements** with the User Organisation.

Justification: Information must be collected and administered in a structured way. Having a pre-existing form will provide clarity on the documentation required on Users and User Organisations by the 1+MG EDIC to register. A seamless process may attract more Users and simplify their experience. For the registration process, the Genome EDIC CC must provide all the relevant information necessary, including the terms of use.

Vetting is important to minimise fraudulent access and create evidence for the lawfulness of downstream data access requests, including the subject-level data discovery.

EHDS Relevance: The EHDS Regulation does not cover any formal registration of health data users before requesting access. However, some kind of database will have to be established at least at the national level, likely under the responsibility of HDABs. A vetting will likely take place as part of the verification of the legal basis that is to be demonstrated by health data users. There will be a partial overlap between User Organisations vetted by the Genome EDIC and health data users applying for access with an HDAB. Arrangements could be made, subject to agreement by the User Organisation/health data holder to share information between the Genome EDIC and the HDAB.

VII.1.3. Prior Registration of the User Organisation – Responsibility: User Organisation

1. **Register** in the Genome EDIC User Registry **before** any **subject-level data discovery** (where applicable) and/or the **submission of any access request** with relevant information on the organisation, such as its legal representatives and other relevant contacts,

administrative information including financial information, its legal setup, the scope of activities that can legally be pursued and the corresponding legal basis the processing will be based on.

2. Agree to the **terms of use**, which include the **joint controller arrangements** with the controller(s) for the data disclosure for subject-level data discovery. *[Research, policy development, QM]*

2. Agree to the **terms of use** and sign a **framework data use agreement for healthcare reuse** covering the principal terms for the later access by healthcare professionals and staff under their supervision. [*Healthcare*]

3. Provide **information on Users as employees** or otherwise persons acting under the supervision of the User Organisation **permitted to pursue subject level data discovery** and/or **access requests** (where applicable). *[Research, policy development, QM]*

3. Provide **information on healthcare professionals or other staff** acting under **medical secrecy** that are authorised to access data under this framework agreement; information must be verifiable real-time.

4. Keep all information **up to date**.

Justification: The User Organisation is the controller for the downstream use and therefore the entity that is legally responsible for the access (request). It is therefore important to have the correct information on the organisation (existence, scope of activities pursued by the organisation, where healthcare activities are pursued; person legally allowed to sign; DPO contacts etc.).

In case of subject-level data discovery, this is already processing of personal data. As it takes place in the interest of both the User Organisation as well as the controllers for the disclosure of the data, this creates joint controllership on this processing. The User Organisation as (joint) data controller must be aware of such processing potentially taking place in advance to an access request and having agreed to it. As Users are acting under the User Organisation, the User Organisation must always be able to confirm that Users are (still) acting on their behalf.

EHDS Relevance: To be found out how health data users/applicants are registered for health data (access) requests in the EHDS. This may take place as part of the access application as no subject-level data discovery takes place.

VII.1.4. Prior Registration of the User Organisation – Responsivity: Genome EDIC Member Country

1. **Enable the vetting** of healthcare organisations and/or health professionals with a **valid license**, **acting under medical secrecy**.

Justification

Medical secrecy is the main safeguard for access to health and genetic data under the GDPR. It is therefore necessary that the Genome EDIC is able to verify that this status is still applicable when the access is sought by a User for healthcare reuse.

EHDS Relevance: This might be limited as healthcare reuse will likely not be pursued under the mechanisms of the EHDS Regulation.

VII.2. DATA DISCOVERY

VII.2.1. Data Discovery – Responsibility: Genome EDIC CC

1. Provide a **1+MG User Portal** with **information** on the Genome EDIC Secondary Use Framework in general as well as on the data collections based on own knowledge or after retrieving such information from the national level through the respective 1+MG NCP. For information on the **secure processing environment (SPE)**, a **direct discussion** of the User should with the **1+MG SPE Provider** should be possible where information **needs go beyond the capabilities of Genome EDIC CC**.
2. Host and make available through the 1+MG User Portal the **central Genome EDIC data catalogue** characterising different data collections with rich aggregated metadata; each collection should include a **mock (synthetic) version of the dataset**.
3. Offer an **option for subject-level data discovery** to be **launched from the Genome EDIC Data Catalogue**. Accompany the discovery tool with **information on the terms of use** relevant for the User. *[Research, Policy Development, QM in Healthcare]*
4. Offer the possibility to find in the central Genome EDIC data catalogue **externally governed datasets** that are **clearly labelled as external** to the Genome EDIC Secondary Use Framework.

Justification: The Genome EDIC CC should act as a contact point for Users to ask any question regarding the Genome EDIC and data access to avoid the user having to retrieve information all over the federated infrastructure. If unable to answer a question, it is therefore also the Genome EDIC CC's responsibility to source the answer, with support from the 1+MG NCP (if required).

For the same user experience reasons, a central data catalogue must be available to users, providing an overview of all the available data, and associated dataset descriptions relevant for their use. A comprehensive data overview and as well as the availability of mock datasets supports the User in understanding the available data, testing their analysis plan, and preparing their analysis workflow. Such information should allow the user as much as possible to assess the usefulness of the data before applying for access.

Subject-level data discovery must be available to users, allowing them to see whether there are any available data (particularly important for rare diseases and rare variants) because many genomic use cases require special patient profiles sourced all over Europe. As this requires the processing of personal data, the Genome EDIC CC should inform Users about their data handling responsibilities as an additional safeguard beyond mere contractual clauses.

The possibility of listing external datasets was introduced after the vote for pursuing a decentralised approach in the decision making because on the national level, there may be 1+MG Data Providers not willing or able to make the data available according to the harmonised data governance and no other 1+MG Data Holder may be available in the country who would take over the responsibility. The extension towards externally governed datasets is supposed to either a) increase the number of datasets that can be made available in the 1+MG SPEs even though they do not come with the advantage of a harmonised data governance or b) make available relevant phenotypic information that can be linked to the 1+MG Genomic data, but held by data providers outside the 1+MG governance framework.

EHDS Relevance: The one-stop-shop implementation reflects the Genome EDIC's status as a single data infrastructure authorised participant within the EHDS even if the underlying data are federated. Where the data are to be made available as an authorised participant in the EHDS, a central data catalogue is required to reflect all the data made available through the Genome EDIC Secondary Use Framework.

The catalogue must be interoperable with the EU dataset catalogue of the EHDS and should comprise at least the same information. However, we expect that the Genome EDIC data catalogue will also contain other elements, such as the mock datasets or the possibility of subject level data discovery.

Externally governed datasets could also be genomic or other health related datasets that are made available through the EHDS mechanisms and could be jointly analysed with data from the Genome EDIC Secondary Use Framework.

VII.2.2. Data Discovery Responsibility: 1+MG Data Holder/Genome EDIC CC [Research, Policy Development, QM in Healthcare]

1. Perform a **data protection impact assessment (DPIA)** for the principal procedures of subject-level data discovery with **prior consultation of the data protection authorities**. The DPIA should also consider that results returned to the User could aggregate information obtained across several 1+MG Data Hosts. Where the Genome EDIC is found to be a joint controller with the 1+MG Data Holder for 1+MG compliant data, the Genome EDIC may prepare and pursue the DPIA for the 1+MG Data Holder.

Justification: The disclosure of the genomic and health data to the User bears in principle a high risk, which warrants a DPIA. A DPIA is always the obligation of the controller responsible for the processing. Subject-level searches always bear the risk of an abuse of the system or to create potential vulnerabilities of the system, which need to be met with suitable measures based on a DPIA to minimise the risk to the right and freedom of the data subjects. As the processing of genomic and health data of potentially vulnerable subjects is a high risk processing, DPAs should be consulted in advance.

Where two entities act as joint controllers, they can determine their respective responsibilities according to Art. 26. In the EDPB Guidelines 07/2020 on the concepts of controller and processor in the GDPR, in para 166 it is suggested that the pursuit of a DPIA is among the responsibilities that can be considered to be distributed to one of the controllers. Delegating to the Genome EDIC can make sense here because it is the Genome EDIC that introduced the subject-level data discovery as part of the services under the Genome EDIC Secondary Use Framework and that defines the common risk management approach.

EHDS Relevance: It could be considered that the subject-level data discovery can be used to find data subjects and becomes the basis to apply for additional data from the EHDS. Such scenarios should also be considered in a DPIA but may require individual risk assessments case by case.

VII.2.3. Data Discovery Responsibility: 1+MG IT Infrastructure Provider [Research, Policy Development, QM in Healthcare]

1. If acting as a 1+MG Host, provide the **IT infrastructure and/or services** necessary for the User to pursue a **subject-level data search** on the relevant scope of data available for the purpose of the User.
2. If acting as a 1+MG Host, deploy a **search engine that is linked to the 1+MG User Portal** and that has **built-in safeguards** in line with the DPIA for sufficient cybersecurity

management and to mitigate the risk of a malicious or accidental compromising of the privacy of data subjects.
3. If acting as a 1+MG Host, **log the subject-level searches** and use suitable measures to **detect potential suspicious behaviour** of the User.

Justification: A subject-level data discovery is an important feature. Many collections within the Genome EDIC comprise data of many people, such as cohorts or population-level datasets brought in by a country. For a project, not all data may be required to answer the question driving the project. In rare diseases, only very specific inclusion criteria apply, typically related to a special genotype or phenotype. The data minimisation principle requires that access is only given to those data necessary for the purpose.
Very important is also the advance discovery to determine if relevant data are available at all. Users may only want to go through the application procedure if they know that there will be relevant data available. The subject-level data search will allow the Users to find out how many records available through the Genome EDIC Secondary Use Framework fit the inclusion criteria for their project and then to decide if or if not to apply for access. This avoids unnecessary access requests and provides the relevant actors on the national level already with the list of data subjects to be informed and relevant records to be disclosed in case of a positive review.

EHDS Relevance [→→→ back]*:* The EHDS Regulation does not provide for any subject-level data discovery. Nevertheless, the data minimisation principle must be complied with also under the EHDS Regulation. Procedures would therefore be different: in the Genome EDIC, the User makes the selection in advance to the access request. Under the EHDS Regulation, the data minimisation is an obligation to be fulfilled by the health data holder and/or the HDAB, so any processing related to selecting the appropriate subset of data has to be done by the 1+MG Data Provider or the HDAB. The tools for finding and selecting relevant data subjects with defined genetic variants may be used also by health data holders or HDABs and therefore provide scope for synergies.

In addition, a User may build on its application in the Genome EDIC Secondary Use Framework and use this as a gateway to identify relevant data available through HDABs as well.

VII.2.4. Data Discovery Responsibility: Users [Research, Policy Development, QM in Healthcare]

1. When pursuing subject-level data discovery, provide **in advance basic information** about the **nature of their access request** (research, policy development or QM), a **brief summary**

of the purpose together with a justification for the need of subject-level data discovery and the **relevant inclusion criteria (which features should data subjects have)**, which determine the search to be performed.

2. **Agree to Terms of Use** for the subject-level data discovery. This includes in particular that the search should **not be used for any other aim than to find out about the availability** of sufficient data for an access request.

Justification: Where the Users determine the number of relevant records for their project, this means that they process data of identifiable natural persons even if no personal data is transmitted to the Users. Such operations are subject to the GDPR, which means that the context of the processing must be captured. The specification of inclusion criteria provides a basis for data minimisation and to check the validity of access requests.

The Users should explicitly acknowledge their rights and obligations when pursuing subject-level data discovery. Where the project involves only simple statistics, Users may be tempted to obtain answers to their questions already on the level of the data discovery tool. Therefore, they must understand that such an approach is prohibited.

EHDS Relevance: Only applicable in the Genome EDIC Secondary Use Framework.

VII.2.5. Data Discovery Responsibility: 1+MG NCPs

1. Act as a **gateway between the Genome EDIC CC and 1+MG Data Providers** where additional information by the 1+MG Data Providers is needed.

Justification
There will be different scenarios with respect to 1+MG Data Providers on the national level. In some countries, a national genome centre is a single 1+MG Data Provider, in other countries, individual universities and hospitals may be 1+MG Data Providers. A national instance will know the situation best and can also communicate with 1+MG stakeholders in the national language if needed. It is more efficient if questions in advance to an access request are solved on the national level where also the awareness of the situation and the language skills are present.

EHDS Relevance: These procedures are entirely internal to the Genome EDIC. However, the 1+MG NCPs could overlap in a country with the HDABs and achieve synergies, as 1+MG Data Providers are likely also health data holders. Therefore, additional information may even be available already at the HDAB.

VII.2.6. Data Discovery Responsibility: 1+MG Data Providers

1. Shall **respond within a reasonable timeframe to questions by the 1+MG NCP** regarding the data collections they have made available through the Genome EDIC Secondary Use Framework if any clarification is required to progress with the data access procedures. However, where data were initially **collected by another entity** than the 1+MG Data Provider, this initial collector **can only be motivated** by the 1+MG Data Provider (or, where relevant the 1+MG NCP) to provide any missing information **but not obliged** to provide the necessary input.

Justification: Not all information can be captured in metadata, which is why the knowledge of 1+MG Data Providers will still be relevant. Also, not all 1+MG Data Providers may give precise enough information at the time of the initial data submission. The obligation to provide answers works therefore also as an incentive to provide as precise and as comprehensive information as possible at the time of submission to minimise later information requests.

There will be cases though where the 1+MG Data Provider will not be able to answer the question because they have not collected the data directly from the data subject. In such cases, they can reach out to the initial collector but as the initial collector is a third party to the data inclusion into the Genome EDIC Secondary Use Framework, they cannot be obliged to respond.

EHDS Relevance: These procedures are entirely internal to the Genome EDIC.

VII.3. DATA ACCESS REQUEST

VII.3.1. Data Access Request Responsibility: Genome EDIC CC

1. Provide **comprehensive online information** and **tutorials** on the data access request procedures.

2. Provide a **help desk** for dealing with the questions of Users; perspectively, consider **offering a chat bot** as a first line of communication.

3. Provide through a **user-friendly central 1+MG User Portal** the possibility to submit a data access request. The 1+MG User Portal shall capture structured and free-text information on the access request in a single access request form for the entire data available in the Genome EDIC. Access request forms must be specific for the different applications of research, policy making, QM and healthcare reuse. The 1+MG User Portal should have features that allow guiding the applying User in the choice of the right form. The access

request form must integrate into the central data access request form of the EHDS but may go beyond if additional assessments are made or if the EHDS provides for a less structured information capture.

4. Provide for an efficient possibility in the 1+MG User Portal that an access request on the same project can be submitted by different User Organisations acting as Joint Controllers, where a User Organisation in the application process "invites" additional User Organisations to its application.

5. Offer a staged access request procedure for healthcare reuse where Users can progressively ask for more detailed and therefore more privacy-critical information.

6. **Detect**, through suitable settings in the 1+MG User Portal, **obvious mistakes** (missing/wrong values) and **prevent submission** in such cases.

7. Where the Genome EDIC CC has the mandate, **reject an access request** based on **obvious mistakes** (e.g. wrong or outdated documents) detected and inform the 1+MG NCP and/or 1+MG Data Holders; **without such a mandate**: at least, **inform applicant** about the likely rejection.

8. Request **confirmation of the validity of an access request** to the User Organisation.

9. Ideally: Use **suitable technology to flag inconsistencies** that need **to be examined** in the data access review by the data access committee(s).

10. **Channel the information** on a **valid data access request** to the **1+MG DAC** as well as the affected 1+MG NCPs ~~or, where applicable, straight to the responsible 1+MG Data Holders~~. *[proposed suggestion: do not include this option]*

11. Provide the possibility for interested **Users** to **contact directly Data Providers** for high value externally governed datasets to request access **outside the Genome EDIC Secondary Use Framework**.

12. The Genome EDIC CC should have **in place suitable IT infrastructure** that allows to support the above tasks.

Justification: The experience of data repositories is that efforts for data access requests and errors related to such procedures can be significantly reduced if proper information is made available, which is why the Genome EDIC CC should invest well into such information. However, not all questions can be answered by online information, which makes an additional helpdesk necessary. The centralisation of the helpdesk with the Genome EDIC CC, a FAQ section on the web and a continuously updated information provision as well as eventually a chat bot will help to quickly answer incoming questions. Where the Genome EDIC CC cannot help. Users may be referred to the 1+MG NCP or a 1+MG IT Infrastructure Provider if such information is not efficient to be retrieved by the Genome EDIC CC for the User.

The Users must transfer easily from the data discovery to the application for the chosen datasets. Structured information capture can help to provide IT support for the assessment of the access request and should cover all information needed for such assessment. Information capture should be comprehensive and, where applicable, specific for the respective nature of the application as different types of use will require different information and different safeguards. Guidance should be provided on this process as much as possible to avoid misunderstanding and mistakes, thus improving the process for both Users and the Genome EDIC CC.

The possibility to submit the same text for a data access request should be foreseen as this streamlines the procedures: it eases the life of collaborators working jointly on a project; it also prevents the Genome EDIC reviewing accidentally the same project twice.

In line with data minimisation, the access for the healthcare reuse is staged. Healthcare professionals may have to understand patient profiles or even have contacts to healthcare professionals for additional data to draw conclusions or their own patients (see below). However, Users should first aim to get the relevant information in the least intrusive manner and will therefore only be able to get more intrusive information in a step-wise approach when other possibilities have been exhausted.

IT-supported features further reduce manual interaction where they can detect missing elements and obvious mistakes and will therefore reduce access related cost. Structured information is essential for an automated or semi-automated support of access request procedures and is vital for scalability. Structured information will also help to identify those access requests that are not eligible or need close scrutiny for quality and consistency.

A rejection in case of missing or inconsistent information is recommended as any feedback asked will lead to significant delays in the access procedure. As timelines have to be met, such mistakes on the side of the applicant will affect the perceived performance of the reviewing process under the Genome EDIC Secondary Use Framework. Such rejections should go along with information for an encouraged re-application. However, it allows for a renewed access request after the clock has been reset rather than creating time pressure on the data access committees involved on the central and local level.

In case of externally governed datasets, the Genome EDIC does not take any responsibility for the access request procedures, which means that interested entities are routed straight to the responsible 1+MG Data Providers. In case of data made available through the mechanisms provided by the EHDS Regulation, this would be an application through the EU dataset catalogue.

The communication on the data access request should be sent to all stakeholders immediately to give as much time as possible to prepare a response. In case of 1+MG Cohort Data, the 1+NG NCP is the natural recipient as it has the overview of who should be involved

on the national level. (see also below). The situation is different for 1+MG Compliant Data where the 1+MG Data Holder is responsible for distributing such information and could theoretically be directly contacted and any downstream information would be down to the 1+MG Data Holder. However, it is likely more efficient and reliable if this is still organised centrally by the 1+MG NCP who informs the 1+MG Data Holder but also any other relevant entity, e.g. the local DAC.

The support by IT tools is strongly recommended for making the operations of the Genome EDIC scale and also to establish reliability of procedures.

EHDS Relevance: This information provision to Users is tailored to 1+MG. Some overlap with information provided on the national level by HDABs can be expected and there is scope for synergies.

Each HDAB and each authorised participant like the Genome EDIC must have a 1+MG User Portal. The 1+MG User Portal could serve as a model for HDAB portals where countries do not want to invest into an own solution and adapt the 1+MG version, thereby creating synergies.

However, the Genome EDIC is free to use IT tools in the assessment of access requests for scalability based on structured information captured. The HDABs may not have the same freedom to collect additional, structured information. This will have to be verified with the Commission and the EHDS Board when Implementing Acts will be adopted.

Another difference is that the EHDS Regulation seems to suggest that additional information is asked, e.g., in case of incompleteness rather than the possibility given to reject access requests, so workflows will be different.

Externally governed datasets that are listed in the Genome EDIC Data Catalogue but available via HDABs could be directed to Commission provided data catalogue and the related access request form for cross-border applications. Where data are made available by 1+MG Data Providers outside the EHDS mechanisms, Users will have to deal with these entities directly to find out how to apply because the EDIC cannot take any role and responsibility for data that are entirely outside its influence.

VII.3.2. Data Access Request Responsibility: 1+MG NCP [Research, Policy Development, QM in Healthcare]

1. Channel the information on the data access request to the relevant stakeholders, e.g., 1+MG Data Holders for 1+MG Compliant Data, where applicable, 1+MG Data Providers and/or a Local DAC as defined on the national level.

Justification: In principle, it is always recommended to involve the 1+MG Data Provider in the access decision as they understand best the kind of data subjects included and where there may be ethical or legal issues. However, in some countries there may also be laws in place that require certain review procedures such as by a national ethics board. The background here is not any legal obligation to do this. The 1+MG Data Holder discloses based on consent, which means legislation that regulates ethics approval as part of the processing without consent does not apply. The same applies to the Genome EDIC as controller for 1+MG Cohort Data; the Genome EDIC is not even subject to the law of 1+MG Member Countries (except for the Host Country). Nevertheless, the possibility to follow established procedures on a voluntary basis should be given as this is an essential element of trust-building. The consultation of relevant entities is then to be arranged on the national level and to be implemented through the 1+MG NCP.
Channelling through the 1+MG NCP in all cases, also for 1+MG Compliant Data has the advantage of keeping a national overview of data access requests received, and ensuring that all relevant bodies are involved and that downstream the 1+MG NCP can also send reminders in case of upcoming deadlines.

EHDS Relevance: These procedures are internal to the Genome EDIC Secondary Use Framework.

VII.3.3. Data Access Request Responsibility: Users [Research, Policy Development, QM in Healthcare]

1. build for their access request on the existing information in the User Organisation Registration database.

2. must provide information in English about the **context of the access request**, the **User Organisation**, the natural and legal **persons** that will access the data under this request; information on their **project as well as the context** (e.g. research project from competitive funding or a policy development project following a concrete governmental request), including a **scientific and a lay summary** and a **controlled-vocabulary** characterisation; a **data analysis plan** and, where applicable, **statements on AI methodologies**; the **inclusion criteria** for data subject profiles and a justification for the **requested data types**; where applicable, information about **data to be imported**, **software needed** including **bespoke/proprietary software** to be imported, not readily available on the platform; **improved datasets** that may be generated; estimated **duration of access**, any **expected sleeping periods** where no access to the data will be needed for a longer time, and requirements on the **stability of the dataset** and/or the need for **longitudinal updates** during the project; justification where for the same question, access will be pursued over a longer timeframe and potentially updated/additional data (this may in particular be

applicable to healthcare QM); an **ethics assessment** by a relevant ethics body about the planned work in case of a research or a policy development project; and, where applicable, information about the **funding organisation** and any **peer-reviews**.

3. may invite collaborators on the same data access request where these are to act as joint controllers on the project and will want to access the data for the same project.

Justification: The User does not have to provide the administrative information anymore as this was already done during the registration and verified in the process. This avoids duplicate efforts on the side of the Users and avoids having to check/correct wrong information provided in the application process.
The information provided must allow assessing if the planned project has been submitted under the right scheme (i.e. research, policy development or QM), the soundness of its methodology, the compliance with data minimisation requirements as well as any potential ethical challenges. In this context, it can be quite relevant for the assessment in which context the access request takes place. E.g. competitive funding based on peer-review will have likely gone already through a thorough scrutiny that can be built on. To assess if the User aligns with the goals of policy development in the health system, it may be relevant to know if the study is made as a general input (e.g. for health technology assessment) or in the context of concrete plans for new policies. Furthermore, where applicable, an ethics review and information on AI methodologies provide insight if the planned project itself is in line with ethical standards. Such proof should be provided by the User if possible and not by the Genome EDIC or the 1+MG Data Holder. The data analysis plan and the data inclusion criteria are needed to justify the data requested and compliance with the data minimisation principle. The lay summary is required for transparency reasons to inform the data subject. Peer-reviews and funding information can support the assessment of the soundness of the methodology.

Information on data import and software needs will provide information for the necessary processing environment and is relevant to assess the IT setup needed, including evaluating potential cost drivers, in terms of software maintenance, licensing fees as well as compliance issues. Information on any requirements of the stability of the dataset will be important to inform the data subjects as it limits their possibility to object to the processing for the project at a later stage.

EHDS Relevance: The requirements for health data access applications under the EHDS Regulation are currently still under discussion. They will be established through implementing decisions by the European Commission. Modifications are possible based on national ethics laws. We expect that it should be possible for authorised participants to go beyond the scope of the access request forms intended in the EHDS where this is necessary for the access decisions of an authorised participant in line with its data governance. A

dialogue should be sought with the stakeholders planning the EHDS how the Genome EDIC can adhere to its standards also within the EHDS.

VII.3.4. Data Access Request Responsibility: Users [Healthcare reuse]

1. Provide information about the **context of the access request**, the **User Organisation**, the **persons** that will access the data under this request; information on **context of their access request** based on a **controlled-vocabulary characterisation** (general information **the medical issue** prompting access); information on **data types** and **categories of data subjects** to be investigated with a **brief justification for the requested data types**; information about **software** needed; estimated **duration of access**.

2. Follow, where applicable a 3-step approach:

    1. Step: apply to gain information based on **statistical data**.
    2. Step: in case statistical information is not sufficient, submit a **follow up request** under the same application to access individual level data to be able to review individual records of patients. Extend the original data access request with a justification to **explain why statistics based on these records are not sufficient** for them to achieve the purpose.
    3. Step: in case the review of individual records does not provide sufficient information, submit a **second follow up request** under the same application to be able to get **information on individual patients** beyond the data available through the Genome EDIC Secondary Use Framework itself. Extend the original data access request with a **justification to explain why the additional information on the patient is needed** to achieve the purpose.

Justification: The information provided must allow assessing that the planned processing falls under preventive or occupational medicine, medical diagnosis, provision of healthcare or treatment as described under Art. 9.2.h GDPR. A balance needs to be found that the information allows a plausibility check by the DAC but at the same time should reveal as little information about the patient as possible to avoid disclosing personal data creating a risk for the patient.
Controlled access descriptions should make it possible to collect information that can be used in records of processing activities (ROPAs) as required by the GDPR and for the transparency measures towards the data subjects without revealing personal information on the patients.
In the context of healthcare data reuse, Users may request different types of access to meet their objectives. Users in the healthcare sector would typically look for statistical analyses or aggregated data (e.g., obtaining survival rates through Kaplan-Meier curves, tree-plots,

hazard ratios, or oncoplots of aggregated mutational data). However, in specific cases, they may still request individualised data, as they need to review individual patients in more detail (e.g., if there are unexpected phenotypes and/or relevant co-morbidity) or cannot obtain aggregate data about variant clinical significance. As the types of genomic and clinical data required to address clinical questions vary, and data contained in the datasets are selective, Users might need to follow up with the treating healthcare professional or another entity mandated to disclose additional information if essential patient information is missing. Therefore, the IT infrastructure should enable Users to request additional data at each stage of their process with a respective justification. This would not be considered a new request, ensuring that previously provided information is not requested again.

EHDS Relevance: the EHDS's information requirement for a data access request is still under development and will be determined in implementing acts. It is not yet clear if healthcare reuse will be considered at all as the mechanisms under the EHDS Regulation are not well suited for the needs of healthcare regarding speed of access and the necessity to find similar patients. In any case, no direct contact of data subjects with health data holders is covered under the EHDS Regulation. The consequences of this difference between the processing under the Genome EDIC Secondary use Framework and the mechanisms covered in the EHDS Regulation will have to be found out. It may well be that in consequence such options can only be offered through the Genome EDIC Secondary Use Framework outside the EHDS framework, i.e. not as an authorised participant.

VII.3.5. Data Access Request Responsibility: User Organisation [Research, Policy Development, QM in Healthcare]

1. Confirm the access request of a User and/or **provide information on whitelisted Users always authorised to submit requests** where such data shall always be kept up to date.

Justification: The User Organisation will become the data controller for the User project and will have to sign the data use agreement. Not all Users working at their host organisation have the authorisation to pursue their own projects and/or handle sensitive information. As financially liable, the User Organisation has a legitimate expectation to understand and sign off on the expected cost related to the project. To avoid reviewing access requests by persons who are not eligible, the coverage of costs and support for the submitted (assumption of controllership) request must first be confirmed by the User Organisation.

EHDS Relevance: The EHDS Regulation does mention such a feedback loop before processing the access application. However, that should not exclude such a request of the Genome EDIC CC to obtain this confirmation in advance to the access request review. Alignment with the stakeholders planning the EHDS should be sought on this aspect.

VII.4. DATA ACCESS REVIEW

VII.4.1. Data Access Review Responsibility: Genome EDIC CC [Research, Policy Development, QM in Healthcare]

1. Provide the personnel for a 1+MG DAC to review the data access request. The review shall cover the appropriateness of the chosen application type(i.e., is it a research project, a policy development project, a QM project or potentially healthcare reuse), the consistency of information provided, the suitability of the chosen data for the intended project and the detection of any vulnerability of subjects that should be considered in case of research or policy development projects, which is not already flagged through the ethics assessment provided.

2. Ensure that the 1+MG DAC finishes the initial assessment in principle within 5 working days and transmit the outcome of the review, which includes an opinion for the decision on the access request to the 1+MG NCPs who can channel it to 1+MG Data Holders and/or relevant bodies on the national level (see below).
[NB: the feasibility of a timeframe of 5 working days needs to be reviewed after some experience with the reviewing process has been collected. In addition, a check of the technical feasibility may lead to a suspension of the review.]

3. Ensure that all entities on the national level and all external experts involved in the review through the 1+MG DAC have signed a confidentiality agreement.

4. Give reminders to 1+MG Data Holders and/or the 1+MG NCP before the deadline for feedback is reached. In exceptional and justified cases an extension of the time frame may be given.

Justification: A central initial review procedure allows economies of scale because a large part of the review takes place independent of the specific data requested such as the validity and consistency of information provided. It can also provide specialised knowledge such as on AI methodologies, which may not be present in all local DACs. Overall, it will save time on the national level if the local DAC can build on "pre-digested" information rather than having to check all details by themselves from scratch.

Having a central review enables also a more coherent interpretation of the review criteria.The implementation of a central and a local DAC (see below) serves as a safeguard ("4-eyes principle"; in this case 2-DAC principle), which combines a professional, dedicated DAC on the central level that interprets the criteria across all countries with a local DAC that may have a better understanding of the local situation and the background under which

data were collected but may not in all cases be a dedicated DAC staff that pursues such reviews as a main part of their job. Such a 4-eye principle therefore improves the quality of the access review, limits the effects of a potential conflict of interest and combines different types of expertise for a more balanced overall recommendation. In case of 1+MG Cohort Data, it will also allow countries to still build on established bodies and not rely on the Genome EDIC's decision making only.

A concise timeframe for the assessment is planned to offer a competitive data infrastructure. If the turn-around time for data access is too long, there will be not enough interest in the infrastructure. Experience from the repositories by the National Institute of Health (NIH) in the US demonstrates that high-quality reviews are possible in a short timeframe where experienced staff is provided for such tasks.

As the Users will disclose confidential data on their research and their methodologies as part of the application process, it is pivotal that anyone who has access to this information is subject to confidentiality obligations.

EHDS Relevance: The data governance of the decision making under the Genome EDIC Secondary Use Framework is not subject to the provisions of the EHDS. However, upcoming Implementing Acts may provide some criteria that have to be fulfilled, such as the exclusion of any conflict of interest in the decision making.

Just as a comparison: the EHDS also has a central review of the data access request on the national level through the HDAB. Health data holders are either not involved in the review of the health data access request at all or, in cases of trusted health data holders, can only give recommendations where the final decision is still with the HDAB.

VII.4.2. Data Access Review Responsibility: Genome EDIC CC [additional for Research, Policy Development]

4. In cases where the 1+MG DAC found the planned project may affect vulnerable subjects or groups, and /or may lead to discrimination, call on an additional review committee for vulnerabilities with relevant experts including a representation of affected vulnerable groups, involving where applicable, the 1+MG Data Providers. Another reason for calling on an ethics review committee is where other special ethical issues are identified, the provided ethics assessment does not give sufficient information, or a User may not have been able to obtain an external ethics review at all. Obtaining an assessment by the additional review committee should not take longer than 10 working days.

Justification: Part of the 1+MG DAC work is to assess if special ethical issues may arise with relevance for the data subjects implicated by the planned project. Additionally, due to the lack of coherence in the expertise of ethics committees across Europe and/or the mandate of such committees in which cases they provide an assessment means that an ethics assessment may not respond to all questions relevant for giving a recommendation or, in particular for policy development, there may be no relevant ethics body at all that would give an assessment. In such cases, relevant experts and/or representatives of affected groups should be involved.

EHDS Relevance: As above: 1+MG should be free in how it achieves its access decision if it adheres to the quality and timeframes of the EHDS. As a side node: also, the EHDS allows the national level to build additional ethics review procedures into the reviewing process.

VII.4.3. Data Access Review Responsibility: Genome EDIC CC [Healthcare reuse]

1. Establish whether the User's healthcare professional status has remained unchanged since the signature of the framework agreement either through the User Organisation or following the verification procedures established by the 1+MG Member Country.

2. Provide the personnel for a 1+MG DAC to review the data access request. This review shall cover the appropriateness of choice to submit the request under healthcare reuse, consistency of information provided and the suitability of the requested data for the described healthcare application. This applies to the entire stepwise approach, where applicable:
the initial request as well as to any request for more information through record-level access and subsequently to reach out to the treating healthcare professional that may subsequently be raised by the User.
3. Ensure that the 1+MG DAC finishes the initial assessment in principle within 1 working day and transmits the outcome of the review to the User and the 1+MG NCPs who can channel it to relevant bodies or databases on the national level (see below). In justified cases, the review can be extended to up to 3 working days, i.e. by 2 days.
4. Even for this "fast track" procedure a "4-eyes" principle must be implemented to prevent corruptibility of the process. This means that in case of 1+MG Cohort Data, it has to be ensured that at minimum two people have to check the appropriateness of the access request and in case of 1+MG Compliant Data arrangements have to be made with the 1+MG Data Holder on who should co-review the data access request in the short timeframe on top a single person in the 1+MG DAC.
[NB: the feasibility of a timeframe of 1-3 working days needs to be reviewed after some experience with the review process has been collected.]

Justification: The verification of the status as a licensed healthcare professional is key to ensure that the User is acting under professional secrecy, which is the main safeguard for access to the data for healthcare purposes, in accordance with Art. 9.2.h GDPR for the permission to process health and genetic data in a healthcare context.

It is to be noted that the reuse in healthcare is currently envisaged to be made available only for Users in Genome EDIC Member Countries, which should allow for related protection measures, which is to oblige Genome EDIC Member Countries to provide for ways of vetting individual Users as healthcare professionals/persons acting under medical secrecy in a healthcare organisation. The "licensed healthcare professional" status verification process will ideally be done through a database of qualified and licensed healthcare professionals provided by the 1+MG Member Country. It is still subject to discussion if/how this can be realised. Where this is not possible, it needs to be explored what alternative approaches are, including a confirmation by the User Organisation. However, such a process may be clumsy as it could require active feedback by the User Organisation and further challenges where the User is a self-employed healthcare professional.

Same as in the other applications, a central review takes place because the 1+MG DAC can do once what otherwise would have to be done in parallel by many different entities on the national level, thus saving time and efforts. is responsible for the data disclosure and has therefore dedicated personnel that is able to review the access request. The review will be limited to a plausibility check of the request provided. No principal ethical problems are expected. A high level of trust is given to the fact that the User is a healthcare professional who by nature of the profession deals with highly sensitive and usually even identified data and who are under professional secrecy. However, to rely not entirely on trust, a plausibility check is still performed. This becomes even more important where record-level data on the individuals become accessible and where there is a direct exchange between the treating healthcare professional and the User. *[Principle: DPbDD; legal feasibility]*

The review time is set to a much shorter time than for the other types of data use because in healthcare, time may be critical and there are cases where healthcare professionals need answers within 24 hours. The shortening seems feasible because the questions are limited to learning from similar patients and therefore less complex, the analysis by the User is limited to statistics of some kind or a review of individual patient records in comparison to an own patient. There is less discretion in the decision making: if an access request from healthcare is genuine and justified, the request should go ahead – for healthcare, there are not different purposes with each access request but rather only different use cases under the same purpose, so the principal approval had already been given where the User Organisation was vetted as a healthcare organisation and the framework data use agreement signed. The Genome EDIC CC should therefore be entitled to provide the User with an answer regarding their requests based on the plausibility checks of the 1+MG DAC

and the User vetting as quickly as possible, which is why a central review procedure is favoured.

EHDS relevance: while the EHDS time frames for health data access request reviews are not suited to the needs of healthcare, the challenge of establishing the continued validity of the User's credentials is applicable to EHDS as well. No information is available yet how this subject will be addressed. However, synergies should be sought.

VII.4.4. Data Access Review Responsibility: 1+MG NCP [Research, Policy Development, QM in Healthcare]

1. Channel the information on the data access request and subsequently the information on the 1+MG DAC opinion to the relevant stakeholders, e.g., 1+MG Data Holders for 1+MG compliant data, 1+MG Data Providers and, where applicable, a Local DAC as defined on the national level.

2. Communicate the feedback received from the stakeholders on the national level to the 1+MG DAC within not more than 1 working day.

3. Send reminders to the relevant stakeholders before the deadline is reached.

Justification: In principle, it is always recommended to involve the 1+MG Data Provider in the access decision as they understand best the kind of data subjects included and where there may be ethical or legal issues. However, in some countries there may also be laws in place that require certain review procedures such as by a national ethics board. Even though these laws should not be applicable (for the disclosure by 1+MG Data Holders is based on consent and therefore not subject to national law on data disclosure or, in case of 1+MG Cohort Data, the controllership is with the Genome EDIC who is not subject to the Member Country's jurisdiction for the disclosure), the possibility to follow established and trusted procedures should be given. The consultation of relevant entities is then to be arranged on the national level and to be implemented through the 1+MG NCP.
The 1+MG may here directly channel all information onwards or pre-prepare/filter information as necessary. However, such measures should not lead to major delays.

EHDS Relevance: These procedures are internal to the Genome EDIC Secondary Use Framework.

VII.4.5. Data Access Review Responsibility: 1+MG NCP [Healthcare reuse]

1. For 1+MG Compliant Data: Channel the information on the data access request to the 1+MG Data Holders immediately.

1. For 1+MG Cohort Data: channel the information on the data access request following the 1+MG DAC decision to the relevant stakeholders or databases within not more than 1 working day.

2. For 1+MG Compliant Data: channel the information on the outcome of the checks performed by the 1+MG DAC immediately.

Justification: In case of 1+MG Compliant Data, the 1+MG Data Holders are responsible for the access decision, even if the necessary checks are performed by the 1+MG DAC. Therefore, they have to be immediately informed on the access request even though the principal "go ahead" does not require a new signature of a contract, contrary to, e.g., a research project where every new research project means that a new question is asked, which means a new purpose is pursued. In case of healthcare reuse, under an established framework contract, the patients for whom access is asked may change but not the principal question(s), which is to provide information into the decision making of the treating healthcare professional of that patient.

In case of 1+MG Cohort Data, 1+MG NCPs are the ones who know about what will be required in terms of information on the secondary use of data – it may be that certain stakeholders such as 1+MG Data Providers should be informed about the ongoing data use or that data are simply made available in the relevant databases. Information may be directly channelled by the 1+MG NCP or prepared/filtered as if desired on the national level. However, such measures should not lead to major delays.

EHDS Relevance: these information flows have no equivalent in the EHDS Regulation as this case is not catered for.

VII.4.6. Data Access Review Responsibility: 1+MG Data Holder [1+MG Compliant Data]/ [Research, Policy Development, QM in Healthcare]

1. Perform a review of the data access request based on an internal or external local DAC that has no conflict of interest, considering, among others, the opinion of the 1+MG DAC and, where applicable, an additional ethics board for vulnerabilities.

2. Communicate the outcome of the data access request review within 15 working days where no ethical challenges requiring extra scrutiny were identified and within 25 working days if additional consulting on ethical challenges is necessary.

Justification: As the 1+MG Data Holder is responsible for the access decision, it is down to the 1+MG Data Holder to determine how this decision is taken on the national level. This is different to the 1+MG Cohort Data where the entities involved in the national review can be determined by the 1+MG Member Country. However, there are requirements on how the review takes place, which is that conflicts of interest have to be avoided. If it is an infrastructure provider, there can be a bias towards making the infrastructure accessible and therefore a certain lenience/risk tolerance when reviewing an access request. If the 1+MG Data Holder is a research organisation, it may not want to disclose data to a competitor - and vice versa an applicant is not likely to want to disclose their research protocol to another research organization competing for the same funding . Such conflicts of interest must be mitigated on the side of the 1+MG Data Holder.

The time frame for the access decision is derived from the previous assumption that the national level should be able to review the access request based on the opinion of the 1+MG DAC within 10 days. However, the "clock" starts counting here as soon as the access request is received as the 1+MG Data Holder may also perform an independent review straight away.

EHDS Relevance: The EHDS Regulation equally requires that conflicts of interest for the decision making on health data access requests are to be avoided. This is typically done by mandating HDABs as the central decision making body and not letting health data holders participate in the decision making. However, also HDABs may have conflicts of interests but Art. 55.3 EHDS addresses explicitly that these have to be mitigated.

VII.4.7. Data Access Review Responsibility: 1+MG Data Providers and/or Local DACs [1+MG Cohort Data only] [Research, Policy Development, QM in Healthcare]

1. Review the recommendation on the access request decision including, where applicable, the opinion of an additional ethics board for vulnerabilities.

2. Communicate, if there is no agreement with the opinion of the 1+MG DAC, a veto together with a justification why the 1+MG DAC opinion is not suitable within 10 working days.

3. Optional: communicate an agreement with the 1+MG DAC opinion within the time frame of 10 working days.

[Consequences of a veto – see below under access decision]

Justification: The possibility to review the access request decision and exercise a veto allows that the access request is effectively reviewed on the national level by the body deemed suitable by the Genome EDIC Member Country. However, it is suggested that the feedback is not obligatory. Background is that some data providers, if nominated to perform the national review, may not have the capacity to properly review an access request. The national review, however, also allows that another body than the 1+MG Data Provider to take responsibility to review the access request. However, these bodies may also be hampered by capacity problems (e.g., data protection authorities, HDABs). This should not lead to delays in the procedure where the Genome EDIC would become unable to respond to a data access request because it must wait for feedback that is not forthcoming. The risk should be limited though – even if nobody on the national level reviews the access request, the review through the 1+MG DAC should still be a good basis for the decision making as it is performed by well trained staff equivalent to a permit authority. In comparison: under the EHDS Regulation, the decision is also taken by a single HDAB. Nevertheless, a country could also think of back-up solutions in case the originally envisaged entity is overloaded and unable to respond.

EHDS Relevance: These procedures are internal to the Genome EDIC.

VII.4.8. Data Access Review Responsibility: User Organisations (where applicable) [Healthcare reuse]

1. Confirm the ongoing validity of the credentials of the requestor affiliated to them within 1 working day or through reliable authorisation tools.

Justification: Since an initial verification of User credentials was conducted before the request submission, the full process does not need to be repeated. The Genome EDIC CC will only coordinate with User Organisations to confirm that there have been no changes since the framework agreement was signed. This involves the confirmation by the User Organisation that Users are still affiliated with the User Organisation and that their credentials remain valid. This may be achieved through IT tools where these are secure and reliable.

EHDS relevance: the challenge of establishing the continued validity of the User's credentials is applicable to EHDS as well. No information is available yet how this subject will be addressed. However, synergies should be sought.

VII.5. DATA ACCESS DECISION

VII.5.1. Data Access Review Responsibility: 1+MG DAC [Research, Policy Development, QM in Healthcare]

1. Integrate the feedback obtained (approvals by 1+MG Data Holders on 1+MG compliant data; national veto on 1+MG Cohort Data, where applicable; consents by data subjects/objections in case of disclosure based on legislation).

2. Aim to find consensus on the data access request in case of conflicting opinions received from the national levels or insufficient justification of exercised vetoes. If still no consensus can be reached, even with a suitable mediation body, the respective data on which no consensus on the disclosure could be achieved will not be released.

3. Inform, where applicable, other 1+MG Data Holders/local DACs on the reason for the change of the original opinion and allow them to reconsider their own opinion.

Justification: The 1+MG DAC is the place where all feedback from the different involved countries comes together and gets integrated to allow the Genome EDIC CC to communicate the result of the access review. A consensus should be sought, ideally directly between the 1+MG DAC and the opposing entity or, if this is not successful, a neutral third party should be brought in for mediation where the Member Countries can determine the composition of that body. Where still no consensus can be found, the no data should be released as there is a high risk that a wrong decision is taken. The principle is that the protection of the data subjects has the highest priority: "If in doubt, no data should be disclosed". This option reflects the responsibility that Member Countries have towards their citizens and should instigate trust of the data subjects in the management of their data within the Genome EDIC Secondary Use Framework.

The information that the initially communicated opinion of the 1+MG DAC has been altered is important for other reviewers who may want to reconsider their own review based on that additional information. Therefore, they should be given additional time for a re-review.

EHDS Relevance: These procedures are internal to the Genome EDIC.

VII.5.2. Data Access Review Responsibility: Genome EDIC Member Country [Research, Policy Development, QM in Healthcare]

1. Enable the possibility of data subjects to consent to or reject the use of their data in the planned project where the legal basis for the disclosure is consent. There may be a

limited time window only within which their feedback can be considered.
Have the possibility to object against the disclosure where the disclosure is based on legislation. Such objections should be communicated within 12 working days.
[Research (except for recruitment into clinical studies), Policy Development, QM in Healthcare]

2. Establish a mediation body to help the 1+MG DAC and a local DAC to find consensus in case of deviating opinions on a data access request.

3. Review annually the quality of the reviews (accuracy, timeframe) and take action where necessary to mitigate bottlenecks and/or lack of expertise.

4. Provide 1+MG IT infrastructure on the national level that allows a reliable linking between the key-coded identifiers used in the context of the Genome EDIC Secondary Use Framework and the identity of the data subject for the subject of information provision to the data subject and the possibility to obtain consent or information on the objection of the data subject against the disclosure in case of disclosure based on legislation.

5. Strive for a legal basis to allow the use of the entity that handles the transfer between key-coded identifiers of the Genome EDIC Secondary Use Framework and identifiers in the EHDS to also serve for the purposes in the context of the Genome EDIC Secondary Use Framework.

Justification: Consent under the GDPR is specific. We have identified that each disclosure for a different question constitutes a different purpose for the disclosure, which determines the risks and surroundings of downstream data use. Each new project in research, policy development or QM in healthcare would therefore require re-consenting as these questions could not be covered in the initial consent. The situation is different for the reuse in healthcare or the recruitment into clinical studies: here, the questions seem sufficiently predefined to be covered by the initial consent without having to reconsent.

These considerations are subject of the upcoming Chatham House Rule workshop after which we will hopefully be able to update the above considerations.

Where the processing is based on a legislation that provides for a task in the public interest, the GDPR requires under Art. 21 that data subjects should have the possibility to object against the processing of their data. The granularity of objection is here equivalent to the consent options discussed above. This right to object should be ensured through the Genome EDIC Secondary Use Framework as much as possible.

The Genome EDIC Member States should be involved when there is any disagreement between experts on the disclosure of their country and are therefore well placed to establish a mediation body to support the consensus finding between the experts.

The idea behind the 1+MG Declaration was to provide an efficient cross-border access to genomic and health data according to a harmonised data governance. It is therefore in the interest of the Genome EDIC Member countries that the procedures are efficient and take action to improve the procedures where necessary. This is particularly relevant as the entire infrastructure can suffer where access requests are hampered based on individual non-performance.

At the time of inclusion of data subjects, key-coded identifiers must be provided to replace direct identifiers; however, it must be possible to revert such key-coded identifiers to allow an information of data subjects on the intended disclosure purposes and to obtain consent or give the possibility to object, as applicable. This is best done on the national level according to suitable information such as national IDs that can be used. Existing services or services created in the context of the EHDS should ideally be used for synergies.

EHDS Relevance: The EHDS Regulation does not cover the possibility of mediation in case of conflicting opinions. The final decision is with the HDAB and no further suggestions are made how the decision is reached. If, from whom and how many opinions on the disclosure are obtained is left to the Member States. Regarding the mediation body – where an HDAB is not involved in the decision making on the disclosure of data through the Genome EDIC Secondary Use Framework on the national level, they may be good candidates to support a mediation body.

A pseudonymisation and de-pseudonymisation will be relevant for the EHDS and therefore the necessity to swap direct identifiers into a key-code, even though the possibility to object is not directly addressed in the EHDS. The right to object is covered in the GDPR under Art. 21 though and might lead to similar procedures for data disclosure through the EHDS as well. In any case, the right of data subjects to be informed about significant findings means that the pseudonymisation methodology that will be deployed for the EHDS needs to allow a re-identification of data subjects. There are therefore synergies that should be used for both 1+MG and HDABs under the EHDS.

VII.5.3. Data Access Review Responsibility: 1+MG NCP [Research (except for recruitment into clinical studies), Policy Development, QM in Healthcare]

1. Inform the data subjects in case of a positive access decision about the planned project providing the lay summary and other key information as required under the GDPR within not more than 1 working day through a reliable information workflows and based

on suitable 1+MG IT infrastructure including a identity management tool/service swapping between key-coded identifiers and identity as well as an information portal where data subject can find the specific information on the use of their data (for more details see below for the Genome EDIC CC).

2. Collect and channel consent information or information on objection to the relevant parties on the national level as well as the 1+MG DAC.

Justification
The transparency principle of the GDPR, namely Art. 13 and/or 14 GDPR, requires that data subjects are informed about the processing of their data in advance of such processing. An exemption is possible under Art. 14 where the information provision is impossible or a disproportionate effort. In case of the Genome EDIC, it is assumed that most data are included as key-coded with a link to the identity and related contacts that make it possible to inform data subjects on incidental findings or to contact them about an inclusion into a research project. Therefore, it cannot be argued that procedures for contacting data subjects about the likely use of their data in a project would be a disproportionate effort. Suitable IT tools such as an information portal can ease the information provision and can allow data subjects to set preferences about information.

In addition to the information obligation, it will also be necessary to obtain consent where the disclosure to the User does not take place based on legislation. Therefore, the communication should happen with as little delay as possible.

However, this information flow may not have to go through the 1+MG NCP but might also be organised otherwise on the national level – to be discussed.

In addition: it is still to be clarified if/that no repeated consent or possibility to object would be necessary in case of healthcare reuse and recruitment into clinical trials. More details on the (re-)consenting or possibility to object see below.

EHDS Relevance: The EHDS Regulation does not need the consenting of data subjects unless this is introduced by Member States for genomic data based on Art. 51.4 EHDS. However, information obligations under Art. 13/14 GDPR should in principle apply and may lead to an interest by Genome EDIC Member Countries to establish IT tools that can serve both 1+MG and EHDS implementation.

VII.5.4. Data Access Review Responsibility: 1+MG Data Holder [1+MG Compliant Data]

1. Perform a principal DPIA and regular updates for the data disclosure procedure and related workflows and IT infrastructure with a prior consultation of the data protection

authorities before the first access is granted. Perform a specific DPIA in cases that exhibit a high risk related to the data disclosure due to the nature of the processing and/or the amount and types of data involved.

[Research, Policy Development, QM in Healthcare]

2. Adopt and document the access decision based on the opinions given by the local and the 1+MG DAC.

3. Distribute the information through the 1+MG NCP to the relevant stakeholders of the Genome EDIC Secondary Use Framework.

Justification: The 1+MG Data Holder is responsible for taking the access decision, which makes it a controller for the disclosure. As the disclosure of genomic and health data is high-risk processing, a DPIA is needed. On the level of the overall procedures, a prior consultation of the data protection authorities will be needed. Such overarching DPIAs need regular revisiting. However, it may still be required that individual access provisions also need to be proceeded by a DPIA, which could influence specific TOMs to be taken beyond the "standard measures".

NB: no formal access decision is needed in case of access for healthcare reuse; here, the purposes do not change where for the same User Organisation, access is sought for different patients or even where different User Organisations request access to support the healthcare provision for their patients. No additional contracts need to be signed for these access requests.

EHDS Relevance: These procedures are internal to the Genome EDIC.

VII.5.5. Data Access Review Responsibility: Genome EDIC CC

1. should offer modular tools to 1+MG Member Countries that support an information portal, a consenting app and a related identification management service. The tools should allow for customisation by the data subjects according to their preferences in notification and include the possibility for giving automated responses for consent requests, subject to suitable safeguards.

2. must perform a DPIA with regular updates on the information portal components and a prior consultation of the data protection authorities on the information portal and consenting app. Where applicable, support the DPIAs and prior consultation of 1+MG Data Holders.

3. must support the 1+MG Data Holder with the DPIA and the prior consultation of DPA in case of 1+MG Compliant Data or perform DPIA with prior consultation of DPA in case of 1+MG Cohort Data.

[Only for 1+MG Cohort data use in Research, Policy Development, QM in Healthcare]

4. must adopt and document the access decision on the disclosure of relevant 1+MG Cohort data based on the opinions given by the local and the 1+MG DAC and distribute the information through the 1+MG NCP to the relevant stakeholders of the Genome EDIC Secondary Use Framework.

5. must notify the User and the User Organisation on the data access decision and, in case of entirely or partially rejected access, the reasons for the rejection with a possibility to resubmit, where appropriate, or, in case the of approved access, the conditions (e.g. prices, technical setup) to the extent they had not been established before submission.

6. if the request is accepted, must issue a data use agreement covering all data sources and [where applicable, after approval by 1+MG Data Holders on 1+MG compliant data], encourage the User Organisation to promptly sign the data use agreement.

7. must have suitable IT infrastructure in place that can automatically store the information about all access requests, request assessments and decision-making steps. In case of 1+MG Compliant Data, local copies should be available for all 1+MG Data Holders through suitable APIs.

8. must publish regularly aggregate statistics on access and maintain a transparent register of access requests and approvals. This register should include only general information about the request (category of purpose; categories of User Organisations; countries of origin of User Organisation; countries from which data are requested; in case of rejection, category of reason for rejection; time to decision making).

Justification: An IT supported and customisable toolset for data subjects is a "data protection by design" solution that should allow data subjects to be informed and have their data used as they wish to without overburden them with too much information or too many requests but also without deciding top-down what information could be sufficient for them. The possibility to give automated feedback to consent requests must be verified as a principle in the next Chatham House Rule workshop. Nevertheless, the actual proposed implementation should still be subject to a DPIA and a prior consultation of the data protection authorities to ensure that a valid consent can be obtained and that the information portal will not lead to any risks to the privacy of the data subject.

As the situation in the different countries is different, tools made up of modular components should be offered through the Genome EDIC CC to minimise efforts in the Genome EDIC

Member Countries when establishing an information portal and the offer of a consenting app.

About the performance of the DPIA for the data disclosure, the Genome EDIC could take over the main work where it acts as a joint controller with the 1+MG Data Holder or could be entirely responsible in case of 1+MG Cohort Data, where it would also be responsible for the access decision.

For downstream User communication, the Genome EDIC CC fulfils its role as one-stop-shop towards the User by communicating the outcome of the review. In case of rejection, the Users have the right to know why an access request was not granted to be able to remedy their access request accordingly. A plethora of different feedback from different sources should be avoided, in line with the principle that the infrastructure should appear to be coherent towards the User.

For issuing the data use agreement, it should be discussed and decided if the Genome EDIC will be mandated to sign on behalf of 1+MG Data Holders to speed up the procedure as another approval step beyond the access approval will once again take time and the submission of the data use agreement will be subject to the slowest party responding.

Documentation is required for accountability reasons under the GDPR and in case of legal proceedings. This must be integrated workflows that do not need manual interactions to allow scaling of operations.

This type of transparent information is important to build trust and promote the initiative's activities. Statistics are seen here as sufficient; in case of a positive access decision, specific information will be provided on each use in accordance with the data use governance. Details of applications should not be provided for confidentiality reasons.

EHDS Relevance: Not all of the above considerations are applicable to the EHDS where the legal basis under the GDPR is provided through the Regulation and HDABs are responsible vis-à-vis the User for the disclosure, not the health data holders.

Regarding documentation requirements and statistics – these are also relevant activities of HDABs. Due to potentially different responsibilities on the national level as well as EHDS specific elements in the reporting, the Genome EDIC solutions could feed into national systems but not replace them.
HDABs also must publish information and statistics on the access requests. Information obligations are more detailed in the EHDS, e.g. it seems like the entire application must be published. However, for the Genome EDIC, we would rather limit the information provision to the absolute necessary if this will be possible to maintain as an authorised participant.

VII.5.6. Data Access Review Responsibility: Genome EDIC General Assembly

1. Shall implement a right to re-dress procedures for User Organisations.
Justification
A re-dress procedure should be offered for the fairness of the process.

EHDS Relevance: Complaints in the EHDS can be raised with the HDABs, or where they relate to data protection issues, with the data protection authorities. Nevertheless, the right to judicial remedy as covered in the Treaty of the Functioning of the EU (TFEU) is still applicable and could be exercised. Genome EDIC Member Countries could consider to involve their HDABs in re-dress procedures for the Genome EDIC unless there is a conflict of interest.

→ IN THE FOLLOWING: DETAILED JUSTIFICATION DATA USE

VIII. DATA USE

VIII.1. TRANSPARENCY AND COMMUNICATION DATA USE

VIII.1.1. Transparency and Communication on Data Use: Users

1. must make all information elements that are required by Art. 14 GDPR available through the 1+MG User Portal at the time of submission of the data access request to ease the retrieval of information to be made available to the data subject.
2. must provide a plain language summary of the results obtained that can be published. In case of healthcare reuse, this should include, at minimum, whether the access was useful for clinical decision making.
3. must report any achievements, based on the results, to the Genome EDIC, such as policies adopted in the health system, based entirely or partially on the access to 1+MG data. The Genome EDIC should be allowed to reference these achievements if published.
4. must report back to 1+MG EDIC the benefits achieved for quality management based on results from data access in 1+MG. Where relevant benchmarking data on diagnosis or treatment is obtained where this evidence could be of interest for other healthcare providers as well, this information should be fed back to the Genome EDIC; the Genome EDIC reserves the right to make this data available as potentially valuable for other countries.
5. must adhere to the rules of the publication policy as also further specified below.
6. must collaborate with Genome EDIC CC on the communication of results.

Justification: The User, as a controller for its data use, is obliged to provide information on the processing for the purpose pursued. However, as they do not have a possibility to contact the data subjects themselves, the information provision should take place through the communication channels provided through the Genome EDIC Secondary Use Framework. This setup reflects a data protection by design approach that requires the design of the processing/the secondary use framework to consider the rights of data subjects. For efficiency and feasibility reasons, the information relevant for the data subject should be provided already in a suitable way at the application stage.

A dedicated information on results is not a legal obligation but rather part of courtesy towards the data subjects who have provided their information as well as an important element to communicate impact towards the general public. The information will also be relevant for reports on achievements of the Genome EDIC. While this is largely an interest of the Genome EDIC, the User, benefitting from the services of the Genome EDIC should support its communication efforts, which are important for the sustainability of the services.
The usefulness of access to the 1+MG EDIC must be made transparent also to demonstrate impact. Policies derived from data in 1+MG EDIC should be made available to other stakeholders in the field. In addition, reports on outcome should be accessible (see next recommendations on publications). In case of quality management, for future Users, accessing benchmarking data from other countries can also be highly beneficial as it provides valuable insights into how different regions address quality management challenges. It allows them to compare their own practices with international standards, identify areas for improvement, and adopt effective solutions that have been successful elsewhere, ultimately enhancing the quality and efficiency of their own work. This effect should be demonstrated.

In addition, any results obtained that can be of use for other stakeholders should be captured and made known, as this contributes to 1+MG's mission to contribute to the improvement of health and healthcare in Europe.

EHDS Relevance: The direct information of data subjects on secondary use in accordance with the GDPR applies in principle under the EHDS Regulation. Health Data Users also need to make the relevant information available, in case of the EHDS to the HDABs.
There is also a general obligation of health data users to make public the results or output of secondary use, which also includes information relevant for the provision of healthcare. A timeframe of 18 months after the completion of the processing in the SPE is given. However, no further specifications have been published yet but more guidance will be given through the JA TEHDAS2.

VIII.1.2. Transparency and Communication on Data Use: Genome EDIC CC

1. must prepare the suitable information capture through the 1+MG User Portal, ideally directly in the application form to capture the information elements required under the GDPR for the information of data subjects.

2. should have End-to-end Information Management Tools, i.e. automated workflows minimising manual intervention. This should include a common European portal with IT tools and APIs that

- captures the relevant information from the User during the access procedures in the 1+MG User Portal and channels the information to both central and national information portals, including the data subject information portal, as well as, where applicable, 1+MG Data Holder's information management systems.
- allows auditable administration and linking of all relevant information related to ongoing and completed projects: approval process, contracts, logging of access etc. Detailed requirements for the information management will be provided through a separate DPbDD characterisation of the IT infrastructure.

3. must allow the User Organisation to fulfil its transparency requirements towards the data subject through the communication tools of 1+MG.

4. must comply with the transparency requirements applicable to controllers for the data disclosure of 1+MG Cohort Data and applicable to processors for the processing in the data use context, where it acts as processor for the User Organisation.

5. must support the transparency requirements of 1+MG Data Holders for 1+MG Compliant Data.

6. should publish a transparent register of all authorised projects.

7. should support feedback collection by Users on output achieved through suitable communication channels and forms that permit a targeted collection of relevant information.

8. should communicate results of data use to the data subjects participating whose data were used for the respective project/data use.

9. should communicate results of data use to the general public, including providing special features of a more detailed reporting of results with the support of communication experts, based on a selection of projects recommended by an advisory board including representatives of patients and the public.

Justification
As the Genome EDIC should have communication channels established, it must, in accordance with DPbDD, plan for the possibility that Users can fulfil their information

obligation through these channels. Manual intervention in the information management should be kept at a minimum, which favours end-to-end solutions. The documentation must be auditable as this is required for accountability reasons under the GDPR and should be administered through APIs between different players rather than having parallel book-keeping.

Apart from legally required information, transparency of data use and outcome is also important for the sustainability of the Genome EDIC as it demonstrates its impact. Therefore, information should be provided in different ways, serving different needs – statistics as an overview; content of all projects and the results obtained. Featured results will gain particular attention and are expected to increase the buy-in of stakeholders.

EHDS Relevance
There is some overlap between the tasks assigned to the Genome EDIC CC and HDABs. Many elements such as featured results will still be defined on Member State level. 1+MG may give inspiration and even provide tools and platforms that could be used in synergy.

VIII.1.3. Transparency and Communication on Data Use: 1+MG NCP

1. must provide translated information to data subjects in the language of choice in addition to the English versions provided by the Genome EDIC CC, at minimum, the choice should cover the official languages of the respective country. Other relevant languages of the country may be offered as a choice as well.
2. should publish information on projects and results of relevant projects on information portals in the country in the relevant language of the country.
3. must consider privacy and quality aspects when using natural language processing tools to support any translation in the context of 1+MG.
4. should strive to improve the performance of natural language processing tools, e.g. through glossaries, where such tools are used for the translations performed in the context of 1+MG.

Justification
Data subjects have the right to be informed in a language they understand and according to the national custom. For impact reasons, also general information on data use should be made available in the language(s) of the respective country. It is likely that translations will be supported by natural language processing tools. This must not lead to any violations of privacy. Also, such automated translations need oversight. Where corrections are fed back to the tool, the quality of the translations can be improved. For efficiency and quality assurance reasons, this is strongly recommended.

EHDS Relevance
Similar tasks are expected to be performed by HDABs as data access requests and result communication in the EHDS will often be in English as well.

VIII.2. Data use agreement

VIII.2.1. Data use agreement - rights and responsibilities of Genome EDIC CC

1. must establish **standard Data Use Agreement templates** that includes also flexible elements for the specific information about the requestor, project, and data requested, including special conditions that can be derived of the specific information, the controller for the disclosure, the fees applicable to the data use as well as general conditions applicable to data use through the Genome EDIC Secondary Use Framework that are derived from the data governance,
2. must sign a data processing agreement with the User specifying the relevant subprocessors and the TOMs established based on the Genome EDIC Secondary Use Framework,
3. must conclude **(sub)processing agreements** with all entities providing the federated 1+MG IT Infrastructure for data use,
4. must make the relevant **SPE audit reports** available to Users/User Organisation,

Justification
The suggestions implement GDPR requirements in accordance with the accountability principle and the provisions in Art. 28 GDPR on processors. The Genome EDIC will act as a processor vis-à-vis the User to avoid loading the responsibility for the federated SPE system on the User Organisation. Where IT infrastructure is subcontracted by the Genome EDIC, it is the EDIC who becomes responsible for auditing them or to check out the report of third party auditors, which makes using the federated IT infrastructure feasible. This means that the Genome EDIC is accountable towards the User Organisation. This role also reflects the Genome EDIC's role as determining the (minimum) requirements under which an SPE can serve in the federated IT infrastructure of the Genome EDIC Secondary Use Framework.

VIII.2.2. Data use agreement - rights and responsibilities of Users

1. have the right to pursue the proposed use as specified in the approved data application request with 1+MG data as long as it complies with the data use agreement.

2. must agree to the conditions of the standard Data Use Agreement, in particular to the condition that the processing takes place in the 1+MG secure processing environment (SPE) or, where applicable, the SPE provided by the European Commission under the EHDS Regulation, to use 1+MG data for the approved purpose only, to respect confidentiality and not make any attempt to trace or match the identification of data subjects, to not pursue any record linkage without explicit permission, not to export any personal data, to report any errors found in the data and to report any data breach in accordance with the contractual provisions to the Genome EDIC CC;

3. may extend a data access request to ask for additional information beyond the already approved request, subject to the Genome EDIC Secondary Use Framework approval mechanisms.

Justification: The User must be able to rely on the data access for its purposes once access is granted.

The conditions in the Data Use Agreement that govern the processing have to be non-negotiable and to be adhered to by the User as they implement our data governance and our safeguards.

EHDS Relevance: The rights and responsibilities of Users are set out in the EHDS Regulation. They are implemented through the health data permits rather than data use agreements, the former being an administrative act based on the EHDS Regulation. However, the principal approach that the permit gives the right to pursue the intended purpose is the same and that rights and responsibilities as set out on a general level are translated into the data permit that then also addresses the more specific elements. Also in the EHDS, the SPE that can be used for pursuing the purpose is provided top down and must be accredited.

VIII.3. Processing in the SPE

VIII.3.1. Processing in the SPE – rights and responsibilities of Users

1. must agree to the Genome EDIC's sub processors to take measures to minimise the risk to the rights and freedoms of data subjects, such as to remove duplicates originating from data linkage, and/or hardening infrastructure and software, i.e. limiting the potential for information breaches, where applicable.

2. should make arrangements for times when access provision is actually needed as Users may benefit from a "sleeping" period, i.e. periods where access to data is ceased

during a certain timeframe when it is not necessary to access the data but access can be resumed on request later on, e.g. for validations;

3. must agree to the access scheme that requires to confirm continued need for access each year.

4. may bring in their own data and/or own software (including commercial and open-source software) subject to sufficient documentation and a security review by the 1+MG IT Infrastructure to prevent data breaches and potential contamination. This includes data obtained through HDABs based on the EHDS Regulation and externally governed high value datasets under other authorisation regimes.

5. may be able to study pseudonymised individual records of data subjects or reach out to the treating healthcare professional where this is necessary for their purpose and where this has been approved by the 1+MG DAC [healthcare only].

Justification: Where datasets are linked between sources, in particular where external data are brought in, this can lead to a risk to the privacy of data subjects. This linkage takes place under the controllership of the User. The reduction of information that may be necessary would normally be performed by the User. To reduce the risk for the data subjects, the User must therefore agree that the measures to reduce the risk can be performed by entities under the Genome EDIC.

The access should be confirmed at regular intervals to ensure that no unnecessary access is given where it may no longer be needed, e.g. where a project is no longer pursued. This is a data protection by design principle that data should be exposed as little as possible. The same accounts for the "sleeping period" – if no access is needed over a longer period, then data should not be accessible; this includes that data could be kept in cheaper cold storage where this is more cost effective. The possibility to temporarily cease access is a feature that was asked by researchers during the 1+MG Use Case Workshops. This may also be more cost effective.

User data and tools should be allowed as otherwise the questions that can be pursued through the Genome EDIC Secondary Use Framework and the data accessible through it may be too limited. This option is particularly relevant to be able to link data that are made available through the EHDS mechanisms or where high value data can be accessed in SPEs provided through national entities of the Genome EDIC, which are not available under the Genome EDIC Secondary Use Framework but still valuable to access.
Special provisions will have to be foreseen to enable the healthcare Users to gather information on the level of individual subjects as may be required in the area of rare diseases.

EHDS Relevance: The rights and responsibilities for the Users are overlapping but not exactly the same as in the EHDS Regulation. Sleeping periods and reconfirmation of continued data use do not apply under the EHDS Regulation. It is not (yet) clear how HDABs deal with increased risks to the privacy of data subjects when linking in external data. This will have to be explored jointly, where data brought in through the Genome EDIC Secondary Use Framework and made available through HDABs should be linked.

VIII.3.2. Processing in the SPE – rights and responsibilities of Genome EDIC CC

1. **where applicable, should orchestrate the provision of SPEs** and **ensure that the conditions** offered in the Data Use Agreement are fulfilled.
2. must provide in the 1+MG User Portal for the possibility to request sleeping periods or stable datasets.
3. has the power to penalise User Organisations for infringements and also **revoke legally the access** given based on the Data Use Agreement if **the conditions of the Data Use Agreement were violated**, in particular purpose limitation or confidentiality. The Genome EDIC CC will also have the power to prohibit future access by the User and/or User Organisation for a certain period depending on the nature of the infringement. The details on potential penalties will be worked out as part of the EDIC contractual and procedural framework.

Justification: The User must be able to rely on the data use agreement and the conditions under which access was given. The provision of a federated SPE may require a certain amount of alignment between the SPE providers on the provision, which should then be organised by the Genome EDIC CC as the central coordination.

It must be possible for the Genome EDIC to act if the User infringes the Data Use Agreement. If the infringement is down to the User rather than the User Organisation, it should be made possible to exclude such Users from access temporarily or permanently. While the 1+MG SPE providers will be the one removing the actual technical access to the data, access must also be removed legally. It would make sense that this is done by the Genome EDIC CC on behalf of all the 1+MG Data Holders.

When access is withdrawn, it may not make sense to exclude the entire User Organisation here, depending on who is behind the origin of the infringement. If that infringement is systemic from that User Organisation, of course the User Organisation should be penalised. If a malicious non-compliant processing was driven by the User though rather than the User Organisation, there must be a possibility to block the User from future access. This should be possible even if the User moves to a new host organisation (such User could likely be

dismissed due to infringements). It must be avoided that Users could try to undermine security again from a new position.

EHDS Relevance: The EHDS Regulation builds on data permits that are defined through the legislation and that are not applicable to Genome EDIC. The permit is based on an administrative act, which can only be issued by HDABs. The Genome EDIC, also as an authorised participant, has its own legal framework and conditions. it cannot issue EHDS health data permits, and it is not an authority that can issue permits. Therefore, data use agreements should be concluded, and they need to be complemented with a data processing agreement as the personal data is not supposed to leave the 1+MG IT infrastructure but remain in the 1+MG SPEs. NB: these SPEs can also serve as SPEs for HDABs.

With respect to penalties in case of infringements, these can be aligned with the EHDS Regulation if required. However, we may want to reserve the right to exclude Users rather than User Organisations from data use in the Genome EDIC even if the EHDS Regulation does not provide for such an option. The penalties and exclusion from the EHDS are based on the User Organisation only.

VIII.3.3. Processing in the SPE – responsibilities of 1+MG IT Infrastructure Providers

1. when acting as 1+MG Data Host, must make data available swiftly on behalf of the 1+MG Data Holder or the Genome EDIC as required based on the Data Use Agreement or any update of the Data Use Agreement,
2. when acting as 1+MG Data Host, must ensure only those selected data, which are covered in the Data Use Agreement, are made available to authorised Users.
3. when acting as 1+MG Data Host, must perform the steps needed for data minimisation or engage a processor who implements data minimisation
4. must provide secure processing environments (SPEs) for the data use meeting the needs of Users as agreed in the data use agreement; the SPE should be compatible with the requirements for authorised participants in the EHDS (see below for more details) and the audit/certification framework set out under the Genome EDIC Secondary Use Framework,
5. when providing an SPE, must enable the import of externally governed data from the User directly, from data providers of externally governed high value datasets or from HDABs,
6. must take steps to limit the risk to the privacy of data subjects (e.g. removing duplicates) resulting from data linkage from different sources or engage a processor who takes care of these steps,

7. when providing an SPE, should allow efficient analysis of data from multiple 1+MG Data Providers/Member Countries, ideally in a federated manner across national secure processing environments.
8. when providing an SPE, offer a suitable IT platform in compliance with the risk and quality management framework of the Genome EDIC Secondary Use Framework; the IT platform is part of a federated system on top of which tools are deployed and into which data can be imported, where applicable to allow Users the pursuit of their purpose in an efficient manner across data from different sources, where applicable; this includes tools for federated data analysis, which are of guaranteed quality, security and suitable licensing and should also plan for high-performance computing to attract Users.
9. when providing an SPE, must technically revoke access in case of a suspected breach of the purpose limitation or confidentiality by the User, inform immediately Genome EDIC CC (as well as other affected 1+MG SPE providers where applicable – this should ideally be enabled automatically through the 1+MG communication infrastructure) in such a case and align with Genome EDIC CC on the communication with the User and User Organisation.

Justification: The requirements follow data minimisation – only those data must be given access to that were recognised on the level of the Data Use Agreement as needed. This data provision should lead to as little delays as possible.

As part of the data provision, there may be additional steps needed for data minimisation (e.g. generalisation). As the data are held by the 1+MG Data Host, this is a task that may be performed by the 1+MG Data Host. However, it may well be that a 1+MG Data Host does not have the expertise to do this. It might be done by any other the 1+MG IT Infrastructure Provider or the 1+MG NCP. As countries should be as free as possible in the distribution of tasks, it is suggested that the initial responsibility is with the 1+MG Data Holder as the "first in line" but that a subcontracting can be done to another entity.

As no download is possible and data are supposed to remain in the Genome EDIC Member Country where possible, suitable SPEs must be provided on the national level so that Users can pursue their purpose within these SPEs. As the SPEs form a federated IT infrastructure, they must comply with the same minimum requirements to provide a coherent service.

When datasets are linked in the SPE, it may turn out that the linked datasets provide clues to the data User on the identity of the data subjects. This should be avoided as much as possible. However, this can only be found once the data linkage has taken place. Therefore, this task can only be performed after data linkage in the SPE.

The 1+MG IT Infrastructure Providers may be the one that detects anomalies. It is important that they can act immediately as they are the controller for the technical and organisational

measures to protect the data, which includes the provision of access to Users. However, as the User Organisation's main contact point is the Genome EDIC and as also several other 1+MG SPE providers may be affected, action should be taken in alignment with Genome EDIC CC, who serves as the interface to the User and User Organisation.

EHDS Relevance: The responsibility of the 1+MG Data Host resembles on the first glance the health data holders in the EHDS. However, the health data holder acts as controller under a legal obligation. The 1+MG Data Host acts only as a processor unless it is identical to the 1+MG Data Holder for the data it is holding. Also, the data held by the 1+MG Data Host are held for secondary use in suitable formats and databases that ease the process of making them available. As under the Genome EDIC Secondary Use Framework, data can be held for secondary use, which is not covered in the EHDS Regulation, the Genome EDIC Member Countries would therefore consider to which extent they want to make use of these mechanisms to increase the availability of improved data in their country where these relate to/complement genomic data. As an example, data that are improved by health data Users can only be made available through EHDS secondary use mechanisms if they are held by a health data holder for certain purposes that do not include secondary use. Where this is difficult to implement, a country could choose to transfer these improved data to a 1+MG Data Holder for secondary use under the Genome EDIC Secondary Use Framework.

Requirements for SPEs in the EHDS are similar. The EHDS Regulation does not address federated computing; it is, however, also not excluded to pursue federated computing under the EHDS, which means that the SPE network provided through the Genome EDIC and its 1+MG SPE providers can form also an important component of the national EHDS implementation and the pursuit of cross-border access under the EHDS rules. The SPEs established through 1+MG should therefore be suited to serve also HDABs in the EHDS.

It is not clear how the HDABs deal with the risk of compromised privacy where data are linked beyond their influence, e.g., once the User brings in their own data or where data from several countries are brought into the Commission provided SPE. Where HDAB provided data are linked with data from the Genome EDIC Secondary Use Framework, the risk could be addressed through the actors under the Genome EDIC.

VIII.3.4. Processing in the SPE – rights and responsibilities of Genome EDIC Member Countries [Research only]

1. may choose to adhere to the approach of federated data use only; in this case the data always remain with the national 1+MG SPE provider(s); alternatively countries allow data streaming and/or enable temporary data pooling for data use where the purpose of

the processing cannot be achieved otherwise or would be too complicated to implement.

2. The choice must be clearly described and communicated as ELSI metadata to the Genome EDIC CC and the national 1+MG NCP to include into the relevant dataset descriptions and to inform the 1+MG Data Providers as this information may be relevant for the data subjects' decision to have their data included. The national policy may differ on the level of categories of sources of data (i.e., healthcare data may follow different rules to data from research).

Justification: Federation is a protective measure – the less often data are transmitted, the fewer data are pooled, the lower the risk. However, where data would never leave the infrastructure provided according to the rules of the Genome EDIC Secondary Use Framework, the temporary pooling in a certified SPE according to the Genome EDIC Secondary Use Framework should come with a very limited increased risk. The real risk that led to the establishment of SPEs is where data are transmitted to Users directly and reliance is on Users to treat the data properly.

Not all research can be pursued in an entirely federated manner and will therefore require pooling at times. The benefit of allowing more complex research questions could likely outweigh the risk of pooling within the Genome EDIC Secondary Use Framework of certified SPEs in many cases. However, some countries may not want to agree to any such pooling, which will have consequences for the usability of their data. This is a decision, which should be taken at country level but may be specific for categories of sources – e.g., healthcare data coming under higher protection than research data.

The decision whether data can be pooled will now rest for 1+MG Compliant Data with the 1+MG Data Holder. This can lead to a much higher complexity where each 1+MG Data Holder may decide differently. It could therefore be decided that, as part of the Genome EDIC Secondary Use Framework, the decision on a possible pooling of data is taken on the level of the Genome EDIC Member Country and form part of the conditions that 1+MG Data Holders must agree to when participating in the Genome EDIC Secondary Use Framework.

EHDS Relevance: The EHDS Regulation does not exclude the possibility to pool data. There will even be an SPE provided by the European Commission for that purpose. However, it is also not excluded to pool data at national SPEs where these are subcontracted by the relevant HDABs. No special conditions are made for SPEs that can receive data from other Member States. The practice of the Genome EDIC providing a central SPE to HDABs through its 1+MG SPE network is therefore an option that could be considered.

VIII.4. Special Processing in [healthcare reuse only]

VIII.4.1. Processing in the SPE – rights and responsibilities of Genome EDIC CC

1. must provide in the implementation of the Genome EDIC Secondary Use Framework for the possibility of Users pursuing healthcare purposes to reach out to treating healthcare professionals and establish through its central and local IT infrastructure to do so in a GDPR compliant way.
2. must ensure through the Genome EDIC Secondary Use Framework that there will be no direct contact data or identities of treating healthcare professionals or data subjects given to Users.

While the overarching possibility for healthcare reuse is defined on the level of the Assembly of Members, it is the Genome EDIC CC that must provide for the practical implementation. An element of that implementation is that for healthcare reuse purposes of the User, Users should never get in contact with data subjects themselves but only with their treating healthcare professionals if such contacting has been consented by the treating healthcare professionals and the data subject. However, also here, no unnecessary disclosure of personal data should happen.

VIII.4.2. Processing in the SPE – rights and responsibilities of Users

1. should have the possibility to request additional information from treating healthcare professionals via a request lodged in the 1+ MG communication infrastructure for update, clarification and/or other additional information if consented by the treating healthcare professionals and the data subjects.
2. must not have direct contact data or identities of the data subjects.

Justification

Given the diverse nature of genomic and clinical data necessary for addressing clinical inquiries and considering that datasets available to the User only contain selective information, Users may find it necessary to engage with the treating healthcare professional should critical patient data be absent. Users should not be provided with the contact details of treating healthcare professionals, but the opportunity should be given to the latter to accept or refuse the provision of additional information to the User, in accordance with the data minimisation principles of the GDPR.

EHDS Requirement
As explained above - this feature is not applicable to the EHDS.

VIII.4.3. Processing in the SPE – rights and responsibilities of 1+MG Data Holders / Genome EDIC CC

1. must enable that data subjects can consent to their treating healthcare professional to disclose additional data to an approved User for purposes of prevention, diagnosis or treatment; this consent should be provided prior to any specific request or access by the User and must be incorporated into the 1+MG metadata.
2. must establish mechanisms in the country that data subjects may revoke the consent given to the treating healthcare professional at any time with effect on any future disclosure.

Justification

The treating healthcare professional discloses special categories of personal data to the User, which are not covered by the legal framework covering data inclusion into the 1+MG EDIC. Such a disclosure constitutes new processing, requiring a new legal basis. The related consent should be given separately for the three different purposes of prevention, diagnosis and treatment. The consent will be given independently of a given access request as it may not be possible to reach data subjects in time when an actual request for more information happens, meaning the consent is given proactively rather than reactively. This should be a "one-off" consent that is given and can be withdrawn again. The 1+MG Data Holders and the Genome EDIC CC must make arrangements that such consents are obtained where the data subjects agree to this kind of information provision.

EHDS Requirement: As explained above - this feature is not applicable to the EHDS.

VIII.4.4. Processing in the SPE – rights and responsibilities of treating healthcare professionals

1. may consent to being contacted for providing additional patient information if their patient has consented to this information;
2. may disclose additional data to Users based on the consent given by their patients, i.e. disclosing information only within the scope covered by the consent;
3. shall not disclose directly identifying information on their patient and strive to minimise indirectly identifying information to what is necessary;
4. should send an information of rejection through the 1+MG communication system if they intend not to reply to the User's request.

Justification: Additional information from the treating healthcare professional has been highlighted as a very important feature for healthcare reuse. Treating healthcare professionals possess the discretion to determine whether they can provide relevant information to the User should have the capacity to decline, should they not be in position to give additional information. However, treating healthcare professionals cannot be obliged to do this even if they may be in possession of such data, which is why any contacting needs to be based already on their consent. Nevertheless, once consented, as a minimum when contacted, they should send a reject to the Users to inform them that no additional information will be forthcoming if they intend not to reply to avoid the User waiting unnecessarily.

Providing consent means the Genome EDIC has a legal basis for holding contact information on the treating healthcare professionals and being able to reach out to them.

EHDS Requirement: As explained above - this feature is not applicable to the EHDS.

VIII.4.5. Processing in the SPE – rights and responsibilities of 1+MG NCPs [healthcare reuse only]

1. must have an information chain to reach out to treating healthcare professionals; in this case the information chain is to inform them about the additional information need of a User and allow the treating healthcare professionals to get into contact with the User for a secure information exchange or for the treating healthcare professionals to reject the information request.
2. must be able to inform the treating healthcare professional about the identity of the respective data subject in accordance with the identity management used by the treating healthcare professional.This may require the possibility to link between the key-coded identifier in accordance with the Genome EDIC Secondary Use Framework and the identifier used on the side of the treating healthcare professional.

Justification: The suggested information chain is to protect the privacy of the treating healthcare professional. This follows the data protection by design and default, and particularly the data minimisation principle.

EHDS Requirement: As explained above - this feature is not applicable to the EHDS.

VIII.5. REACHING BACK TO DATA SUBJECTS

VIII.5.1. Reaching Back to Data Subjects – Rights and Obligations: Users

1. [Research only] should have the right to reach out to data subjects via the 1+ MG communication infrastructure to recruit them into research projects in accordance with the preferences expressed by the data subjects.
2. should have the right to reach out to 1+MG Data Providers for additional data on data subjects beyond the data available under the Genome EDIC Secondary Use Framework [if agreed by 1+MG Data Providers].
3. [Healthcare reuse only] should have the right to study individual records of data subjects (that have been de-identified as much as possible) where this has been approved by the 1+MG DAC.
4. [Healthcare reuse only] should have the right to reach out to the treating healthcare professional for additional information on the data subject [if agreed by treating healthcare professional and consented by the data subject; the additional information must be limited to the information covered in the consent by the data subject and approved by the 1+MG DAC]
5. must not have any direct contact data with the data subjects.
6. Must mandate the Genome EDIC and/or its national entities to perform a risk analysis and limit any additional information provision on data subjects that increases too much the risk to the data subjects.

Justification: The 1+MG will be a unique source that can allow to find and recruit data subjects/patients with a certain genetic constellation. This possibility is particularly relevant for rare diseases. Many platforms that are currently supporting such a findability for rare disease patients require that data are being made available openly on the internet. The Genome EDIC can therefore create real added value for Europe, for both patients and Users. In addition, the possibility to clarify information or to gain additional data (e.g. through generation from biosamples) can boost the possible research and the attractiveness and potential impact of the Genome EDIC. However, such additional data generation needs its own legal basis and must be transparent to the data subjects with the possibility to object, where applicable.

Healthcare: The nature of evidence-based healthcare may make it necessary to have information beyond statistics, for which Users should be allowed to extend their request. These needs were communicated by the Use Case Working Groups. Therefore, the option should be offered but is of course subject to relevant safeguards and should apply only where there is no other less intrusive option.

However, it is important that the data subjects themselves are never contacted directly. They receive information through the information portal/the established communication channels and can act based on that information. Where they decide for example to engage with the User on a clinical trial, this subsequent information exchange takes place outside the

Genome EDIC framework, based on contact information provided to the data subjects. Such an approach follows the data protection by design approach as the least intrusive processing.

Where the User wants to bring in additional data not yet planned for at the level of the access application or not yet obtainable at the beginning of the project, the User must allow that the risk associated with data linkage can be addressed before access is given to the extended dataset.

EHDS Relevance: The EHDS Regulation does not cover that data subjects can be informed in a use context. Therefore, the recruitment into projects is not covered at all under secondary use because HDABs can only make pseudonymised or anonymised data available to the Users. There is no legal basis provided in the EDHS Regulation to contact data subjects. As such, the Genome EDIC can provide real added value with these options on top of the EHDS framework.
Options for sourcing additional information are available in the EHDS through an extended application to the same or, where applicable, a different HDABs in the country. Data from different sources can be brought in but HDABs only support obtaining electronic health data from health data holders. Here, again the Genome EDIC can bring additional value.

VIII.5.2. Reaching Back to Data Subjects – Rights and Obligations: 1+MG NCP/1+MG Data Providers

1. [see also data access governance]: must have an information chain to reach out to data subjects with a reliable linking of project-specific identifiers and real identity link to a contacting possibility (ideally through a data subject information portal that can also be used for information about individual projects and the possibility for data subjects to exercise rights); in this case the information chain is to inform them about a possible participation in observational studies and clinical trials;
2. should also enable going back to 1+MG Data Providers, or, where applicable, original data collectors, with the possibility to gain additional information outside 1+MG Compliant Data or 1+MG Cohort Data (updated data; new data; clarifications). This may require the possibility to link between a project specific identifier and an identifier on the 1+MG Data Provider's side. Any provision of additional information/data that the subject was not aware may be shared on them will also require a renewed information of data subjects and likely consent to provide a legal basis for this data sharing.
3. must establish with relevant actors on the national level a communication chain to reach the treating healthcare professional and enable a communication exchange

between the treating healthcare professional and the User without disclosing the identity of the treating healthcare professional to the User. [Healthcare reuse only]

Justification: The suggested features are important to meet the needs of important use cases and increase the impact of 1+MG. Reaching out to the data subjects should be realised on the national level. These arrangements have to be secure and follow data minimisation.

EHDS Relevance: The situation for creating a communication chain between the data use level and the data subjects and/or the health data holder as part of the data use is not addressed. However, the EHDS mechanisms always allow extending a project with a request for additional data from the health data holder where this is justified. No conflict is expected and ideally, synergies will be found.

VIII.5.3. Reaching Back to Data Subjects – Rights and Obligations: Genome EDIC CC

1. Must provide for an efficient and reliable information handling on consents obtained from data subjects and treating healthcare professionals on the option for a User to reach out to the treating healthcare professional.
2. Must build into the 1+MG User Portal a possibility for Users to reach out and communicate with treating healthcare professionals, linking to communication channels established on the national level.
3. Must not disclose the identity of the treating healthcare professional to the User.
4. Should consider to structure exchanges with treating healthcare professionals and Users as much as possible to minimise the risk of accidental data disclosures beyond the information covered by the consent.

Justification: For healthcare reuse purposes of the User, Users should never get in contact with data subjects themselves but only with their treating healthcare professionals if such contacting has been consented by the treating healthcare professionals and the data subject. As the 1+MG User Portal is the main tool for interaction for the User, the Genome EDIC CC must make sure that this can link to the relevant communication channels on the national level.
Even through the exchange between the treating healthcare professional and the User takes place outside the Genome EDIC Secondary Use Framework based activities as the treating healthcare professional is not acting as a Genome EDIC national entity, the Genome EDIC offering this option of reaching out, should engage in making the exchange as secure as possible, minimising risks to the privacy of the data subject.

EHDS Relevance: This option of reaching to health data holders in healthcare in relation to secondary use of electronic health data is not covered in the EHDS Regulation.

VIII.5.4. Reaching Back to Data Subjects – Rights and Obligations: Genome EDIC Member Countries

5. may make legal provisions to oblige 1+MG Data Providers or, where these have not collected the data from data subjects directly, original data collectors to provide feedback on certain questions related to the data (i.e. additional metadata);

Justification: Some information about the collection and/or the data subject context can only be provided by the entity that has or had the direct relation with the data subject. Where the 1+MG Data Provider is not that original collector, it may not have the relevant information nor any means to oblige the original collector to provide the information. Equally, the Genome EDIC cannot make any provisions for third parties such as the original collector. The only one being able to provide for these obligations is the legislator. Countries may want to establish such obligations for the efficiency and the impact of the 1+MG genomic infrastructure.

EHDS Relevance: There is no obligation for health data holders to provide additional information beyond the dataset description. There is merely an obligation to provide sufficient documentation to verify the accuracy of the data quality and utility label.

VIII.6. IP AND COMMERCIALISATION

VIII.6.1. IP and Commercialisation – Rights and Obligations: User Organisations [research only]

1. May pursue IP claims on downstream discoveries or inventions;
2. Must not pursue IP claims that limit the use of 1+MG data itself;
3. Must inform the Genome EDIC CC if any IP claims were made in the context of the data use.

Justification: It is recommended to allow Users the pursuit of IP for their projects to stimulate innovation. However, such innovation must not compromise the data available in 1+MG. For the demonstration of impact and for the communication of results, the Genome EDIC should be informed about all results that may move forward towards commercialisation.

EHDS Relevance: The situation is similar under the EHDS Regulation.

VIII.6.2. IP and Commercialisation – Rights and Obligations: 1+MG Data Providers

1. [research only] Must agree not to make any IP claims on results derived from the data.
2. Must agree not to pursue IP protections that would prevent or block access to or use of any data or results drawn directly from data.

Justification: There must be no limitation of the usability and in particular the commercialisation potential of data by the 1+MG Data Providers.

EHDS Relevance: The EHDS Regulation follows the same approach that no rights in the results achieved by health data users can be claimed by health data holders (i.e. by not mentioning any such rights). Contrary to the 1+MG approach, however, the EHDS Regulation covers inclusion of data that are IP protected in the first place (Art. 52.2 EHDS) and introduces measures that such IP is not infringed by the use of the protected electronic health data. Only where those measures cannot offer enough protection, the data do not have to be shared.

VIII.6.3. IP and Commercialisation – Rights and Obligations: Genome EDIC CC [research only]

1. Should record information about the results commercial exploitation of results pursued though IP protection and publish this information;

Justification: Demonstrating benefit will be important for an impact assessment of the Genome EDIC. *[Principle: impact demonstration]*

EHDS Relevance: The EHDS Regulation makes provisions that measures should be taken by HDABs that access provision should not infringe existing intellectual property rights, trade secrets or the regulatory data protection right. However, this would not apply to the entities of the Genome EDIC as only data that can be utilised without "IP strings attached" are included.

VIII.7. PUBLICATIONS

VIII.7.1. Publications – Rights and Obligations: Users

1. must publish their findings where these result from a research project pursued with 1+MG data or where the results are compiled into a reports (e.g., case report for healthcare reuse, best practice findings in QM or a report to inform policy making) unless there are good reasons that a publication does not make sense or the report would be protected (to be specified in a publication policy);
2. are permitted to extend their data use agreement for the access of peer-reviewers from outside the User Organisation where necessary to pursue publications.
3. must respect the 1+MG publication policy for publishing, which comprises among others the elements listed below.
4. must provide a lay summary of the publication in English as well as the language of the publication if not published in English.
5. must respect any publication embargos and not publish during the embargo period without written confirmation of a justified derogation from the embargo (as part of the use conditions in the contract or as amendment to the contract).
6. must acknowledge all data sources as well as the Genome EDIC, where applicable, the 1+MG Data Holder.
7. must report the results of completed projects (e.g., publications) to Genome EDIC CC to add to the access register, the public information portal and, where applicable, data subject information portals.
8. must publish manuscripts or reports on the results with 1+MG data as open access.
9. must not use the publication of results from a QM project as advertising for their own organisation and/or its services; [Healthcare QM only]
10. must support the Genome EDIC in the communication of results that are specially featured for the communication with the public.

Justification: The work supported by the Genome EDIC should be respecting full transparency as part of the public interest justification. The outcome should therefore be communicated in a way that is understandable for the general public and all the support necessary for that should be given alongside the actual publication. Publication should further support and in implementation of research integrity and ethical standards. This includes publication as transparency towards the scientific community as good scientific practice. While publications are not an intrinsic part of healthcare reuse, some healthcare professionals still want to publish on the experience made. In findings of studies supporting policy development are often published as a report to provide justification for policies derived from the findings or organisations supporting policy making will compile their

findings into a report to inform policy makers. These reports should then be made available to all policy makers. Also the results of a QM study may be useful for other healthcare providers. However, in particular where benchmarking exercises were pursued, these must not be used as advertising.

An embargo for publications is needed to allow Genome EDIC CC to check the publications for compliance with the policies.

The Users should inform about the final placement of publications to ease the reporting requirements of 1+MG on its impact. Acknowledgement of the sources and Genome EDIC is good practice and also reflects the support received through the Genome EDIC.

An open access publication policy is chosen as a benefit sharing, reflecting the public funds that will have subsidised access and as an acknowledgement towards the contribution of data subjects whose data were used to generate the result.

EHDS Relevance: While some requirements are identical (e.g., acknowledgement of sources), others such as open access publication and publication embargo go beyond the requirements of the EHDS Regulation. However, no conflict should arise from these deviating practices. Rather, they offer important controls and more benefit sharing, both being in the interest of the Member States in their pursuit of the objectives of 1+MG.

VIII.7.2. Publications – Rights and Obligations: Genome EDIC Assembly of Members

1. must adopt a publication policy and make it public.
2. must adopt a legal framework allowing peer reviewers' access as required by the respective journal where the work is published.

Justification: The rules for publishing must be transparent for Users upfront.

When reviewing manuscripts based on data analysis, reviewers often check the work of the researchers by re-running or slightly changing the analysis to test the robustness of the results. This is required for research integrity and a must for researchers to offer. *[Principle: competitiveness]*

EHDS Relevance: No information on these matters yet.

VIII.7.3. Publications – Rights and Obligations: Genome EDIC CC

1. should follow up with Users and User Organisations on their obligation to publish and inform on publications.

2. should ensure compliance of the publication with 1+MG rules e.g. open access, acknowledgement, respect for embargos, public reporting of results.

3. should perform regular publication searches and report on published findings to identify publications that may not have been notified to it.

4. should record information about the publications in relevant databases and publish this information on the portals.

5. should provide professional communication expertise on a representative selection of publications for the public in English, based on the recommendation of a communication advice panel involving citizens.

Justification: Publications are an important impact of the Genome EDIC. As not all Users may inform about the (final) publication, the reporting of Users should be complemented by an active check through the Genome EDIC CC.

Providing professional communication on selected output will help to better inform and also engage the public.

EHDS Relevance: These obligations of the Genome EDIC CC go beyond the obligations of HDABs under the EHDS Regulation (apart from the recording of publications). However, it does not lead to incompatibilities.

VIII.7.4. Publications – Rights and Obligations: 1+MG NCP

1. Provide translated communication into official languages of the country at least for those cases where data of the respective country were used.

Justification: To efficiently reach out to the general public, native languages in the country should be used. The 1+MG NCPs are best placed to do this or at least oversee/verify this effort.

EHDS Relevance: The EHDS Regulation expects the HDABs to publish the results by the health data users. However, no "featuring" of results with special communication efforts are mentioned in the Regulation. Such communication efforts are entirely left to the Member States. 1+MG may serve here as good practice.

VIII.8. ARCHIVING [research and healthcare reuse only]

VIII.8.1. Archiving – Rights and Obligations: Users

1. Have the right to the archiving of data used to generate the results used for publication or IP protection where archiving is necessary for reproducibility reasons.
2. Must provide information on any required archiving periods.
3. Must archive any own data used in the project.

Justification: Publications and, in case of scientific research also IP claims, will lead to needs to demonstrate how the results were obtained and that they can be preproduced. As Users are not allowed to download the data used, that data must remain available through the Genome EDIC and its national entities to re-run the analyses. The Users are the ones who have the information on what precisely is needed.

As storage is costly, the Users should make their own arrangements for the archiving of their own data. This may well be with a Genome EDIC national entity, but any such arrangements would come outside the direct Genome EDIC services.

EHDS Relevance: Archiving is not explicitly addressed in the EHDS Regulation. However, health data users in the EHDS may ask the HDAB for data to remain available for a longer period beyond the mere data analysis but the timeframe for that is limited. Other than that, only the formula of how the dataset was assembled is kept. This may not be enough for reproducibility though as the data are extracted from the health data users' own use, thus the data can change over time, and the original data may no longer be available in the future. Here, the data governance of the Genome EDIC Secondary Use Framework offers advantages for scientific research.

VIII.8.2. Archiving – Rights and Obligations: Genome EDIC CC

1. Must develop strategies together with the Genome EDIC national entities for how an efficient archiving of data and software for reproducibility can be offered to Users to make up for the prohibition of a download of personal data These strategies must strive to avoid having to store multiple copies of datasets derived from the same core data resources. .
2. Must provide the contractual implementation of data processing for reproducibility on behalf of the User Organisation.

Justification: Archiving is required not to undermine research ethics and integrity and allow good scientific practice. As the Genome EDIC Secondary Use Framework prohibits the download of personal data, there must be solutions offered through the Genome EDIC and its national entities. Here, the Genome EDIC CC should be the one to come up with a strategy that can work across all participating countries. However, the local situation must be accounted for – such a strategy cannot be developed in isolation where such archiving data takes place in a federated, changing environment. This archiving takes place for the Users' needs and is directly related to the individual projects for which they are archived.

EHDS Relevance: As described above.

VIII.8.3. Archiving – Rights and Obligations: Genome EDIC Member Countries

1. Genome EDIC Member Countries must accept that archiving of data used for research projects will be required for a certain period after the completion of the analysis (e.g., 10 years) even if they withdraw the data from (future) availability through the Genome EDIC Secondary Use Framework, e.g. when withdrawing from the Genome EDIC.

Justification: The possibility for reproducibility must be guaranteed to Users to allow research integrity. This creates a contractual obligation where the Genome EDIC national entities act as a subcontractor and (sub-)processor to the User Organisation who is a controller for archiving.

EHDS Relevance: See above: this is not explicitly covered under the EHDS Regulation.

VIII.8.4. Archiving – Rights and Obligations: 1+MG IT Infrastructure providers

1. Must have technical data and information management systems in place to create project-specific data archives or have otherwise suitable data management that allow reproducibility. Preference should be given to referencing data used rather than storing the data in an archive. This will require explicit dataset versioning.
2. Must ensure that, where data were held for project archiving only, these are deleted at the end of the last archiving period.

3. Must make provisions that analyses can be re-run with the same software also up to 10 years after the pursuit of a project. This includes archiving of analyses environments and backward-compatibility of the system.

Justification: The IT infrastructure providing the processing environment and managing the data will also be the one that needs to enable the possibility for archiving and reproducibility. Versioning means that there is no necessity to keep entire project specific datasets but only the information on the relevant data used is stored project-specifically. This minimises the number of copies needed.

Data retention times as required under the GDPR must be monitored.

Keeping the data is not enough, the IT infrastructure providing the processing environment will also have to enable the possibility for reproducibility. This will be the responsibility of the platform provider for the analysis.

EHDS Relevance: (see above) this is not explicitly covered under the EHDS Regulation, which means that there are no obligations for health data holders who have provided the data.

VIII.9. ENRICHED AND IMPROVED DATA

VIII.9.1. Enriched and Improved Data – Obligations: Users

1. must specify in access requests any enriched or improved data (e.g., annotated, harmonised, corrected) that will be generated as part of the data use that could have value for future Users.

2. must have the possibility to report mistakes in the data to ask for correction even where no improvement was planned for.

3. if lawfulness can be established, must legally transfer enriched or improved data of potential value for future Users to the Genome EDIC or 1+MG Data Holders for further sharing free of charge at the end of the project.

4. must provide suitable metadata as required by the Genome EDIC Secondary Use Framework that describes the changes on the data if enriched or improved data are made available in the Genome EDIC.

Justification: Efforts invested into the data, which makes them more usable for secondary use should not be lost. In exchange for benefitting from access to the subsidised 1+MG, the Users should allow the improvements made to be reutilised.

Having the possibility to pursue rectification is particularly relevant where Users operate directly on a dataset without having its own copy. In this case typically, the User will not be able to modify the data and would have to request explicitly any changes that would have to be done. This will be the case for healthcare reuse. Where healthcare professionals as Users can review individual records of patients, errors in the data may be detected. Such errors would have to be reported to be fixed. The rectification can take place under the legal basis provided through the GDPR Article 5.d and could be performed directly on the dataset used. Rectification also allows the Genome EDIC and 1+MG Data Holders to modify their data in accordance with Article 5.d GDPR.

However, where Users modify an own copy of data themselves, e.g. introducing further harmonisation, such transfers are only possible if the enriched or improved data do not constitute personal data or if there is a lawful way to do so. This would require a consent to be obtained or a legislation to which the User is subject that permits the transfer of improved data back to the entity that disclosed the data to it for secondary use. However, so far this is only a hypothetical assumption.

EHDS Relevance: The EHDS Regulation envisages that enriched data should not be lost. The details of how this will be implemented is left to Member States and will require Member State law. Such Member State law, however, can only oblige the HDABs. They cannot oblige the health data user who is subject to the jurisdiction of its own country.

VIII.9.2. Enriched and Improved Data – Obligations: Genome EDIC CC

1. Must include specific details in the data use agreement templates addressing the return of specific enriched or improved data including a reference to the legal basis for such return, where applicable.

Justification: The obligations must be transparent and clear for Users.

EHDS Relevance: See above.

VIII.10. DATA SUBJECT RELATED DATA [healthcare only]

VIII.10.1. Data Subject Related Data – Obligations: Users

1. Must report back to the Genome EDIC their observations related to additional variants based on their own patient for a potential further use through the Genome EDIC (only if the connection between the User and the observation can be severed, ensuring that the

observation no longer qualifies as personal data related to the patient; otherwise consent has to be asked).

2. Must offer to their patients the option of becoming part of the 1+MG initiative and making their data available if the data can meet the necessary inclusion criteria.

Justification: The requirements above form some kind of benefit sharing. When benefitting from the Genome EDIC services, Users from healthcare should ideally also give something back. Any additional information or knowledge drawn from data outside the 1+MG data could be useful to other Users.

The inclusion of additional variant observations made by Users on their own patient could aid in the identification of rare variants and support the validation of clinical significance, as well as enhance diagnostic accuracy and variant interpretation for other Users. However, this is only easily possible where this does not constitute collection of personal data by the Genome EDIC. Where the link between the User and the reported observations can be severed on the side of the Genome EDIC, the information should no longer constitute data related to an identifiable data subject and could therefore be handled by the Genome EDIC without the requirement of a legal basis.

Where Users want to benefit from the Genome EDIC's services, they should also be ready to contribute, i.e. go through the exercise of becoming a 1+MG Data Provider.

EHDS Relevance: Health data users for a healthcare reuse under the EHDS mechanisms are likely not feasible; however, it is also not excluded. A health data user from healthcare would always also qualify as a health data holder for own patient data. Therefore, the inclusion into data that are made available under the EHDS Regulation is automatic unless the patient opts out. There is no for the above rules in the EHDS. However, the situation for the Genome EDIC is different.

VIII.10.2. Data Subject Related Data – Obligations: Genome EDIC CC

1. Must include specific details in the data use agreement templates addressing the reporting of observations and the obligation to ask their patient for data inclusion into the Genome EDIC Secondary Use Framework.

2. Must have a dedicated communication channel in place on the 1+MG User Portal to allow the reporting of the observations from healthcare secondary use.

Justification: Same as above, the Users must be aware of the special conditions that go along with the use of the Genome EDIC services.

A suitable reporting form for observations would minimise the collection of information and help the anonymisation of data on the side of the Genome EDIC.

EHDS Relevance: See above.

VIII.11. INCIDENTAL FINDINGS

VIII.11.1. Incidental Findings – Rights and Responsibilities: User

1. Must report certain clinically actionable incidental findings as defined within 1+MG to Genome EDIC if a flag for the return of incidental findings is set.

Justification
This follows ethical principles and a respect towards the data subjects giving the data for the secondary use and the data subjects' welfare. The flag will inform the User if findings the return is consented by the data subject and the country will actually return such findings.
NB: the legal basis for the User to return this information is not entirely ensured yet. This is subject to discussion with the data protection authorities for the interpretation of the GDPR.

EHDS Relevance
The EHDS Regulation provides for a return of incidental findings ("significant findings"). However, the provisions are in the scope of access provided through HDABs, i.e. referring to HDABs and health data holders. This indicates that the return of significant findings under the EHDS Regulation does not apply to findings made for data processing outside a data permit.

VIII.11.2. Incidental Findings – Rights and Responsibilities: Genome EDIC CC

1. Must establish, based on relevant guidance, a list of findings to be reported by Users and regularly update it.
2. Must provide for a possibility for Users to report incidental findings and engage with the 1+MG NCPs on the best possible solution to establish a communication chain.
3. Should implement the possibility of a "no return" flag that interrupts or even prevents the reporting where reporting is not wanted.

Justification: What counts as clinically actionable must be agreed upon based on relevant expert opinion and updated over time. The User will need clear instructions which findings fall under the reporting obligations. As the same rules should apply across all the activities, the Genome EDIC CC should oversee this process.

As the Genome EDIC CC is the main contact for the User, the Genome EDIC should provide for the possibility to report incidental findings. The information chain should be established according to DPbDD considerations.

Where no reporting should take place, the information chain should be immediately terminated, to avoid unnecessary burden and also a processing without a sufficient legal basis.

EHDS Relevance: The HDAB oversees further communicating the incidental findings downstream, which may even include direct information of data subjects. However, the Genome EDIC Secondary Use Framework needs to identify an own legal basis for the reporting as well as its own reporting channels where incidental findings are to be communicated. This envisaged communication, however, is limited to reaching the relevant entity on the national level that would be mandated based on local or national arrangements to inform about such findings.

VIII.11.3. Incidental Findings – Rights and Responsibilities: 1+MG NCP

1. Must establish jointly with the Genome EDIC CC and relevant entities on the national level a communication chain informing 1+MG Data providers on any requirements for incidental findings in the context of 1+MG on the national level.

Justification: Rather than predetermining an information flow, it is recommended to establish an approach for the communication jointly with the IT infrastructure providers as well as the Genome EDIC CC who serves as the main contact to the Users and is also in charge of the 1+MG User Portal.

EHDS Relevance: In the EHDS, the HDAB serves as the gateway for the communication of significant findings. The situation is less straightforward for 1+MG as the Genome EDIC serves as the contact towards the User, 1+MG NCPs serve as contact for the country and the processing takes place also within the country. Roles and responsibilities have to be further investigated to establish a communication, which meets a DPbDD approach. This should also be done in view of potential legislation passed by the Genome EDIC Member Countries in the context of the EHDS – where legislation deals with significant/incidental findings on a

general level and is not limited to the communication by HDABs, this will have an influence on responsibilities as well.

It is recommended that Member Countries include in the legislation on significant findings that these do not remain limited to the EHDS cases but open up a more general practice for incidental findings that also the Genome EDIC with its national entities could make use of.

VIII.11.4. Incidental Findings – Rights and Responsibilities: 1+MG Data Provider

1. should establish policies and processes to responsibly handle the incidental findings, subject to national policies and to the individual's consent/express wishes where this is not regulated by law. (see 1+MG Incidental Findings Policy). Where the 1+MG Data Provider is not in direct contact with the data subject, e.g. when data were obtained from another entity, the entity that collected the data directly from the data subject may have to take over some of the above tasks.
2. obtain consent by data subjects on the communication of incidental findings.

Justification: The 1+MG Data Provider is the one who collected the data from the data subject or is at least in direct contact with the data subjects and obtained the consent by the data subject and therefore the one to establish how, in accordance with local policies, incidental findings should be handled. However, where such findings are regulated by law, there may be no need to set up relevant processes and responsibilities.

It will nevertheless be needed to obtain the preferences of the data subjects if or if not, such findings should be communicated. Even where a legislation is in place, a consent will still be needed to create a legal basis for the User to report the finding.

EHDS Relevance: The same considerations as above apply.